Ionospheric Scintillation Monitoring with the USRP X310

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Last update: October 28, 2026

Abstract

This report describes the monitoring of ionospheric scintillation with GNSS signals recorded by a USRP X310 device. The receiver is placed in a fixed position in São José dos Campos, SP, Brazil, in the GPS/GNSS Laboratory at Aeronautics Institute of Technology (ITA)[1], close to the southern crest of the equatorial ionization anomaly. Every evening, in the post-sunset hours when equatorial scintillation occurs, raw I/Q samples of several GNSS bands are recorded and processed with a software receiver. From its correlator outputs, the amplitude (S_4) and phase (σ_ϕ) scintillation indices are computed, and only the recordings that show moderate or strong scintillation are stored permanently. The report is updated as new recordings are made.



Figure 1: Location of the GNSS antenna.

[1]https://github.com/ita-gnss-lab

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Part I
Methodology

This part describes how the data is collected: the recording methodology, which keeps only the recordings that contain ionospheric scintillation, and how each recording is checked with the GNSS-SDR software receiver.

1 Data collection routine

The recording is made automatically by a Linux service that follows these guidelines:

- Every day, the program starts within the time range [9PM, 11PM] (local time), i.e., the post-sunset hours in which equatorial plasma bubbles cause scintillation at low latitudes.
- It senses whether the USRP is detectable via `uhd_find_devices --args=''addr=192.168.10.2''`.
- If not found, it exits silently and senses again in intervals of 15 min.
- Once found, two recordings are made, each with a duration of 5 minutes (300 s). The first recording (RC0) happens right after the USRP is detected, and the second recording (RC1) is made 30 min after the first one. After that, no more recordings are made for that day.
- Every recording has two channels (ch0 and ch1), one per GNSS band. The two daily recordings, however, cover distinct band combinations and operate their own sampling rate, as shown in Table 1. Every channel uses a gain of 25 dB.
- RC0 and RC1 are saved as separated SigMF archives. See Section 2 for details.
- Each recording is processed by the GNSS-SDR[2] software receiver right after it is made. The final PVT solutions obtained from each recording are listed in Table 2.
- Before the GNSS-SDR processing, some preprocessing steps described in Section 3 are performed.
- The recorded data during the last 15 days are kept in NAS's temporary directory, while their respective receiver stage outputs (from acquisition to PVT) are appended to Part III. From the prompt correlator samples, the scintillation indices are also computed according to Section II, and each recording is classified as either weak, moderate or strong scintillation severity.
- This report is updated every day and automatically, with the daily recordings moved to the NAS storage and their GNSS-SDR outputs appended to Part III.

[2] https://gnss-sdr.org, v0.0.20.

- Every 15 days, right after the periodic report update is triggered, the following steps are performed:
  1. the report is sent via WhatsApp to the head of the Laboratory, Prof. Dr. Felix Antreich, together with a summary of the recorded metadata.
  2. the recordings of the past 15 days classified with moderate or strong scintillation are moved to the permanent NAS storage, and only their respective scintillation indices are appended to Part IV;
  3. the leftover recordings in NAS's temporary directory is deleted, and Part III is cleared so that the next 15 days of recordings can be appended to it.

Table 1: Channel configuration of the two daily recordings. The → indicates the preprocessing steps made on the recorded data before the GNSS-SDR processing.

Recording	**Channel**[3]	**Band**	**Freq. (MHz)**	**Rate (MS/s)**	**I/Q format**
First (RC0)	ch0	L1	1575.42	4 → 20	16 → 8-bit
	ch1	L5	1176.45	20	16 → 8-bit
Second (RC1)	ch0	L1	1575.42	4	16-bit
	ch1	L2	1227.60	4	16-bit

[3]In the context of the TwinRX daughterboard, ch0 and ch1 are referred to as RX0 and RX1, respectively.

Table 2: Final PVT solutions of each daily recording.

Recording	**Solution**	**Signals**	**GNSS-SDR configuration file**	**Comments**
First (RC0)	sol0	GPS L1 C/A (ch0)	`l1ca8bits4msps_l5_8bits20msps.conf`	RC0-ch1 (tuned to L5) is ignored
	sol1	GPS L1 C/A (ch0) + L5 (ch1)		Because GNSS-SDR's dual-frequency solver was unstable with few L5-capable GPS satellites in view, this PVT was computed using `gnss_spp_lib.py` from the observables and decoded ephemeris of the GNSS-SDR run. L1 C/A is first upsampled from 4 MS/s to 20 MS/s before being processed by GNSS-SDR. See Section 3.2
	sol2	Galileo E1-B/C (ch0) + E5a (ch1)	`e1_8bits4msps_e5a_8bits20msps.conf`	E1-B/C is first upsampled from 4 MS/s to 20 MS/s before being processed by GNSS-SDR. See Section 3.2
Second (RC1)	sol0	GPS L1 C/A + Galileo E1-B/C (both from ch0)	`l1ca16bits4msps_l2c16bits4msps.conf`	RC1-ch1 (tuned to L2) is ignored
	sol1	GPS L1 C/A (ch0) + L2C (ch1)		Since GNSS-SDR[4] has no PVT solution using L2C, it was computed by `gnss_spp_lib.py` from the observables and decoded ephemeris of the GNSS-SDR run.

[4]As of GNSS-SDR v0.0.20, the version used in this work.

Note For the sake of conciseness, the PVT solutions will be referred to as "RC0-sol0", "RC0-sol1", "RC0-sol2", "RC1-sol0" and "RC1-sol1" hereafter.

2 SigMF format

The recordings are stored in the Signal Metadata Format (SigMF)[5], an open standard for sharing recorded I/Q samples together with the information needed to interpret them. A SigMF recording consists of two files: a binary `.sigmf-data` file holding only the raw samples, and a JSON `.sigmf-meta` file describing them. The metadata is split into a `global` object (sample format, sampling rate, hardware, checksum, etc.), a `captures` list (center frequency and start time of each contiguous block of samples), and an `annotations` list (optional labels over sample ranges). Related recordings, such as the channels of one multichannel acquisition, are grouped by a `.sigmf-collection` file, and all these files can be bundled into a single `.sigmf` archive, which is a plain `tar` file.

Each dual-channel recording is therefore stored as one archive with one recording per channel. For example, the contents of the first recording of 26/10/2026 are listed below:

```
tar -tvf recordings/x310_20261026T190001Z.sigmf | numfmt --field=3 --to=iec-i --suffix=B

-rw-r--r-- tapyu/tapyu    572B 2026-10-26 16:05 x310_20261026T190001Z/x310_20261026T190001Z.sigmf-collection
-rw-r--r-- tapyu/tapyu  1.5KiB 2026-10-26 16:05 x310_20261026T190001Z/x310_20261026T190001Z_ch0.sigmf-meta
-rw-r--r-- tapyu/tapyu  2.1KiB 2026-10-26 16:05 x310_20261026T190001Z/x310_20261026T190001Z_ch1.sigmf-meta
-rw-r--r-- tapyu/tapyu  2.3GiB 2026-10-26 16:05 x310_20261026T190001Z/x310_20261026T190001Z_ch0.sigmf-data
-rw-r--r-- tapyu/tapyu   12GiB 2026-10-26 16:05 x310_20261026T190001Z/x310_20261026T190001Z_ch1.sigmf-data
```

The collection file ties the two channels together, and each channel has its own metadata and data files. The data file sizes follow from the 300-s duration and the channel configuration of Table 1: channel 0 (L1) holds 4 MS/s $\times$ 300 s $\times$ 2 bytes = 2.4 GB, and channel 1 (L5) holds 20 MS/s $\times$ 300 s $\times$ 2 bytes = 12 GB.

3 Preprocessing steps

After the recording is made, the following preprocessing steps are performed before the GNSS-SDR processing:

3.1 Requantization from 16 to 8 bits

On the one hand, the OTW data format for USRP X310 is sc16, *i.e.*, 16-bit signed integers for each I/Q component, totaling 32 bits (4 byte) per complex baseband sample. On the other hand, the UHD (USRP Hardware Driver) library determines the CPU data format, which is internally handled by the Python's `uhd` package. However, UHD 4.6 (the one used in the recording) also has no sc16 $\rightarrow$ sc8 converter on the host

[5] https://sigmf.org.

side. Therefore, it is manually done in `x310usrp_record.py` for RC0 (see Table 1).

The requantization is applied right after the USRP recording, so that the stored data file is kept as small as possible.

3.2 Upsampling for multi-rate channels

When reading from a file, GNSS-SDR expects all channels to run at the same rate. `Multichannel_File_Signal_Source` (in the GNSS-SDR's config file) ensures this by reading several raw I/Q files through a single block with one output stream per file. By using the terminal output shown in Section 2 example, they are `x310_20261026T190001Z_ch0.sigmf-data` for ch0 and `x310_20261026T190001Z_ch1.sigmf-data` for ch1. On each call, that block delivers the same number of samples to every stream, so the two bands advance in lockstep[6].

When the channels run at the same rate, this procedure is straightforward. However, the data source for RC0-sol1 and sol2 operate at different rates: 4 MS/s for ch0 and 20 MS/s for ch1. Therefore, `upsample_iq.py` upsamples via zero-order hold the file `x310_20261026T190001Z_ch0.sigmf-data` to 20 MS/s, generating a new temporary file. Then, the upsampled version of ch0 and `x310_20261026T190001Z_ch1.sigmf-data` is used as input to GNSS-SDR. The upsampling factor is 5, so every sample of ch0 is repeated 5 times in the new file.

The upsampling is done right before the GNSS-SDR processing, generating a temporary file that is deleted right after the processing is finished.

4 Outputs for the temporary GNSS data

Before looking for scintillation-disturbed data, one must check whether the temporary recordings hold usable GNSS signals. Otherwise, an amplitude or phase fluctuation could come from a faulty recording (*e.g.*, dropped samples, a wrong gain or a missing reference clock) rather than from scintillation. Therefore, RC0-sol0, RC0-sol1, RC0-sol2, RC1-sol0, and RC1-sol1 are all validated at every stage of the GNSS-SDR receiver chain, *i.e.*, signal acquisition and tracking, navigation message decoding, and PVT computation. In addition, the scintillation indices must also be computed, which should be near to zero for nominal operation. In this way, high scintillation indices can be trusted to come from scintillation, rather than a recording problem.

Figure 2 shows the data chain of GNSS-SDR. Each channel (one per satellite signal) runs its own acquisition, tracking and telemetry decoder; observables and PVT combine all of them.

[6]With one independent source per band, one band runs ahead of the other and its measurements are discarded.

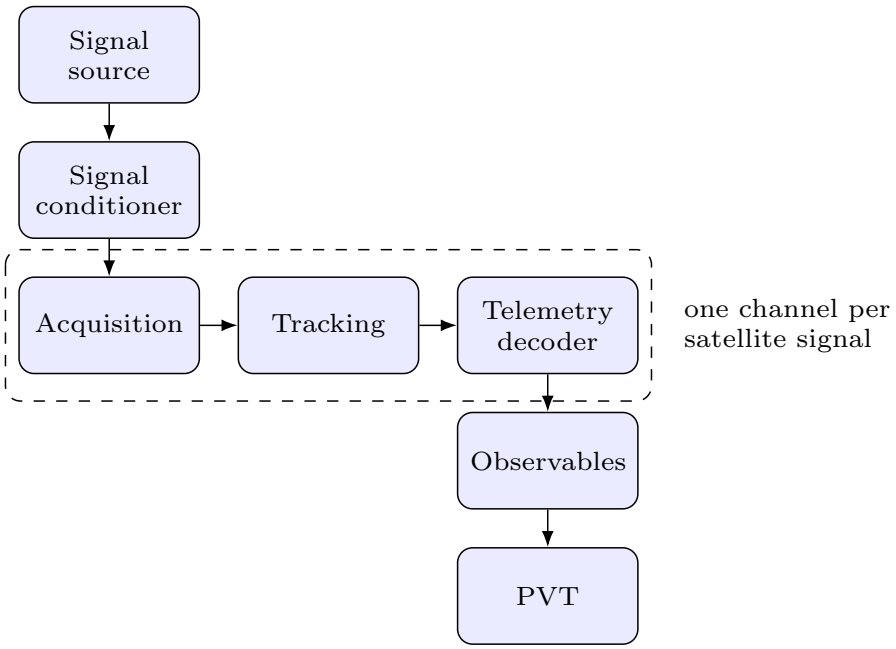


Figure 2: GNSS-SDR receiver chain.

4.1 Signal source

Reads the recorded I/Q samples (here, coming from the data files) and delivers them as a stream, one per channel. GNSS-SDR has no output of its own here, so the recording itself is examined.

- **Power spectral density** per channel: the shape of the receiver passband over a flat thermal-noise floor. GNSS signals are 15–20 dB below the noise, so they are invisible here; a narrow peak would be interference.
- **I/Q histograms**: should be Gaussian, centred on zero, and far from the integer limits (no clipping).
- **Table**: format, rate and gain per channel, I/Q standard deviation and mean, percentage of clipped values, and samples lost during the recording (overflows, zero-filled).

4.2 Signal conditioner

Adapts the input samples (from the binary file) to what the channels process, *i.e.*, converts the file format (signed 8- or 16-bit interleaved I/Q) to complex floating point (data type adapter for C++), optionally filters (input filter) and resamples (resampler).

- **Table** per channel: file format, data type adapter, filter, resampling, and rate before and after.

4.3 Acquisition

Detects which satellites are present and gives a first, coarse estimate of each one's code delay and Doppler shift, by correlating the signal with the satellite's spreading code over a two-dimensional grid of delays and Doppler frequencies, and comparing the peak with a detection threshold (set by the probability of false alarm). With dual-frequency assistance, the second band (GPS L5 or L2C, Galileo E5a) is only searched for satellites already tracked on L1/E1, around their L1/E1 Doppler scaled to the second frequency. Galileo E1 uses a 4-ms code (BOC(1,1) modulation), so its correlation peak is narrower and has side peaks.

- **Delay–Doppler map** (surface plot) of the strongest detection per band: a single sharp peak well above the floor.
- **Table**: search attempts, detections and detected satellites, and the best test statistic relative to the threshold. Attempts on satellites that are not in view, or that do not broadcast that band (L5 is only on Block IIF and III satellites, L2C on IIR-M, IIF and III), are rejected.

4.4 Tracking

Follows each detected signal with a delay-lock loop (code) and a phase-lock loop (carrier), producing every 1–20 ms the prompt correlator output (which carries the navigation data bits), the carrier-to-noise density ratio C/N_0, the Doppler, the code, and carrier phases.

- C/N_0 **and Doppler versus time** per satellite and band: C/N_0 of roughly 35–50 dB-Hz for an open-sky antenna, and smooth Doppler curves following the satellite motion.
- **Prompt I/Q scatter** of the strongest satellite per band: with the carrier phase locked, the energy sits on the in-phase axis in two clusters (the ±1 data bits), or in for the dataless pilot components that Galileo is tracked on (E1-C, E5a-Q), the ±1 chips of their secondary code; without phase lock it would be a ring. Each point is one prompt correlator output, the received samples $x[n]$ multiplied by the local code replica c aligned at the estimated code delay $\hat{\tau}$ and by the local carrier replica (estimated Doppler $\hat{f}_D$ and phase $\hat{\phi}$), summed over the N samples of one integration period (T_s: sampling period):

$$P = \sum_{n=1}^{N} x[n]\, c[n-\hat{\tau}]\, e^{-j(2\pi \hat{f}_D n T_s + \hat{\phi})} = I + jQ. \tag{1}$$

- **Loop errors versus time** per satellite and band: the PLL carrier-phase error (degrees) and the DLL code-delay error (chips), i.e. the discriminator outputs the loops drive to zero. Noise centred on zero means clean tracking; jumps would mark cycle slips or a loop losing lock.
- **Table** per satellite: median C/N_0, percentage of time phase-locked, losses of lock, time tracked, and the standard deviation of the PLL and DLL errors.

4.5 Telemetry decoder

Synchronizes to the navigation-message bits and frames, checks their parity/CRC (Cyclic Redundancy Check), decodes the ephemeris (orbit and satellite clock), and time-stamps every measurement with the satellite's transmit time (time of week). GPS L1 C/A carries the legacy LNAV message (6-s subframes); L2C and L5 carry CNAV (12-s and 6-s messages). Galileo E1-B carries I/NAV (2-s pages) and E5a-I F/NAV (10-s pages).

- **Timeline** per band and satellite of when a valid transmit time was available.
- **Table**: messages decoded, ephemeris obtained, and when the first valid time of week came.

4.6 Observables

Aligns all channels on common receiver epochs and forms, for each satellite and band, the pseudorange (from transmit and receive times), the accumulated carrier phase and the Doppler. These are the measurements a positioning algorithm uses; GNSS-SDR also writes them to a RINEX observation file.

- **Availability** of valid pseudoranges per band and satellite over the recording.
- **Second band minus L1/E1 pseudorange** per satellite: the same constant for every satellite (the receiver's inter-band hardware bias) plus the slowly varying, satellite-dependent ionospheric delay difference (metres). It proves that both bands measure the same satellites consistently.
- **Table** per satellite: common epochs, mean, standard deviation and drift of that difference.

4.7 PVT

Computes position, velocity and receiver clock (PVT) from the pseudoranges and the decoded ephemeris. The antenna position was surveyed: −23.209433°, −45.874875°, 596.42 m (WGS-84), so the errors below are true errors, not just scatter.

- **Horizontal error scatter and east/north/up error versus time** for GNSS-SDR's own solution of each run and, for the GPS runs, `gnss_spp_lib.py`'s dual-frequency ionosphere-free solution. Single-frequency single-point positioning is expected to stay within a few metres horizontally and ∼10 m vertically; the iono-free combination removes the ionospheric bias but amplifies the code noise about three times.
- **Sky plot** of the GPS satellites used by `gnss_spp_lib.py`.
- **Table**: mean error (accuracy), standard deviation (precision), 2-D and 3-D RMS, and 95% horizontal error.

Part II
Scintillation index computation

This part introduces the signal model assumed for the recorded data and the procedure to compute the amplitude (S_4) and phase (σ_ϕ) scintillation indices from the correlator outputs of the software receiver, following [4–6]. It ends with the three scintillation severity classes that decide which recordings are stored permanently.

5 Signal model

The signals of the K satellites in view on the carrier frequencies $f_{c,l}$, $l = 1, \ldots, L$, are well separated by their spreading codes and frequencies. Therefore, it is sufficient to consider the signal of the kth satellite on the lth carrier on its own [6]:

$$y_{\mathrm{RF},kl}(t) = A_{kl}(t)\, c_{kl}\big(t-\tau_{kl}(t)\big)\, d_{kl}\big(t-\tau_{kl}(t)\big) \cos\big(2\pi f_{c,l}\, t + \phi_{kl}(t)\big) + \eta(t), \quad (2)$$

where $A_{kl}(t)$ is the signal amplitude, $c_{kl}(t)$ and $d_{kl}(t)$ are the spreading code and navigation data, $\tau_{kl}(t)$ is the signal delay, $\phi_{kl}(t)$ is the carrier phase (which includes the Doppler shift), and $\eta(t)$ is zero-mean white Gaussian noise. The TwinRX daughterboards of the X310 filter and downconvert each band directly to baseband and sample it at $f_s = 1/T_s$, so that each channel of a recording holds the complex baseband samples (subscripts k, l omitted from now on)

$$y[n] = A[n]\, e^{j\phi[n]}\, c(nT_s - \tau[n])\, d(nT_s - \tau[n]) + \eta[n] = y_{\mathrm{I}}[n] + j y_{\mathrm{Q}}[n], \quad (3)$$

with the short notations $A[n] = A(nT_s)$, $\tau[n] = \tau(nT_s)$, $\phi[n] = \phi(nT_s)$. The phase $\phi[n]$ also holds the instabilities of the X310's local oscillator, which is disciplined by an external 10 MHz reference. Ionospheric scintillation shows up as fast fluctuations of the amplitude $A[n]$ (amplitude scintillation) and of the phase $\phi[n]$ (phase scintillation).

Phase and delay model. The carrier phase and the signal delay can be decomposed into [6]

$$\phi(t) = \frac{2\pi}{\lambda_{\mathrm{c}}}\Big(r(t) + c\big(\tau^{\mathrm{clk,s}}(t) - \tau^{\mathrm{clk,r}}(t)\big) + \tau^{\mathrm{tro}}(t) - \tau^{\mathrm{ion}}(t)\Big) + \eta^{\phi}, \quad (4)$$

$$\tau(t) = \frac{1}{c}\Big(r(t) + c\big(\tau^{\mathrm{clk,s}}(t) - \tau^{\mathrm{clk,r}}(t)\big) + \tau^{\mathrm{tro}}(t) + \tau^{\mathrm{ion}}(t)\Big) + \eta^{\tau}, \quad (5)$$

where $\lambda_{\mathrm{c}} = c/f_{\mathrm{c}}$, c is the speed of light, $r(t)$ is the line-of-sight distance from the satellite to the receiver, $\tau^{\mathrm{clk,s}}(t)$ and $\tau^{\mathrm{clk,r}}(t)$ are the satellite and receiver clock offsets, $\tau^{\mathrm{tro}}(t)$ and $\tau^{\mathrm{ion}}(t)$ are the tropospheric and ionospheric delays, and η^{ϕ}, η^{τ} are higher-order noise processes. The ionosphere advances the phase and delays the code, hence the opposite signs.

To isolate the scintillation, the slow terms (geometry, clocks, troposphere and the mean ionospheric delay) must be removed: this is the purpose of the detrending steps below.

6 Correlator outputs

The tracking loops of the receiver (Section 4) wipe off the carrier with their numerically controlled oscillator (NCO) and correlate the result with a local code replica, over a coherent integration time $T_{\mathrm{coh}} = M_s T_s$ of M_s samples. The mth prompt correlator output reads [6]

$$I[m] = \frac{M_s \bar{A}[m]\, d[m]}{2} R_c(\tau_{\mathrm{e}}[m]) \cos\left(2\pi f_{\mathrm{e}}[m]\, m T_{\mathrm{coh}} + \phi_{\mathrm{e}}[m]\right) + \eta_{c,\mathrm{I}}[m], \tag{6}$$

$$Q[m] = \frac{M_s \bar{A}[m]\, d[m]}{2} R_c(\tau_{\mathrm{e}}[m]) \sin\left(2\pi f_{\mathrm{e}}[m]\, m T_{\mathrm{coh}} + \phi_{\mathrm{e}}[m]\right) + \eta_{c,\mathrm{Q}}[m], \tag{7}$$

where $\bar{A}[m]$ is the mean amplitude within the integration interval, $d[m] \in \{-1, 1\}$ is the data symbol, $R_c(\cdot)$ is the code autocorrelation function, $\tau_{\mathrm{e}}[m]$, $f_{\mathrm{e}}[m]$ and $\phi_{\mathrm{e}}[m]$ are the code-delay, frequency and phase errors of the loops, and $\eta_{c,\mathrm{I}}[m]$, $\eta_{c,\mathrm{Q}}[m]$ are zero-mean Gaussian noises of variance $\sigma_n^2/2$ each, with $\sigma_n^2 = N_0/T_{\mathrm{coh}}$. These are the prompt I/Q outputs dumped by GNSS-SDR's tracking. GNSS-SDR integrates over 1 ms during pull-in and over up to 20 ms once bit-synchronized (10 ms for the L5 and E5a pilots). The scintillation indices are computed from M consecutive outputs such that $M\,T_{\mathrm{coh}} = 20$ ms, i.e., a 50 Hz signal intensity that never straddles a GPS L1 C/A data-bit transition [3].

7 Amplitude scintillation index S_4

The amplitude scintillation index is the standard deviation of the signal intensity normalized by its mean [1]:

$$S_4 = \sqrt{\frac{\mathrm{E}\left(\rho^2[b]\right) - \mathrm{E}\left(\rho[b]\right)^2}{\mathrm{E}\left(\rho[b]\right)^2}}, \tag{8}$$

where $\rho[b]$ is the signal intensity of the bth 20-ms interval and $\mathrm{E}(\cdot)$ is the expectation, replaced, under ergodicity, by the time average over 60 s.

Signal intensity. Under scintillation, the loop errors are not negligible and the signal energy spreads over I and Q, so both are used. The narrow-band and wide-band powers of the bth interval are [5, 6]

$$P[b] = \Big(\sum_{m=1}^{M} I[m]\Big)^2 + \Big(\sum_{m=1}^{M} Q[m]\Big)^2, \qquad \tilde{P}[b] = \sum_{m=1}^{M} \left(I^2[m] + Q^2[m]\right), \tag{9}$$

and their difference $\rho[b] = P[b] - \tilde{P}[b]$ is the raw signal intensity, whose expectation is proportional to $\bar{A}^2[b]$, without the noise-power offset that $\tilde{P}[b]$ alone would carry.

Detrending. The intensity also varies because of the satellite motion (antenna gain, path loss) and multipath, which are not scintillation. Therefore, $\rho[b]$ is filtered by a sixth-order Butterworth low-pass filter $h[b]$ with a 0.1 Hz cut-off frequency, and normalized by the result [5]:

$$\bar{\rho}[b] = h[b] * \rho[b], \qquad \rho_{\mathrm{det}}[b] = \frac{\rho[b]}{\bar{\rho}[b]}. \tag{10}$$

$\rho_{\mathrm{det}}[b]$ replaces $\rho[b]$ in the S_4 estimate below.

Ambient noise removal. Even with a constant amplitude, the noise makes the estimated intensity fluctuate, which biases S_4 upwards. For a constant amplitude, this noise-induced index is [6]

$$S_{4,N_0} = \sqrt{\frac{2}{M T_{\mathrm{coh}}\, C/N_0}\left(1 + \frac{1}{2(M-1)\, T_{\mathrm{coh}}\, C/N_0}\right)}, \tag{11}$$

with C/N_0 in linear units (Hz). For $T_{\mathrm{coh}} = 1$ ms and $M = 20$, it reduces to the well-known $S^2_{4,N_0} = \frac{100}{C/N_0}\left(1 + \frac{500}{19\, C/N_0}\right)$ [5]. The corrected index, using the 60-s average of the C/N_0 estimated by the tracking loops, is

$$S_4 = \sqrt{\frac{\mathrm{E}\left(\rho^2_{\mathrm{det}}[b]\right) - \mathrm{E}\left(\rho_{\mathrm{det}}[b]\right)^2}{\mathrm{E}\left(\rho_{\mathrm{det}}[b]\right)^2} - \frac{2}{M T_{\mathrm{coh}}\, C/N_0}\left(1 + \frac{1}{2(M-1)\, T_{\mathrm{coh}}\, C/N_0}\right)}, \tag{12}$$

set to zero when the radicand is negative (no scintillation at all). One value is computed every 60 s per satellite and band, i.e., five per satellite and band in a 5-min recording.

8 Phase scintillation index σ_ϕ

The phase scintillation index is the standard deviation of the detrended carrier phase, over 60 s [4]:

$$\sigma_\phi = \sqrt{\mathrm{E}\left(\phi^2\right) - \mathrm{E}\left(\phi\right)^2}. \tag{13}$$

The phase measured by the ith tracking channel holds, besides the scintillation phase $\phi_{\mathrm{sci},i}$, the phase due to the satellite motion $\phi_{\mathrm{sat},i}$, the oscillator instability ϕ_{clk} (common to all channels), and white noise $\eta_{\phi,i}$ [6]:

$$\phi_{\mathrm{e},i} = \phi_{\mathrm{sat},i} + \phi_{\mathrm{clk}} + \phi_{\mathrm{sci},i} + \eta_{\phi,i}. \tag{14}$$

These terms are removed in three steps [6]:

1. **Clock error removal.** The oscillator phase noise is not removed by filtering and would dominate σ_ϕ, which is why scintillation monitors use oven-controlled oscillators [5]. Since ϕ_{clk} is common to all channels, it cancels in the difference to a reference channel whose satellite shows no phase scintillation (e.g., the highest one):

$$\phi_{\mathrm{e},i\text{-ref}} = \phi_{\mathrm{e},i} - \phi_{\mathrm{e,ref}} = \phi_{\mathrm{sat},i} - \phi_{\mathrm{sat,ref}} + \phi_{\mathrm{sci},i} + \eta_{\phi,i} - \eta_{\phi,\mathrm{ref}}. \tag{15}$$

2. **Geometry error removal.** For a static receiver, $\phi_{\mathrm{sat},i} - \phi_{\mathrm{sat,ref}}$ is well modelled by a third-order polynomial over a few minutes. Subtracting a third-order polynomial fit leaves $\phi_{\mathrm{det},i} \approx \phi_{\mathrm{sci},i} + \eta_{\phi,i} - \eta_{\phi,\mathrm{ref}}$.
3. **Low-frequency component removal.** The remaining slow variations, which are not due to scintillation, are removed by a sixth-order Butterworth high-pass filter (a cascade of three second-order sections) with a 0.1 Hz cut-off frequency, and σ_ϕ is computed from its output.

9 Scintillation severity classes and storage

The scintillation indices are compared with preset thresholds, which are empirical and vary with the application [2, 4]. Following the usual amplitude-based detection, every 60-s S_4 value is assigned to one of the three severity classes of Table 3 [2]. To exclude multipath and noisy measurements, only satellites above a 30° elevation mask with $C/N_0 \geq 37$ dB-Hz are considered, and the index must stay within the class thresholds for at least 30 s. A recording takes the highest class reached by any of its satellites and bands. σ_ϕ is reported alongside S_4, but it does not decide the class, since it depends on the stability of the receiver's oscillator.

Class	S_4 **(60 s)**	**Storage of the recording**
No (or weak) scintillation	$S_4 \leq 0.2$	deleted after the 15-day cycle
Moderate scintillation	$0.2 < S_4 \leq 0.5$	permanent storage (Part IV)
Strong scintillation	$S_4 > 0.5$	permanent storage (Part IV)

Table 3: Scintillation severity classes and what is done with each recording at the end of its 15-day cycle.

Thus, only the recordings with moderate or strong scintillation are kept: they are moved to the permanent storage, while the recordings with no scintillation are deleted (Methodology, Part I).

Part III
Temporary GNSS data

This part holds the recordings of the last 15 recording days (days with at least one recording), with all the outputs derived from them so far. This data is temporary: at the end of each 15-day cycle, only the recordings with moderate or strong scintillation (Section 9) are moved to the permanent storage (Part IV), the others are deleted, and this part is overwritten with the new 15 days. Older days are kept in the repository history.

Table 4 shows the days of this 15-day window when recordings were made. The color of each day indicates how many recordings were made on that day: green means 2 recordings, blue means 1 recording, and no color means no recording was made.

October

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Table 4: Temporary recording calendar.

10 Records on 08/10/2026

10.1 First record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-08 22:26:37 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 5: Metadata for recording `x310_20261008T222630Z`, captured starting 2026-10-08T22:26:37.215640Z.

11 Records on 13/10/2026

11.1 First record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-13 21:57:56 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 6: Metadata for recording `x310_20261013T215749Z`, captured starting 2026-10-13T21:57:56.746726Z.

11.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-13 22:08:41 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 7: Metadata for recording `x310_20261013T220835Z`, captured starting 2026-10-13T22:08:41.589855Z.

12 Records on 22/10/2026

12.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-22 19:00:06 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	2912	14924
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`, `e1_8bits4msps_e5a_8bits20msps.conf`	

Table 8: Metadata for recording `x310_20261022T190000Z`, captured starting 2026-10-22T19:00:06.973177Z.

12.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

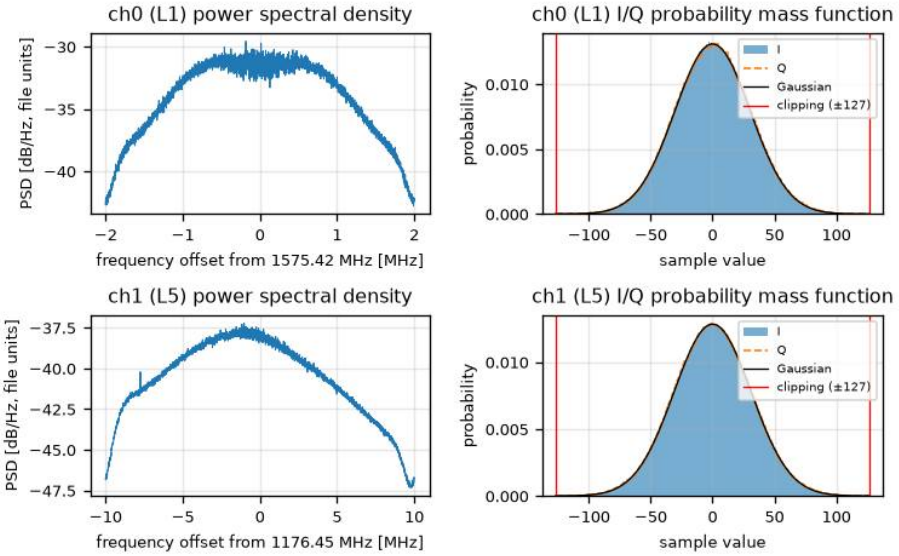


Figure 3: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Bytes per complex sample	2 (1 for I, 1 for Q)	2 (1 for I, 1 for Q)
Full scale	±127	±127
Gain [dB]	25	25
Std I / Q	30.4 / 30.5	31.1 / 31.1
Mean I / Q	-0.26 / 0.20	0.26 / -0.07
Min / max I	-127 / 127	-127 / 127
Min / max Q	-127 / 127	-127 / 127
Headroom[7]	4.2	4.1
Clipped [%]	0.003	0.004
Lost samples	2,912	14,924

Table 9: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[7]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

12.1.2 Signal conditioner

Channel	ch0 (L1 C/A)	ch1 (L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 10: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

12.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

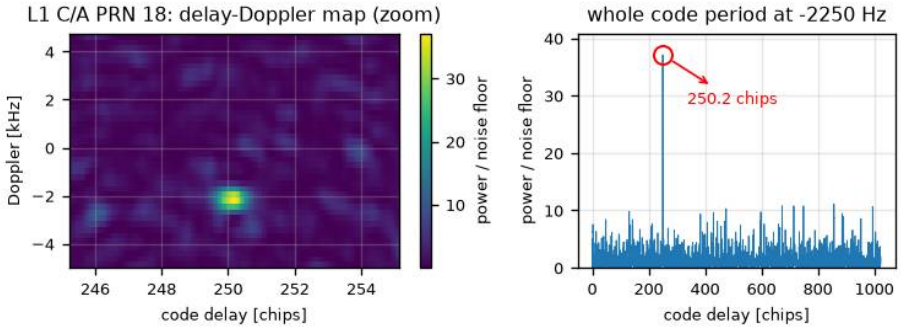


Figure 4: Acquisition, L1 C/A: delay–Doppler search of PRN 18; one sharp peak (test statistic 50 vs threshold 47) at -2250 Hz and 250.2 chips.

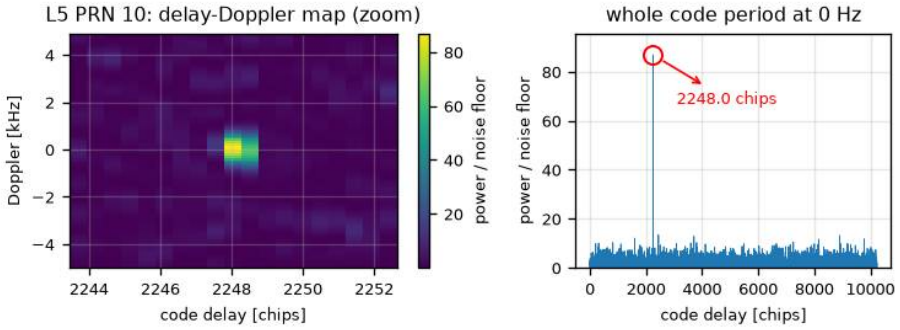


Figure 5: Acquisition, L5: delay–Doppler search of PRN 10; one sharp peak (test statistic 122 vs threshold 49) at -1491 Hz and 2248.0 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[8]
L1 C/A	9	1	18	1.1
L5	1	1	10	2.5

Table 11: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[8]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

12.1.4 Tracking

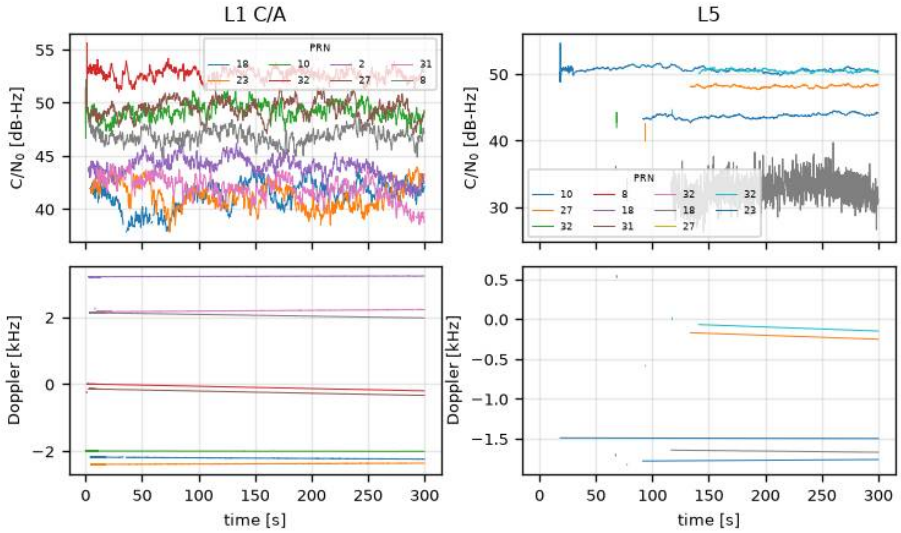


Figure 6: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

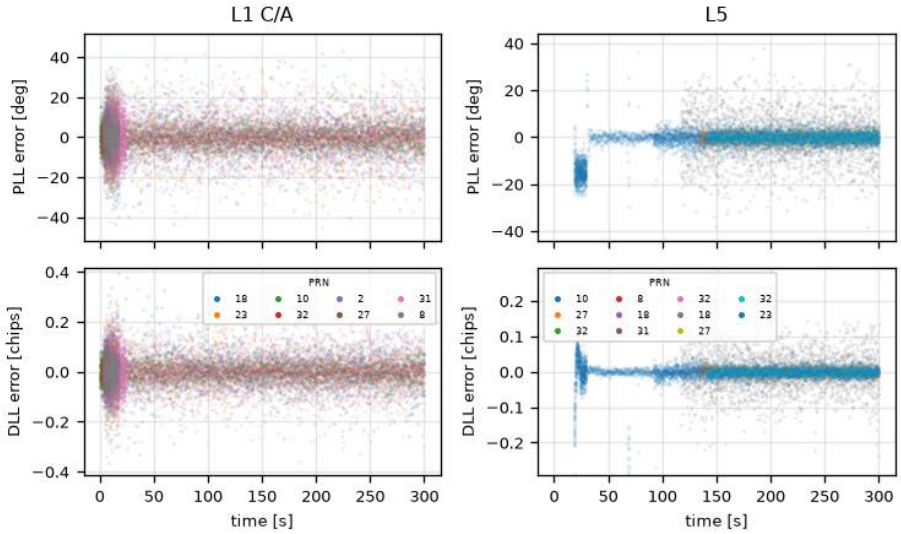


Figure 7: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

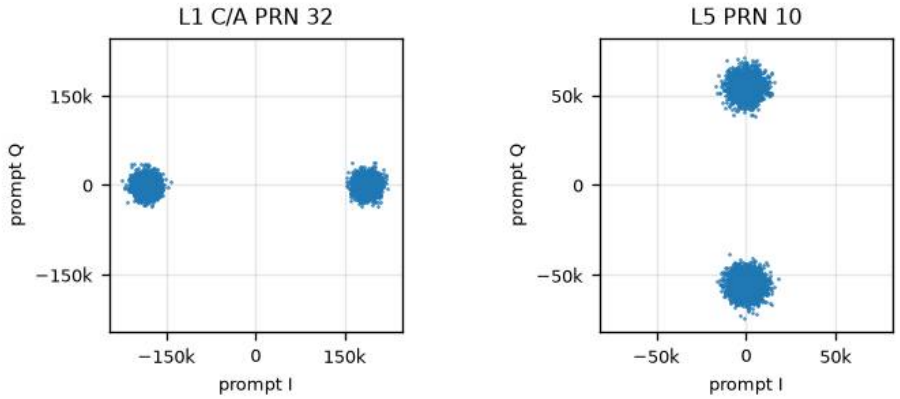


Figure 8: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	18	41.7	93	0	298	12.1	0.090
L1 C/A	1	23	41.9	93	0	297	12.1	0.088
L1 C/A	2	10	49.2	100	0	300	5.0	0.037
L1 C/A	3	32	52.9	100	0	299	3.4	0.025
L1 C/A	4	2	43.9	95	0	299	9.1	0.067
L1 C/A	5	27	49.1	94	0	299	5.7	0.035
L1 C/A	6	31	42.7	94	0	292	11.0	0.080
L1 C/A	7	8	47.0	94	0	298	6.5	0.049
L5	8	10	50.7	100	0	282	7.2	0.028
L5	10	27	48.0	37	2	206	2.5	0.018
L5	10	32	29.0	0	2	16	52.4	0.071
L5	11	8	28.4	0	2	16	–	–
L5	11	18	33.5	0	2	16	27.8	0.200
L5	12	31	29.0	0	2	16	–	–
L5	12	32	28.7	0	2	16	–	–
L5	13	18	32.8	60	1	206	10.8	0.056
L5	13	27	28.8	0	1	16	–	–
L5	14	32	50.3	36	1	198	2.4	0.015
L5	15	23	43.7	63	0	232	3.3	0.019

Table 12: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

12.1.5 Telemetry decoder

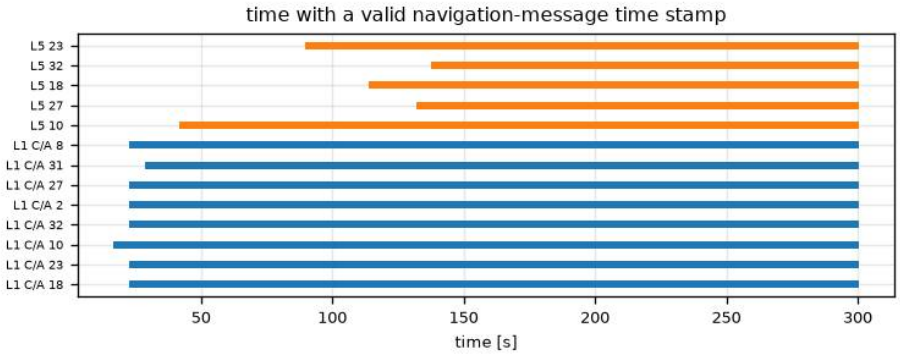


Figure 9: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
L1 C/A	2	47	yes	23.1	414048.0
L1 C/A	8	45	yes	23.1	414048.0
L1 C/A	10	43	yes	17.1	414042.0
L1 C/A	18	47	yes	23.1	414048.0
L1 C/A	23	47	yes	23.1	414048.0
L1 C/A	27	47	yes	23.1	414048.0
L1 C/A	31	46	yes	29.1	414054.0
L1 C/A	32	47	yes	23.1	414048.0
L5	10	26	yes	42.2	414067.1
L5	18	24	yes	114.4	414139.3
L5	23	27	yes	90.0	414114.9
L5	27	17	yes	132.3	414157.2
L5	32	16	yes	138.2	414163.1

Table 13: Telemetry decoder: L1 C/A LNAV subframes, L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

12.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

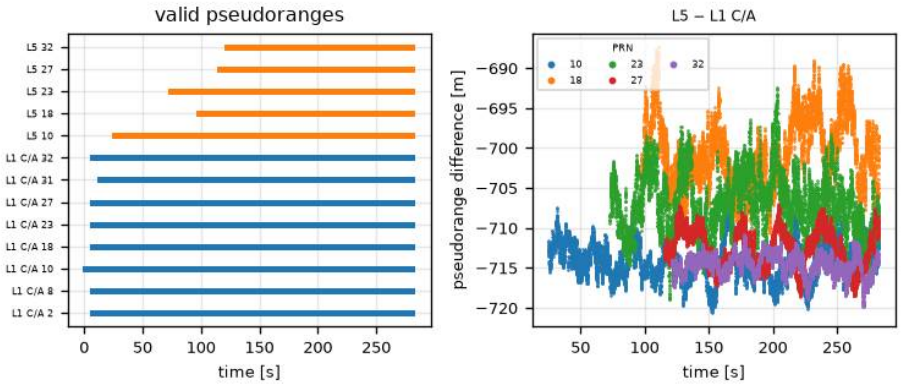


Figure 10: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L5 − L1 C/A	10	12831	86	-714.11	2.26	0.0005
L5 − L1 C/A	18	9267	62	-700.79	4.40	0.0122
L5 − L1 C/A	23	10483	70	-706.92	3.69	-0.0064
L5 − L1 C/A	27	8373	56	-712.59	2.40	-0.0092
L5 − L1 C/A	32	8078	54	-714.63	1.36	-0.0046

Table 14: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

12.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

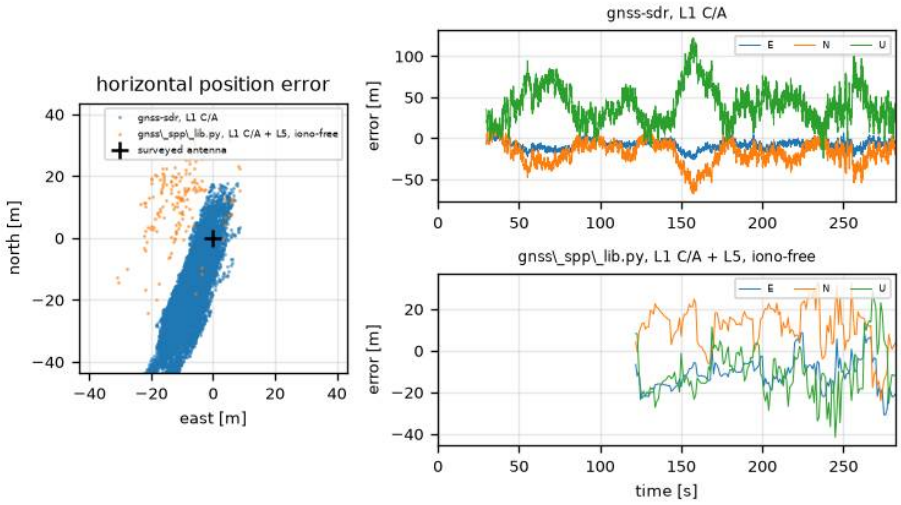


Figure 11: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

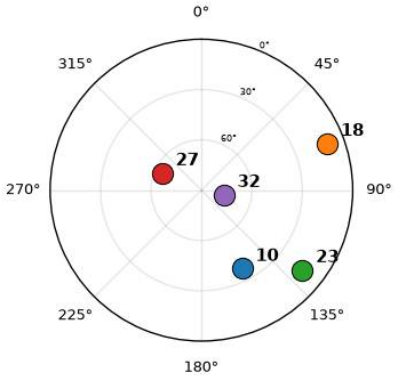


Figure 12: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	gnss-sdr, L1 C/A	gnss_spp_lib.py, L1 C/A + L5, iono-free
Solutions	13135	161
Run with fix [%]	90	57
Mean E / N / U [m]	-7.6 / -18.0 / 40.8	-11.3 / 11.2 / -10.4
Std E / N / U [m]	5.5 / 14.2 / 22.4	7.6 / 10.5 / 12.0
RMS 2-D [m]	24.8	20.5
RMS 3-D [m]	52.8	26.0
95% horiz. [m]	46.5	29.7

Table 15: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

12.1.8 Signal conditioner (Galileo E1 + E5a)

Channel	ch0 (E1)	ch1 (E5a)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 16: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

12.1.9 Acquisition (Galileo E1 + E5a)

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

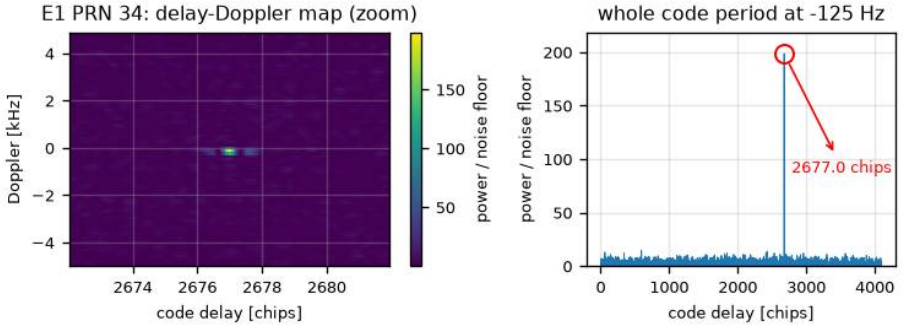


Figure 13: Acquisition, E1: delay–Doppler search of PRN 34; one sharp peak (test statistic 272 vs threshold 52) at -125 Hz and 2677.0 chips.

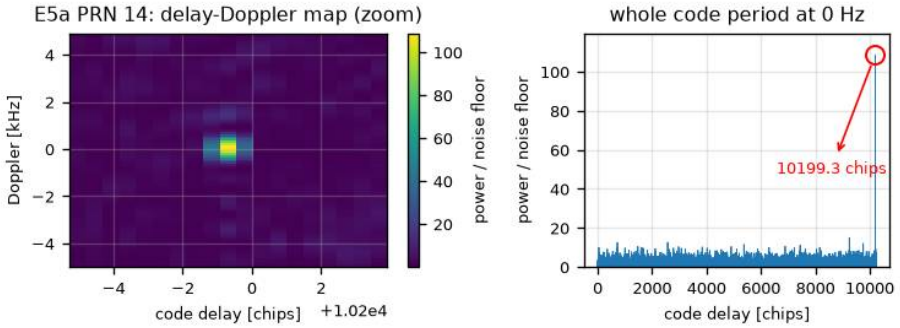


Figure 14: Acquisition, E5a: delay–Doppler search of PRN 14; one sharp peak (test statistic 149 vs threshold 49) at -1427 Hz and 10199.3 chips.

Band	**Attempts**	**Detections**	**Detected PRNs**	**Best statistic / threshold**[9]
E1	5	1	34	5.3
E5a	2	2	14	3.0

Table 17: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[9]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

12.1.10 Tracking (Galileo E1 + E5a)

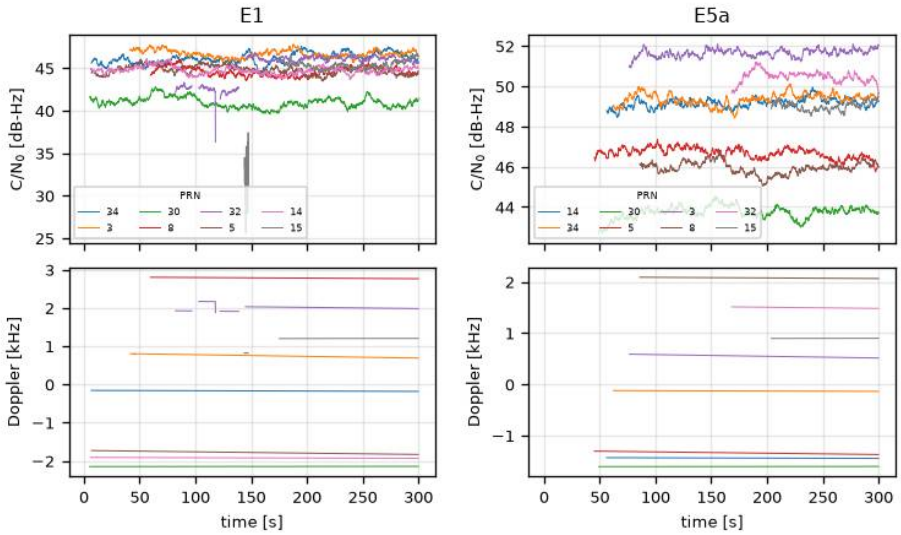


Figure 15: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

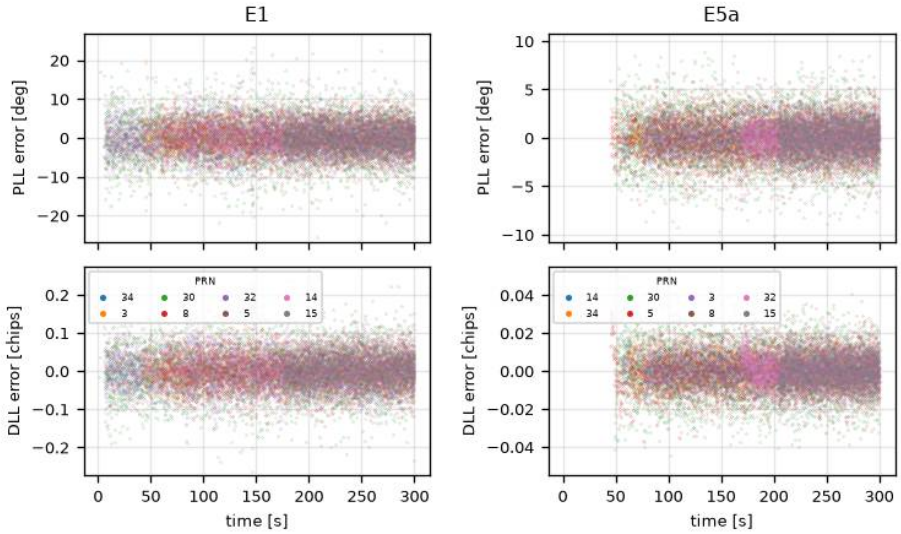


Figure 16: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

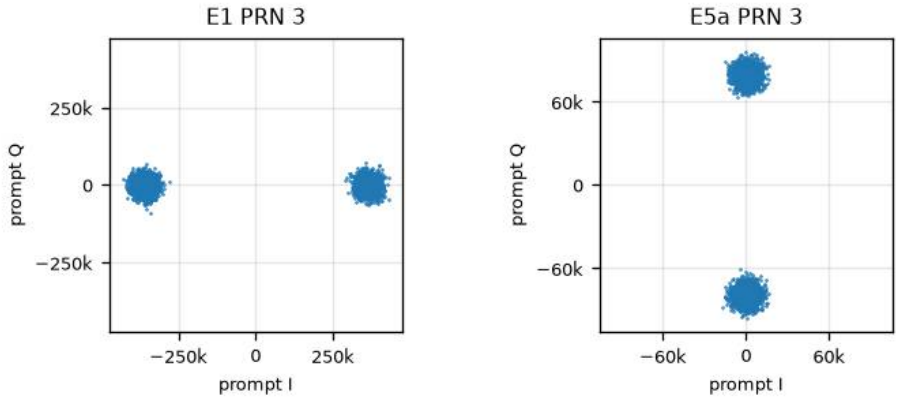


Figure 17: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
E1	0	34	46.1	98	0	299	3.7	0.039
E1	1	3	46.6	87	2	298	3.6	0.036
E1	2	30	40.9	99	0	299	6.7	0.071
E1	3	8	44.4	81	2	298	4.3	0.047
E1	4	32	45.7	68	7	299	4.1	0.043
E1	5	5	44.8	98	0	298	4.4	0.046
E1	6	14	44.7	98	0	300	4.0	0.042
E1	7	15	44.5	43	8	300	4.6	0.048
E5a	8	14	48.9	31	1	275	1.6	0.008
E5a	9	34	48.9	30	1	274	1.6	0.008
E5a	10	30	43.5	52	0	272	3.0	0.015
E5a	11	5	46.8	53	0	274	2.2	0.011
E5a	12	3	51.6	49	0	244	1.4	0.006
E5a	13	8	45.9	48	0	235	2.3	0.011
E5a	14	32	50.4	37	0	144	2.0	0.011
E5a	15	15	49.0	30	0	114	1.7	0.008

Table 18: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

12.1.11 Telemetry decoder (Galileo E1 + E5a)

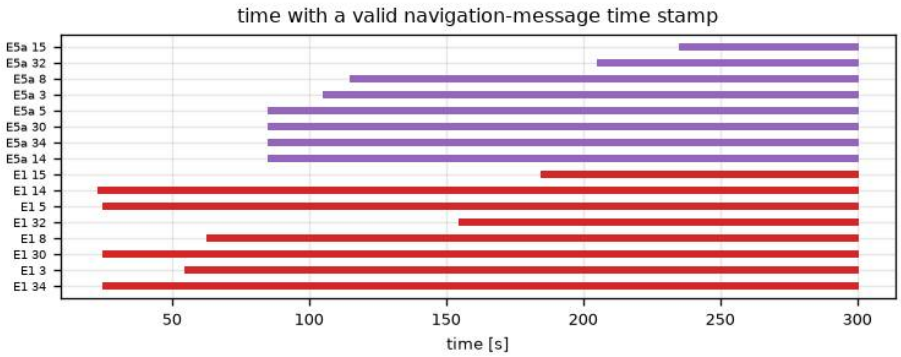


Figure 18: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	3	24	yes	55.2	414080.0
E1	5	26	yes	25.2	414050.0
E1	8	23	yes	63.2	414088.0
E1	14	26	yes	23.2	414048.0
E1	15	10	yes	185.2	414210.0
E1	30	26	yes	25.2	414050.0
E1	32	13	yes	155.2	414180.0
E1	34	26	yes	25.2	414050.0
E5a	3	11	yes	105.4	414130.3
E5a	5	12	yes	85.4	414110.3
E5a	8	10	yes	115.4	414140.3
E5a	14	11	yes	85.4	414110.3
E5a	15	3	yes	235.4	414260.3
E5a	30	12	yes	85.4	414110.3
E5a	32	5	yes	205.4	414230.3
E5a	34	12	yes	85.4	414110.3

Table 19: Telemetry decoder: E1 I/NAV words, E5a F/NAV pages decoded, ephemeris obtained, and the first valid time of week.

12.1.12 Observables (Galileo E1 + E5a)

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

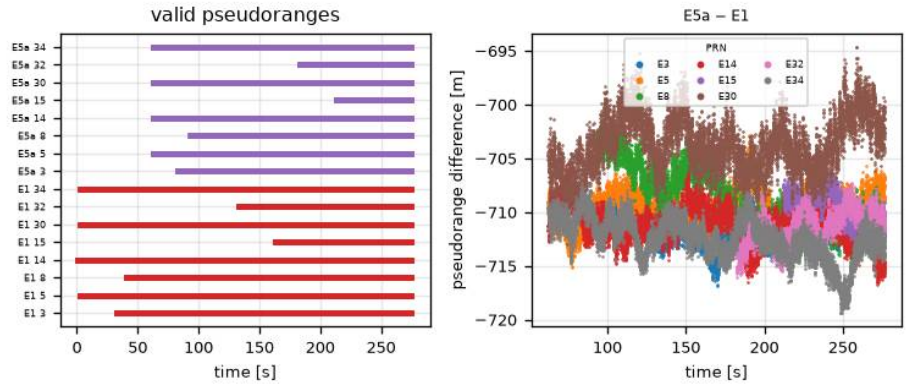


Figure 19: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
E5a – E1	3	9563	64	-711.43	1.45	-0.0075
E5a – E1	5	10437	70	-709.61	1.55	0.0065
E5a – E1	8	9128	61	-708.81	2.89	-0.0451
E5a – E1	14	10437	70	-711.16	1.64	-0.0082
E5a – E1	15	3147	21	-709.62	1.46	-0.0504
E5a – E1	30	10457	70	-703.72	2.59	0.0079
E5a – E1	32	4647	31	-711.39	1.51	0.0347
E5a – E1	34	10437	70	-712.45	1.88	-0.0177

Table 20: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

12.1.13 PVT (Galileo E1 + E5a)

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

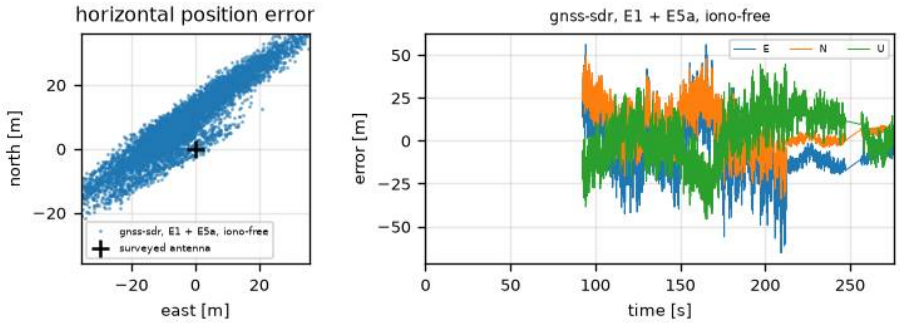


Figure 20: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution.

Solution	**gnss-sdr, E1 + E5a, iono-free**
Solutions	8642
Run with fix [%]	64
Mean E / N / U [m]	-4.8 / 6.7 / 2.8
Std E / N / U [m]	15.5 / 11.9 / 14.2
RMS 2-D [m]	21.2
RMS 3-D [m]	25.6
95% horiz. [m]	40.3

Table 21: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

12.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-22 19:30:07 UTC	
Frequency	1575.420 MHz (L1)	1227.600 MHz (L2)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	4.000 MS/s
Sample type	signed complex 16-bit samples (I+Q), little-endian, as received from the X310 (sc16)	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	4800.0 MB	4800.0 MB
Samples written	1,200,000,000	1,200,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca16bits4msps_l2c16bits4msps.conf`	

Table 22: Metadata for recording `x310_20261022T193000Z`, captured starting 2026-10-22T19:30:07.040026Z.

12.2.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

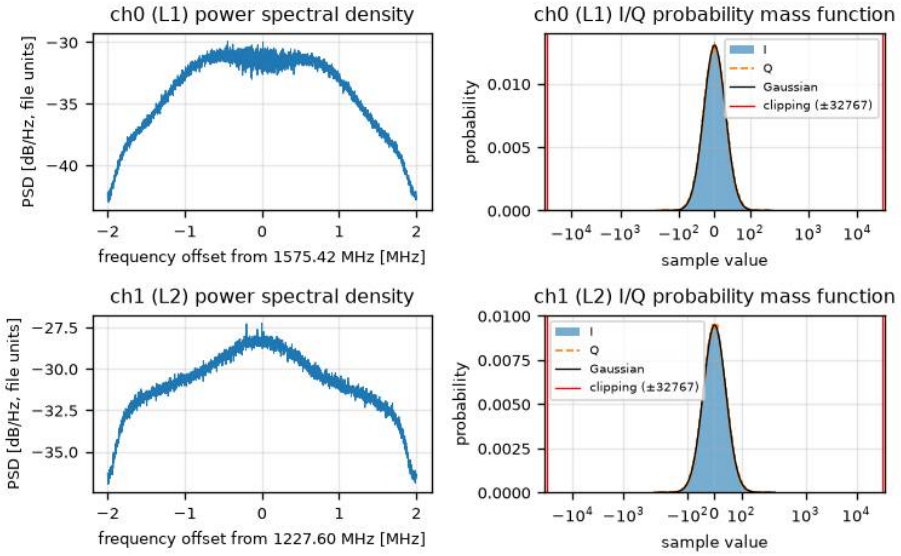


Figure 21: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1227.60
Rate [MS/s]	4	4
Format	ci16_le	ci16_le
Bytes per complex sample	4 (2 for I, 2 for Q)	4 (2 for I, 2 for Q)
Full scale	±32767	±32767
Gain [dB]	25	25
Std I / Q	30.2 / 31.1	42.1 / 42.1
Mean I / Q	-0.04 / -0.13	0.28 / 0.05
Min / max I	-151 / 154	-227 / 208
Min / max Q	-150 / 164	-208 / 210
Headroom[10]	1069.3	778.5
Clipped [%]	0.000	0.000
Lost samples	0	0

Table 23: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[10]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

12.2.2 Signal conditioner

Channel	ch0 (L1 C/A, E1)	ch1 (L2C)
File format	ci16_le	ci16_le
Source item type	ishort	ishort
Data type adapter	Ishort_To_Complex	Ishort_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	Pass_Through	Pass_Through
Rate [MS/s]	4 → 4	4 → 4

Table 24: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

12.2.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

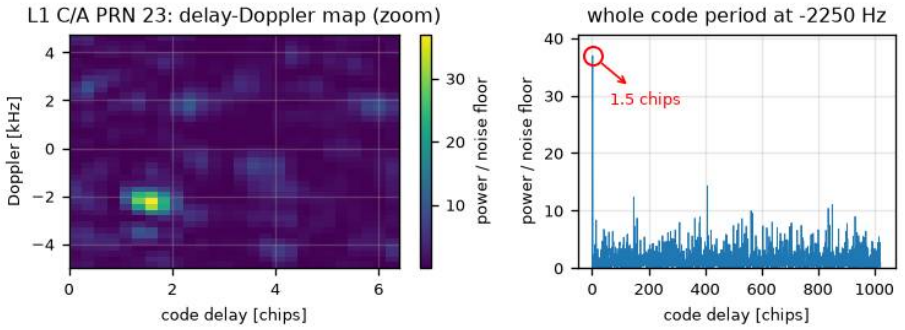


Figure 22: Acquisition, L1 C/A: delay–Doppler search of PRN 23; one sharp peak (test statistic 52 vs threshold 44) at -2250 Hz and 1.5 chips.

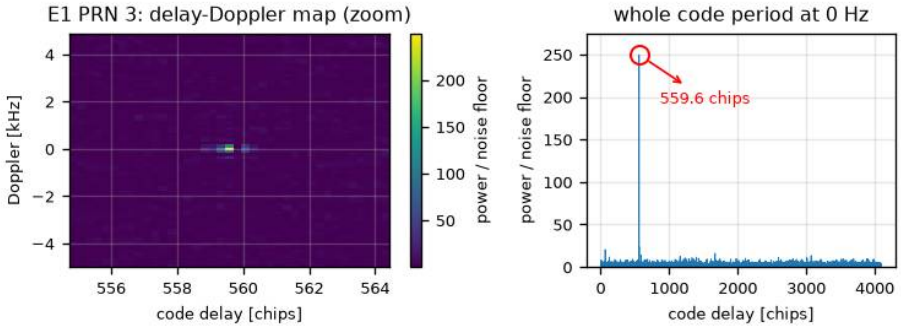


Figure 23: Acquisition, E1: delay–Doppler search of PRN 3; one sharp peak (test statistic 338 vs threshold 48) at 0 Hz and 559.6 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[11]
L1 C/A	3	1	23	1.2
L2C	not dumped (search grid too large); acquired satellites: see the tracking table			
E1	6	1	3	7.0

Table 25: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[11] Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

12.2.4 Tracking

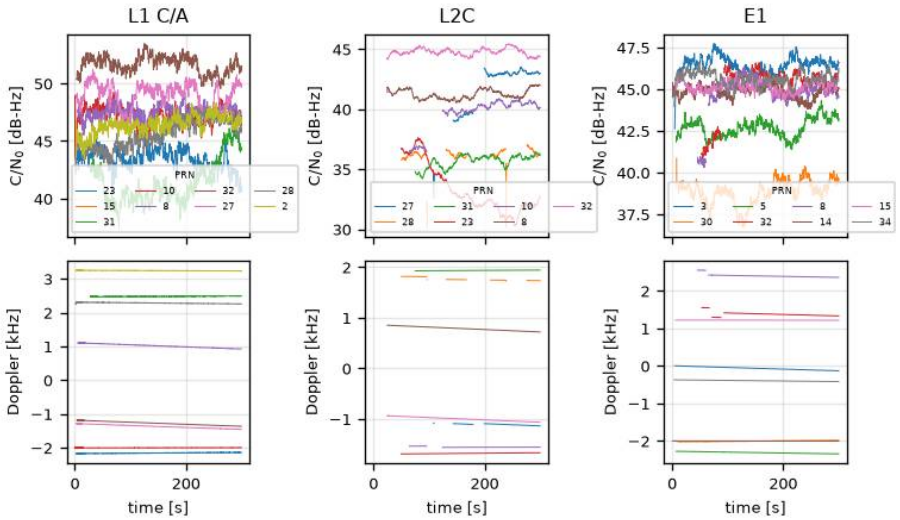


Figure 24: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

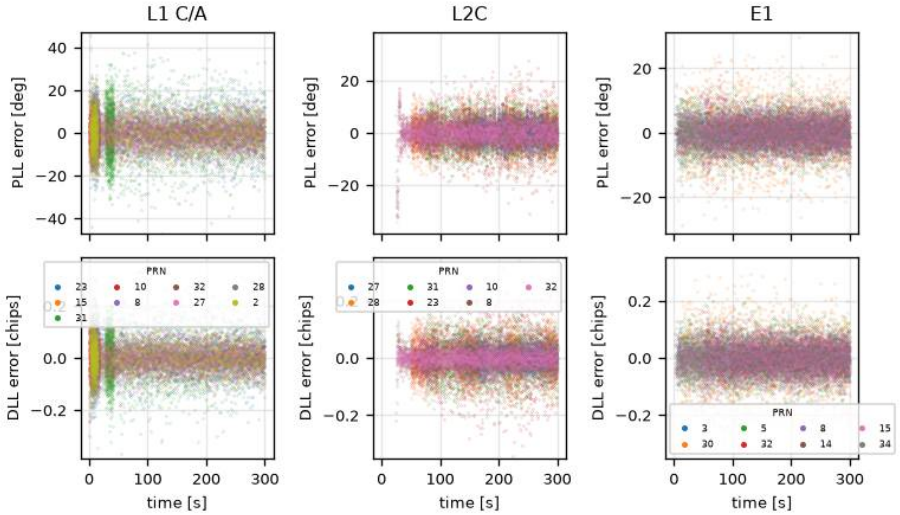


Figure 25: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

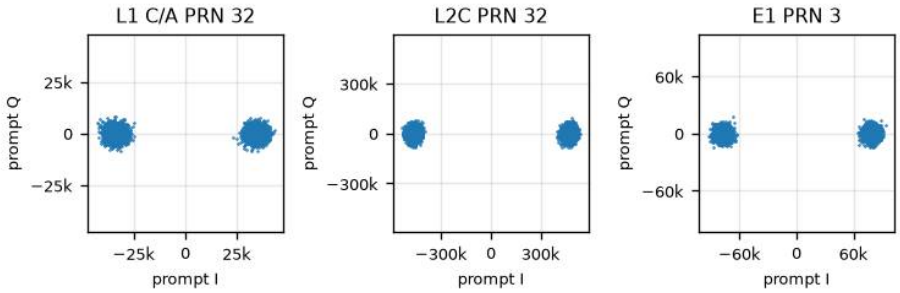


Figure 26: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	23	43.6	94	0	298	9.7	0.076
L1 C/A	1	15	28.7	0	1	16	–	–
L1 C/A	1	31	42.2	95	1	273	11.9	0.091
L1 C/A	2	10	46.9	100	0	300	6.4	0.052
L1 C/A	3	8	47.4	92	0	297	6.2	0.050
L1 C/A	4	32	51.4	95	0	298	3.9	0.028
L1 C/A	5	27	48.7	96	0	298	5.2	0.038
L1 C/A	6	28	45.0	96	0	298	8.6	0.059
L1 C/A	7	2	45.8	100	0	298	7.3	0.057
L2C	8	27	39.5	50	3	275	4.6	0.031
L2C	9	28	36.3	49	3	275	6.8	0.058
L2C	10	31	35.8	90	0	251	5.8	0.061
L2C	12	23	32.7	91	0	276	7.7	0.087
L2C	13	10	40.0	75	1	276	4.1	0.039
L2C	14	8	41.1	100	0	275	4.3	0.033
L2C	15	32	44.8	100	0	275	3.7	0.022
E1	16	3	46.5	100	0	297	3.6	0.040
E1	17	30	38.9	99	0	292	8.6	0.096
E1	18	5	42.7	99	0	297	5.5	0.063
E1	19	32	45.5	80	4	293	3.9	0.046
E1	20	8	44.8	84	3	297	4.5	0.050
E1	21	14	44.9	100	0	299	3.8	0.045
E1	22	15	45.1	98	0	299	4.0	0.044
E1	23	34	45.6	100	0	297	3.9	0.044

Table 26: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

12.2.5 Telemetry decoder

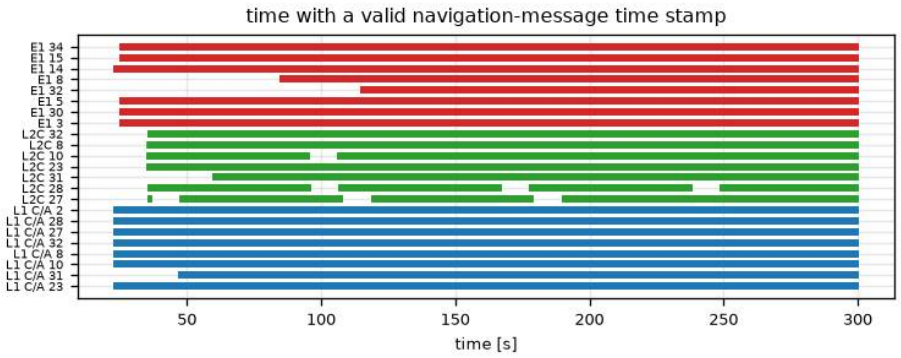


Figure 27: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	3	27	yes	25.1	415850.0
E1	5	26	yes	25.1	415850.0
E1	8	21	yes	85.1	415910.0
E1	14	26	yes	23.1	415848.0
E1	15	27	yes	25.1	415850.0
E1	30	27	yes	25.1	415850.0
E1	32	18	yes	115.1	415940.0
E1	34	27	yes	25.1	415850.0
L1 C/A	2	42	yes	23.0	415848.0
L1 C/A	8	47	yes	23.0	415848.0
L1 C/A	10	43	yes	23.0	415848.0
L1 C/A	23	47	yes	23.0	415848.0
L1 C/A	27	47	yes	23.0	415848.0
L1 C/A	28	47	yes	23.0	415848.0
L1 C/A	31	43	yes	47.0	415872.0
L1 C/A	32	47	yes	23.0	415848.0
L2C	8	13	yes	35.5	0.0
L2C	10	9	yes	35.4	0.0
L2C	23	15	yes	35.4	0.0
L2C	27	5	yes	35.7	0.0
L2C	28	–	no	35.8	0.0
L2C	31	12	yes	59.8	0.0
L2C	32	13	yes	35.6	0.0

Table 27: Telemetry decoder: L1 C/A LNAV subframes, L2C CNAV messages, E1 I/NAV words decoded, ephemeris obtained, and the first valid time of week.

12.2.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

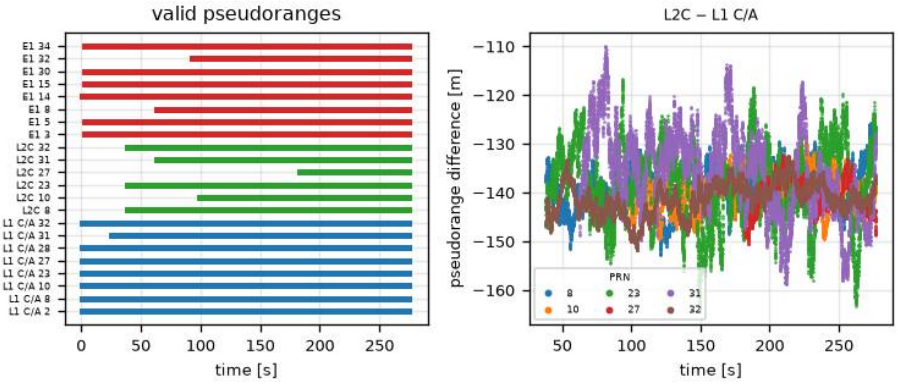


Figure 28: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L2C – L1 C/A	8	11932	80	-138.80	4.22	0.0218
L2C – L1 C/A	10	8950	60	-139.45	3.99	0.0142
L2C – L1 C/A	23	11937	80	-138.23	7.93	-0.0048
L2C – L1 C/A	27	4756	32	-140.52	3.36	-0.0064
L2C – L1 C/A	31	10717	72	-135.85	8.56	-0.0721
L2C – L1 C/A	32	11928	80	-141.29	3.29	0.0180

Table 28: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

12.2.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

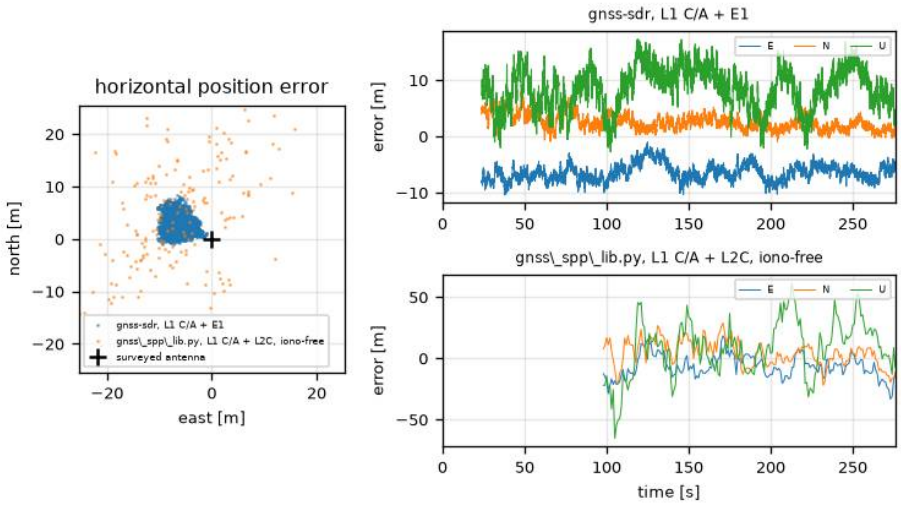


Figure 29: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

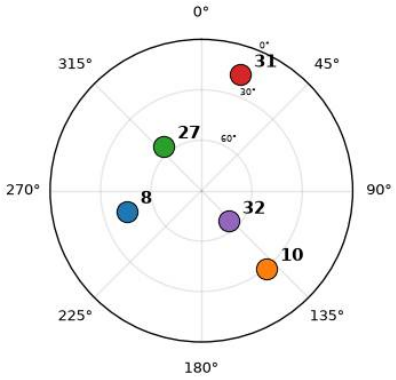


Figure 30: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, L1 C/A + E1**	**gnss_spp_lib.py, L1 C/A + L2C, iono-free**
Solutions	13139	179
Run with fix [%]	92	65
Mean E / N / U [m]	-6.5 / 2.6 / 8.4	-6.7 / 4.2 / 8.1
Std E / N / U [m]	1.5 / 1.4 / 3.7	9.5 / 10.6 / 22.9
RMS 2-D [m]	7.3	16.3
RMS 3-D [m]	11.7	29.2
95% horiz. [m]	9.5	27.9

Table 29: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

13 Records on 23/10/2026

13.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-23 19:00:06 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`, `e1_8bits4msps_e5a_8bits20msps.conf`	

Table 30: Metadata for recording `x310_20261023T190000Z`, captured starting 2026-10-23T19:00:06.932068Z.

13.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

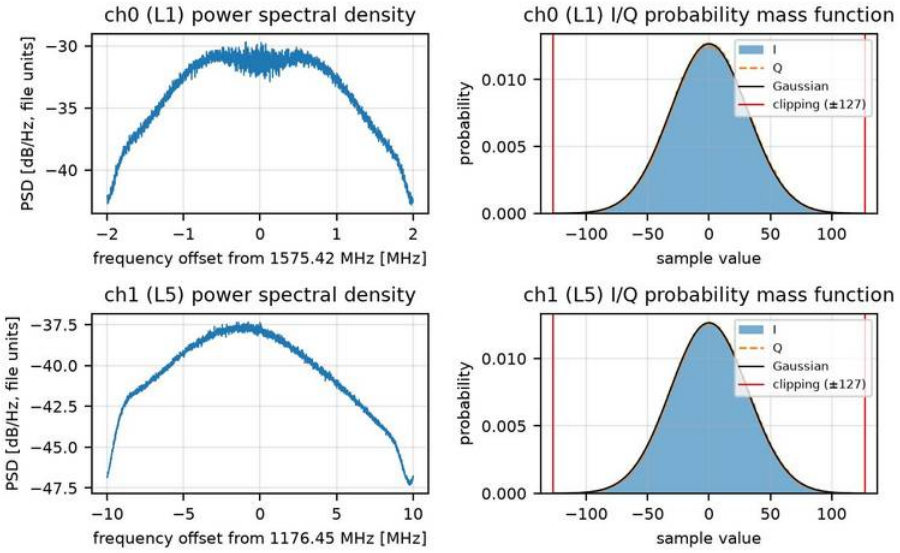


Figure 31: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Bytes per complex sample	2 (1 for I, 1 for Q)	2 (1 for I, 1 for Q)
Full scale	±127	±127
Gain [dB]	25	25
Std I / Q	31.5 / 31.5	31.5 / 31.5
Mean I / Q	0.04 / 0.07	0.04 / 0.01
Min / max I	-127 / 127	-127 / 127
Min / max Q	-127 / 127	-127 / 127
Headroom[12]	4.0	4.0
Clipped [%]	0.006	0.006
Lost samples	0	0

Table 31: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[12]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

13.1.2 Signal conditioner

Channel	ch0 (L1 C/A)	ch1 (L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 32: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

13.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

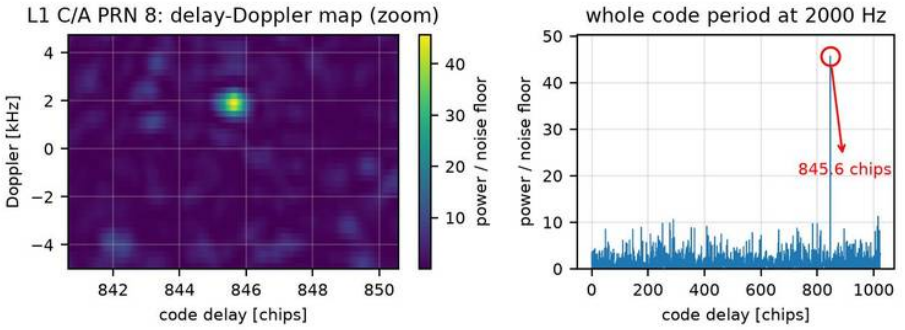


Figure 32: Acquisition, L1 C/A: delay–Doppler search of PRN 8; one sharp peak (test statistic 59 vs threshold 47) at 2000 Hz and 845.6 chips.

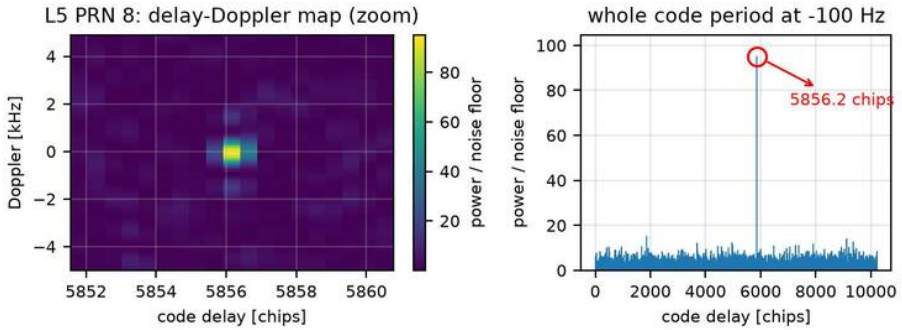


Figure 33: Acquisition, L5: delay–Doppler search of PRN 8; one sharp peak (test statistic 134 vs threshold 49) at 1392 Hz and 5856.2 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[13]
L1 C/A	6	1	8	1.2
L5	2	2	8	2.7

Table 33: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[13]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

13.1.4 Tracking

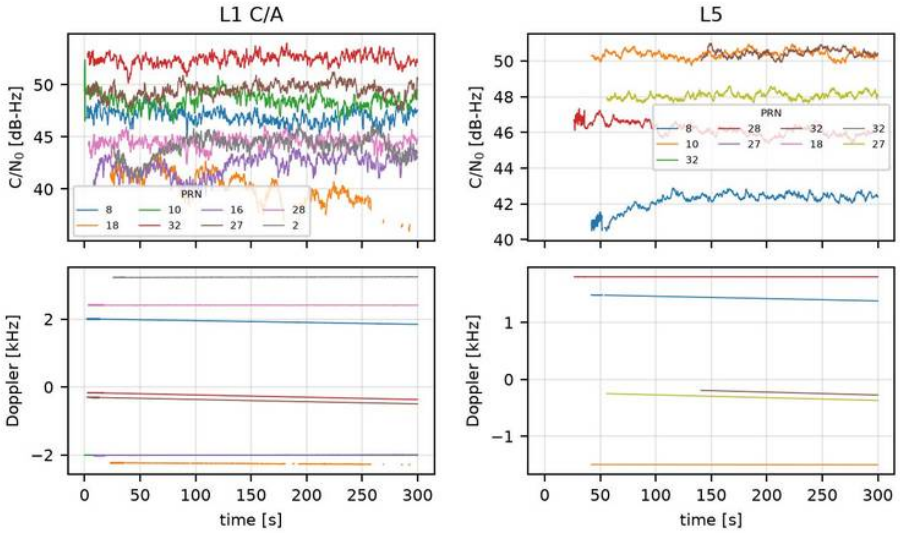


Figure 34: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

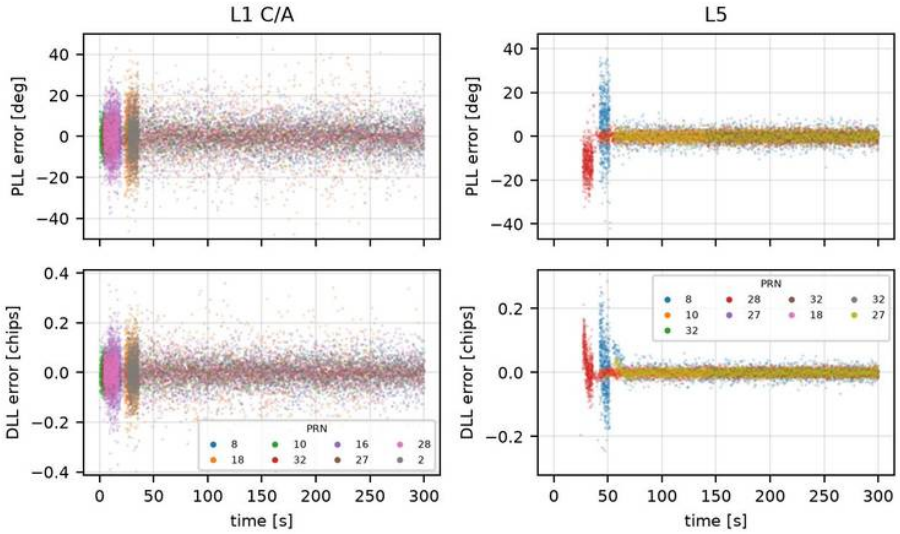


Figure 35: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

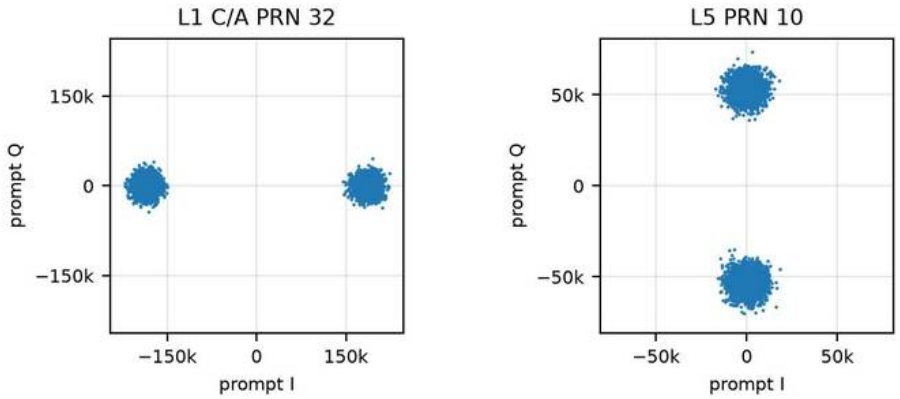


Figure 36: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	8	47.2	99	0	298	6.3	0.048
L1 C/A	1	18	41.1	86	0	277	13.2	0.099
L1 C/A	2	10	49.1	100	0	300	5.2	0.038
L1 C/A	3	32	52.4	95	0	299	3.5	0.025
L1 C/A	4	16	42.3	95	0	293	11.1	0.085
L1 C/A	5	27	49.5	95	0	299	4.8	0.036
L1 C/A	6	28	44.4	91	0	299	8.9	0.063
L1 C/A	7	2	43.6	96	0	275	9.5	0.068
L5	8	8	41.2	66	1	276	7.8	0.052
L5	9	10	50.4	68	0	282	1.5	0.007
L5	10	32	29.1	0	2	32	–	–
L5	11	28	46.3	95	0	275	6.7	0.032
L5	13	27	28.9	0	3	16	–	–
L5	13	32	47.7	29	3	210	1.5	0.007
L5	14	18	28.6	0	3	16	–	–
L5	14	32	29.0	0	3	32	–	–
L5	15	27	48.0	68	0	260	1.9	0.010

Table 34: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

13.1.5 Telemetry decoder

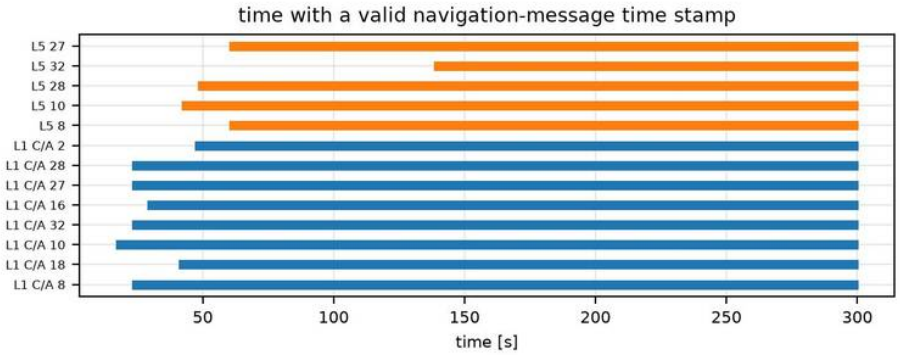


Figure 37: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
L1 C/A	2	43	yes	47.1	500472.0
L1 C/A	8	47	yes	23.1	500448.0
L1 C/A	10	48	yes	17.1	500442.0
L1 C/A	16	46	yes	29.1	500454.0
L1 C/A	18	44	yes	41.2	500466.0
L1 C/A	27	47	yes	23.1	500448.0
L1 C/A	28	47	yes	23.1	500448.0
L1 C/A	32	47	yes	23.1	500448.0
L5	8	25	yes	60.3	500485.1
L5	10	26	yes	42.1	500467.0
L5	27	25	yes	60.3	500485.2
L5	28	32	yes	48.4	500473.3
L5	32	16	yes	138.5	500563.3

Table 35: Telemetry decoder: L1 C/A LNAV subframes, L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

13.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

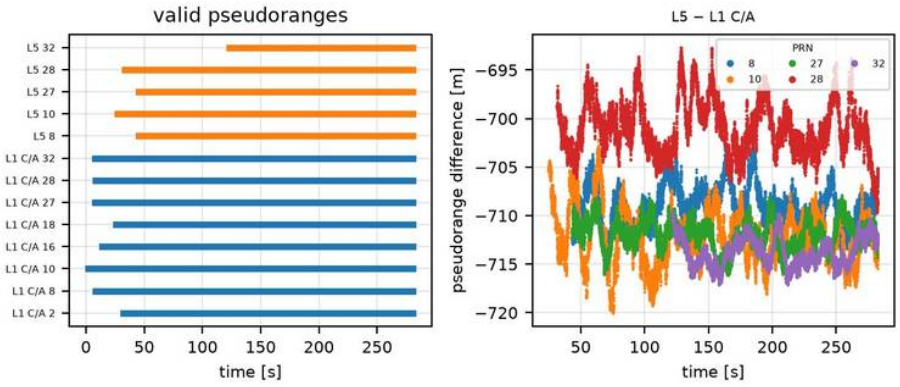


Figure 38: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L5 − L1 C/A	8	11972	80	-709.04	2.07	-0.0001
L5 − L1 C/A	10	12837	86	-712.22	2.77	-0.0056
L5 − L1 C/A	27	11970	80	-711.92	1.52	-0.0025
L5 − L1 C/A	28	12563	84	-700.94	2.75	-0.0052
L5 − L1 C/A	32	8062	54	-713.71	1.50	0.0021

Table 36: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

13.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

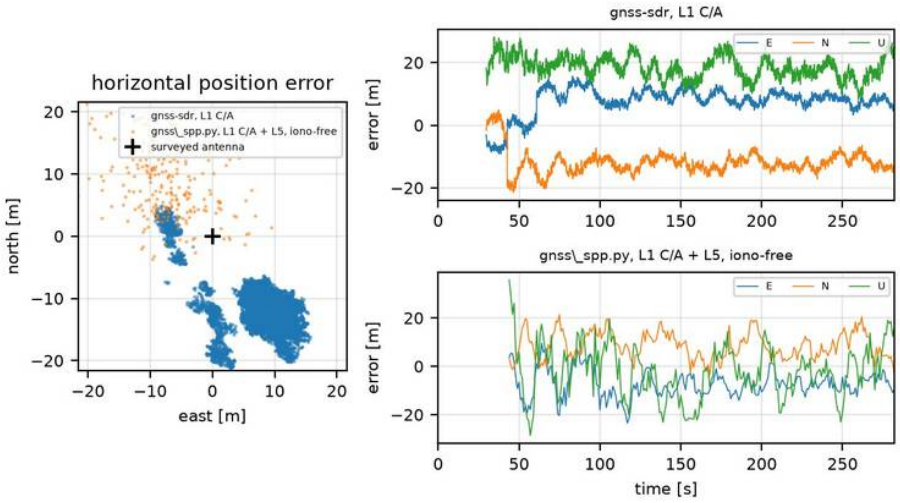


Figure 39: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

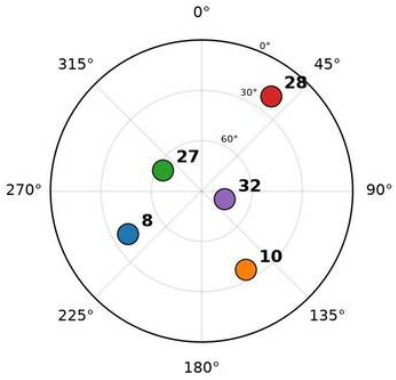


Figure 40: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	gnss-sdr, L1 C/A	gnss_spp_lib.py, L1 C/A + L5, iono-free
Solutions	13133	239
Run with fix [%]	90	85
Mean E / N / U [m]	7.7 / -11.8 / 18.5	-7.3 / 8.5 / -2.1
Std E / N / U [m]	4.5 / 3.8 / 3.5	5.7 / 5.5 / 11.1
RMS 2-D [m]	15.3	13.8
RMS 3-D [m]	24.3	17.8
95% horiz. [m]	20.1	22.2

Table 37: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

13.1.8 Signal conditioner (Galileo E1 + E5a)

Channel	ch0 (E1)	ch1 (E5a)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 38: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

13.1.9 Acquisition (Galileo E1 + E5a)

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

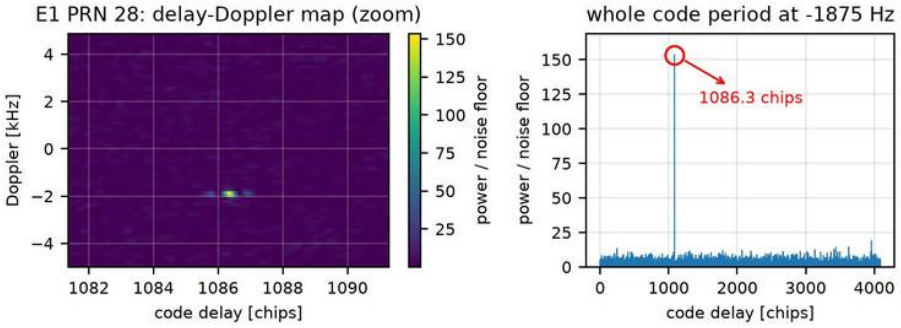


Figure 41: Acquisition, E1: delay–Doppler search of PRN 28; one sharp peak (test statistic 215 vs threshold 52) at -1875 Hz and 1086.3 chips.

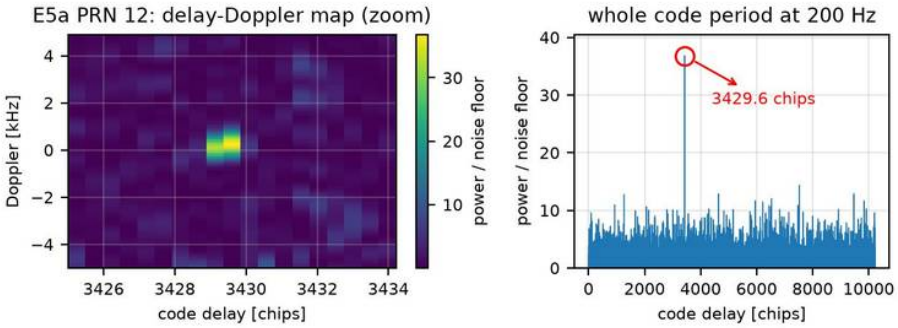


Figure 42: Acquisition, E5a: delay–Doppler search of PRN 12; one sharp peak (test statistic 51 vs threshold 49) at -492 Hz and 3429.6 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[14]
E1	9	4	28	4.1
E5a	2	1	12	1.0

Table 39: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[14]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

13.1.10 Tracking (Galileo E1 + E5a)

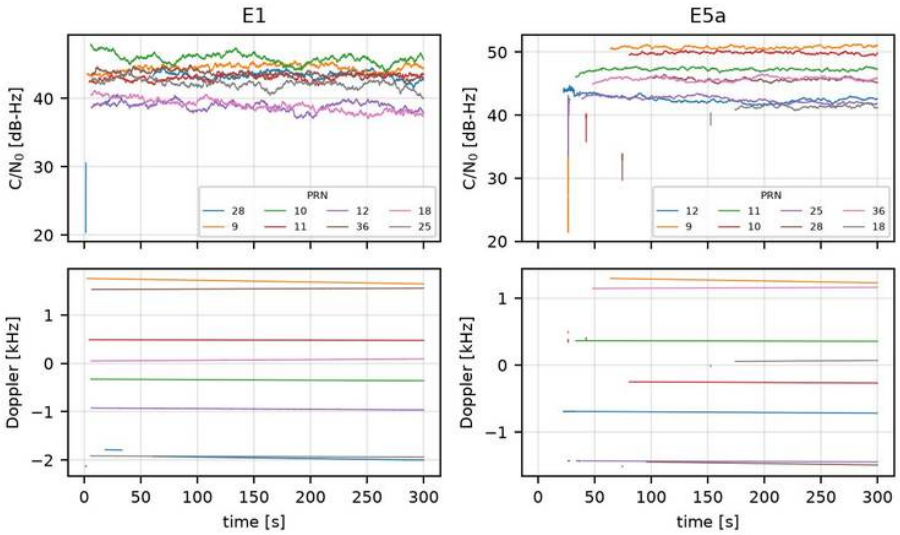


Figure 43: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

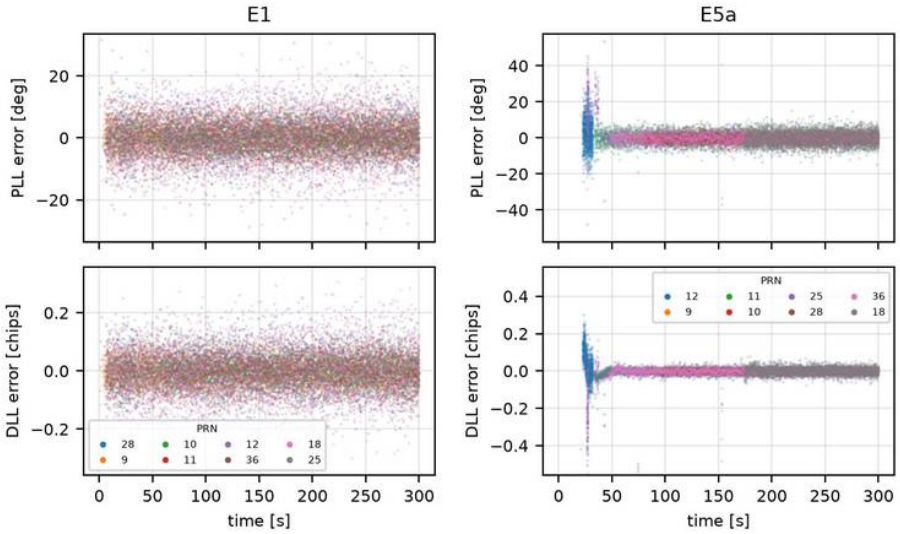


Figure 44: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

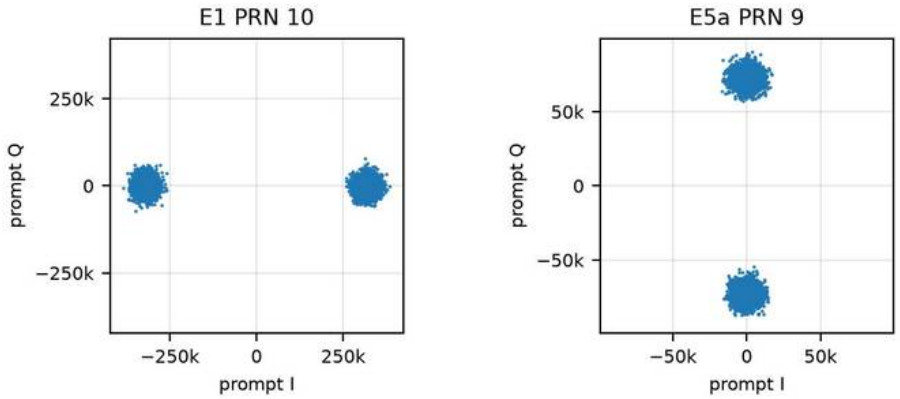


Figure 45: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
E1	0	28	43.4	85	3	299	5.2	0.052
E1	1	9	44.6	99	0	300	4.2	0.044
E1	2	10	45.9	98	0	300	3.9	0.041
E1	3	11	43.1	99	0	300	5.5	0.053
E1	4	12	39.0	98	0	300	8.2	0.088
E1	5	36	43.6	98	0	298	4.8	0.052
E1	6	18	39.0	98	0	299	8.4	0.089
E1	7	25	42.2	98	0	299	5.8	0.062
E5a	8	12	42.9	92	0	279	6.5	0.054
E5a	9	9	50.4	30	1	273	4.6	0.023
E5a	10	11	47.0	54	0	280	3.6	0.014
E5a	11	10	29.8	20	2	273	3.6	0.013
E5a	12	25	42.9	60	0	273	7.3	0.069
E5a	13	28	45.7	47	0	226	3.4	0.024
E5a	14	36	45.4	52	0	273	2.4	0.012
E5a	15	18	41.2	35	0	147	4.7	0.025

Table 40: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

13.1.11 Telemetry decoder (Galileo E1 + E5a)

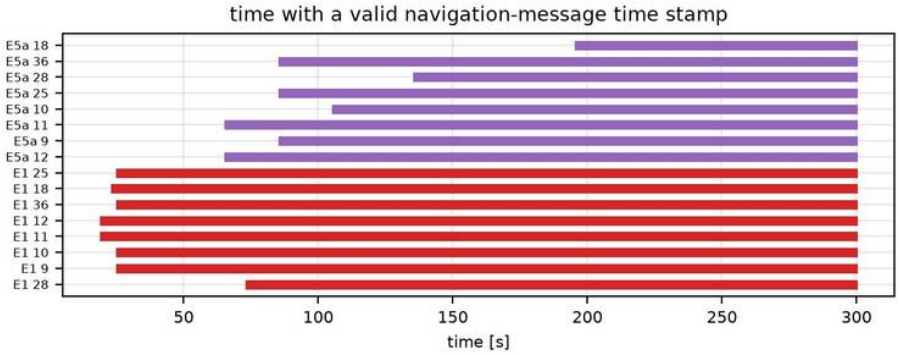


Figure 46: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	9	26	yes	25.2	500450.0
E1	10	27	yes	25.2	500450.0
E1	11	27	yes	19.2	500444.0
E1	12	27	yes	19.2	500444.0
E1	18	26	yes	23.2	500448.0
E1	25	26	yes	25.2	500450.0
E1	28	22	yes	73.2	500498.0
E1	36	27	yes	25.2	500450.0
E5a	9	12	yes	85.4	500510.3
E5a	10	11	yes	105.4	500530.3
E5a	11	13	yes	65.4	500490.3
E5a	12	13	yes	65.4	500490.3
E5a	18	4	yes	195.4	500620.3
E5a	25	12	yes	85.4	500510.3
E5a	28	9	yes	135.4	500560.3
E5a	36	12	yes	85.4	500510.3

Table 41: Telemetry decoder: E1 I/NAV words, E5a F/NAV pages decoded, ephemeris obtained, and the first valid time of week.

13.1.12 Observables (Galileo E1 + E5a)

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

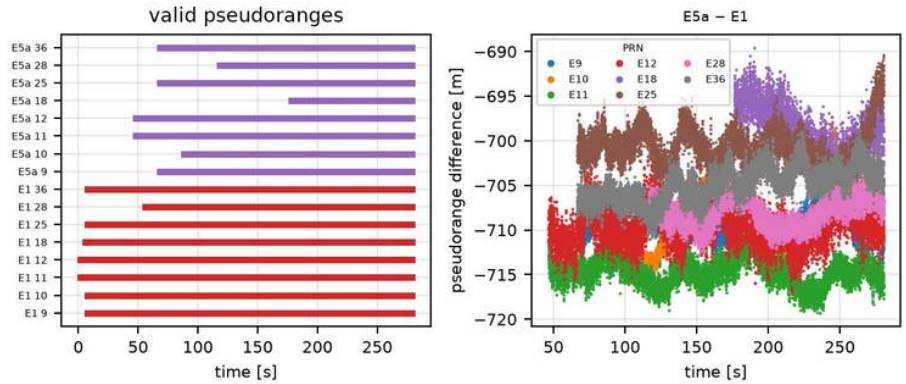


Figure 47: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
E5a − E1	9	10631	71	-709.38	1.88	-0.0007
E5a − E1	10	9645	64	-709.26	3.55	0.0208
E5a − E1	11	11505	77	-714.67	1.53	-0.0035
E5a − E1	12	11520	77	-709.41	2.18	0.0012
E5a − E1	18	5144	34	-698.88	2.95	-0.0589
E5a − E1	25	10637	71	-700.71	2.13	0.0007
E5a − E1	28	8150	54	-708.06	1.63	0.0079
E5a − E1	36	10632	71	-704.86	1.92	0.0213

Table 42: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

13.1.13 PVT (Galileo E1 + E5a)

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

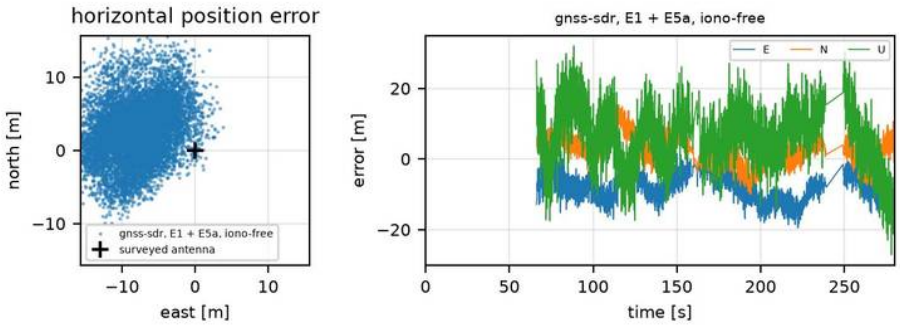


Figure 48: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution.

Solution	**gnss-sdr, E1 + E5a, iono-free**
Solutions	9278
Run with fix [%]	73
Mean E / N / U [m]	-8.1 / 2.7 / 6.0
Std E / N / U [m]	3.8 / 4.0 / 8.4
RMS 2-D [m]	10.2
RMS 3-D [m]	14.5
95% horiz. [m]	15.0

Table 43: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

13.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-23 19:33:16 UTC	
Frequency	1575.420 MHz (L1)	1227.600 MHz (L2)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	4.000 MS/s
Sample type	signed complex 16-bit samples (I+Q), little-endian, as received from the X310 (sc16)	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	4800.0 MB	4800.0 MB
Samples written	1,200,000,000	1,200,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca16bits4msps_l2c16bits4msps.conf`	

Table 44: Metadata for recording `x310_20261023T193310Z`, captured starting 2026-10-23T19:33:16.661684Z.

13.2.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

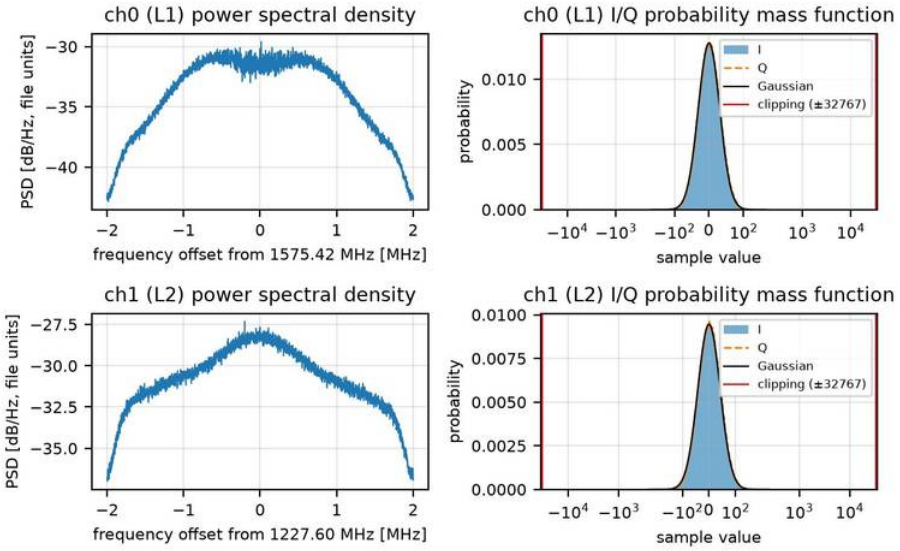


Figure 49: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1227.60
Rate [MS/s]	4	4
Format	ci16_le	ci16_le
Bytes per complex sample	4 (2 for I, 2 for Q)	4 (2 for I, 2 for Q)
Full scale	±32767	±32767
Gain [dB]	25	25
Std I / Q	31.2 / 31.2	42.1 / 42.2
Mean I / Q	-0.04 / 0.06	0.03 / -0.02
Min / max I	-164 / 155	-205 / 216
Min / max Q	-155 / 165	-203 / 212
Headroom[15]	1048.7	777.0
Clipped [%]	0.000	0.000
Lost samples	0	0

Table 45: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[15]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

13.2.2 Signal conditioner

Channel	ch0 (L1 C/A, E1)	ch1 (L2C)
File format	ci16_le	ci16_le
Source item type	ishort	ishort
Data type adapter	Ishort_To_Complex	Ishort_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	Pass_Through	Pass_Through
Rate [MS/s]	4 → 4	4 → 4

Table 46: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

13.2.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

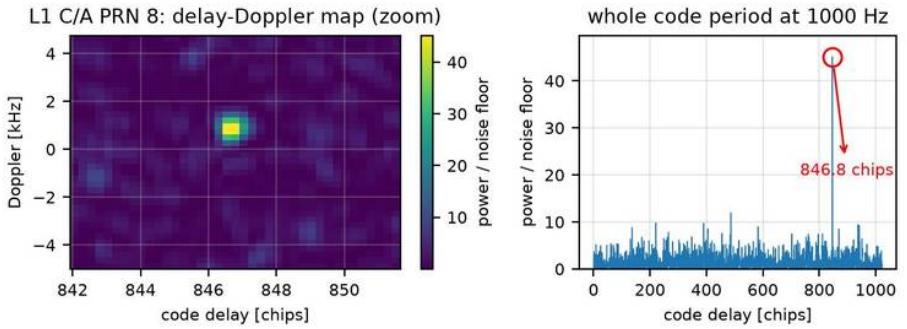


Figure 50: Acquisition, L1 C/A: delay–Doppler search of PRN 8; one sharp peak (test statistic 62 vs threshold 44) at 1000 Hz and 846.8 chips.

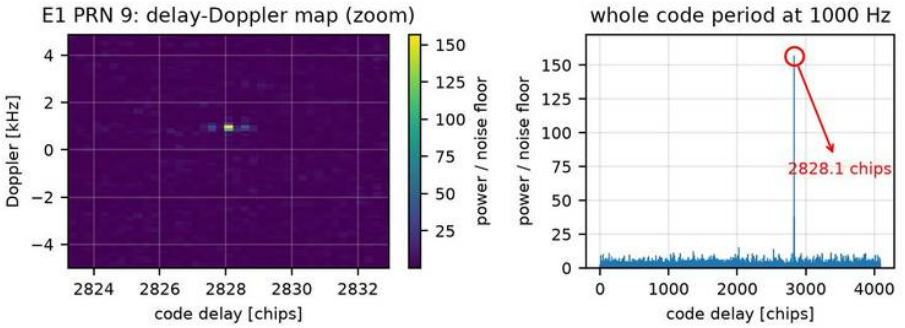


Figure 51: Acquisition, E1: delay–Doppler search of PRN 9; one sharp peak (test statistic 211 vs threshold 48) at 1000 Hz and 2828.1 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[16]
L1 C/A	7	1	8	1.4
L2C	not dumped (search grid too large); acquired satellites: see the tracking table			
E1	1	1	9	4.4

Table 47: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[16]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

13.2.4 Tracking

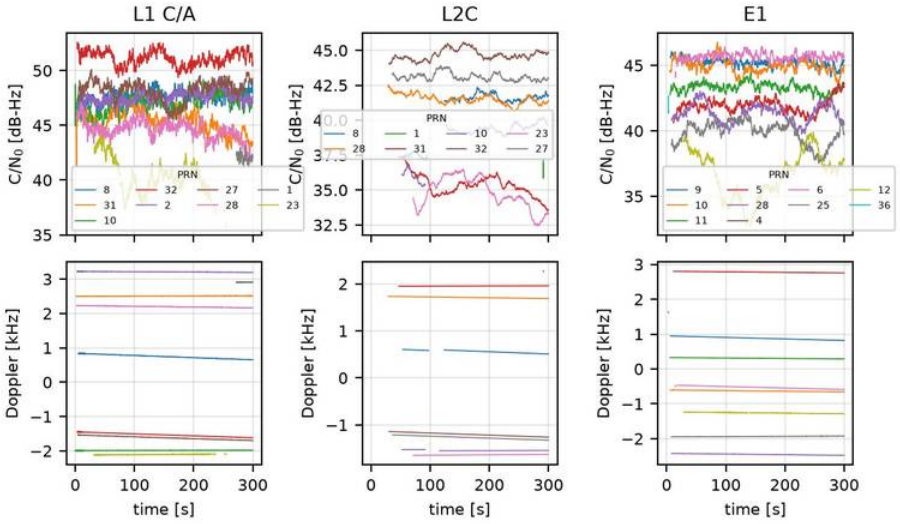


Figure 52: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

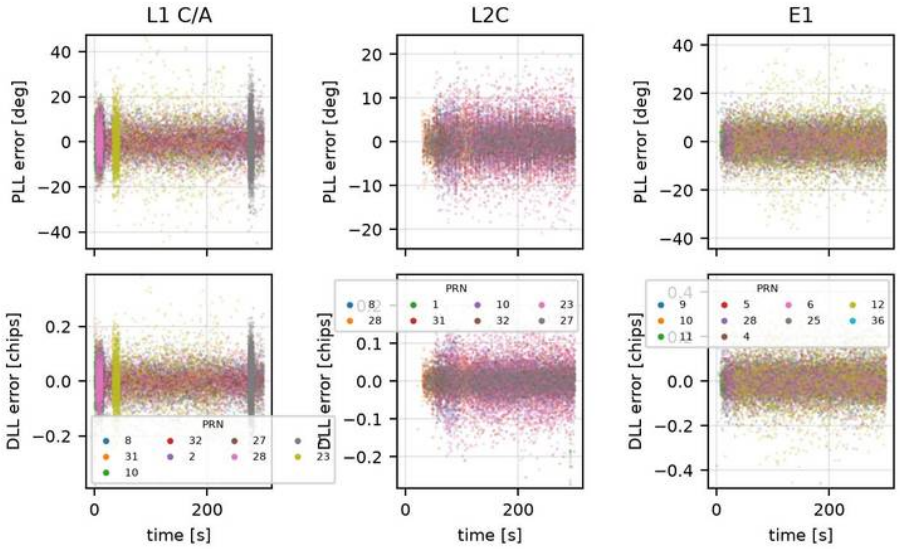


Figure 53: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

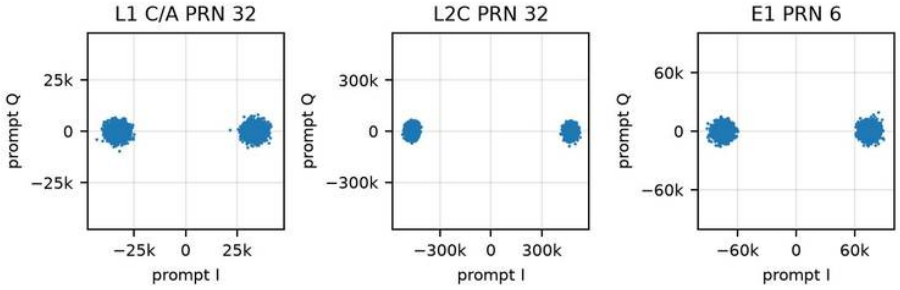


Figure 54: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	8	47.3	91	0	297	6.1	0.049
L1 C/A	1	31	45.8	100	0	298	7.6	0.061
L1 C/A	2	10	46.7	100	0	300	6.7	0.053
L1 C/A	3	32	51.5	94	0	298	3.9	0.030
L1 C/A	4	2	47.3	95	0	298	6.1	0.047
L1 C/A	5	27	48.2	88	0	299	5.7	0.043
L1 C/A	6	28	45.3	96	0	299	8.1	0.061
L1 C/A	7	1	42.2	79	1	30	11.1	0.085
L1 C/A	7	23	42.7	83	1	232	12.3	0.095
L2C	8	8	41.4	80	1	274	3.5	0.035
L2C	9	28	41.5	97	0	279	3.3	0.031
L2C	10	1	36.7	9	0	9	52.8	0.059
L2C	11	31	35.4	91	0	280	5.9	0.065
L2C	12	10	39.3	80	1	280	4.4	0.045
L2C	13	32	44.6	97	0	279	2.7	0.022
L2C	14	23	34.8	92	0	249	6.1	0.070
L2C	15	27	43.1	95	0	279	2.9	0.026
E1	16	9	45.2	98	0	300	3.7	0.043
E1	17	10	44.9	98	0	300	4.3	0.048
E1	18	11	43.3	98	0	300	5.4	0.055
E1	19	5	41.9	97	0	297	5.8	0.068
E1	20	28	41.3	98	0	298	6.0	0.070
E1	21	4	26.4	1	1	11	–	–
E1	21	6	45.7	99	1	286	4.5	0.042
E1	22	25	40.0	98	0	298	7.5	0.086
E1	23	12	37.1	97	1	276	11.2	0.125
E1	23	36	44.3	0	1	21	48.0	0.107

Table 48: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

13.2.5 Telemetry decoder

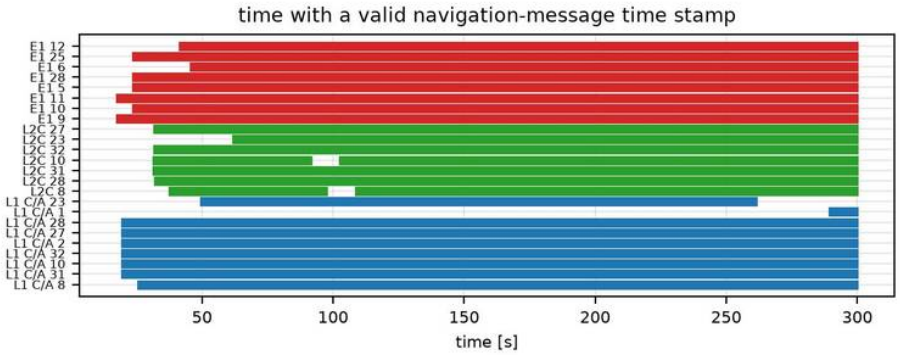


Figure 55: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	5	27	yes	23.5	502438.0
E1	6	23	yes	45.5	502460.0
E1	9	26	yes	17.5	502432.0
E1	10	27	yes	23.5	502438.0
E1	11	27	yes	17.5	502432.0
E1	12	24	yes	41.5	502456.0
E1	25	27	yes	23.5	502438.0
E1	28	27	yes	23.5	502438.0
L1 C/A	1	2	no	289.4	502704.0
L1 C/A	2	46	yes	19.4	502434.0
L1 C/A	8	45	yes	25.4	502440.0
L1 C/A	10	45	yes	19.4	502434.0
L1 C/A	23	36	yes	49.4	502464.0
L1 C/A	27	43	yes	19.4	502434.0
L1 C/A	28	45	yes	19.4	502434.0
L1 C/A	31	47	yes	19.4	502434.0
L1 C/A	32	45	yes	19.4	502434.0
L2C	8	8	yes	37.4	0.0
L2C	10	8	yes	31.5	0.0
L2C	23	13	yes	61.6	0.0
L2C	27	12	yes	31.6	0.0
L2C	28	16	yes	31.8	0.0
L2C	31	12	yes	31.4	0.0
L2C	32	12	yes	31.5	0.0

Table 49: Telemetry decoder: L1 C/A LNAV subframes, L2C CNAV messages, E1 I/NAV words decoded, ephemeris obtained, and the first valid time of week.

13.2.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

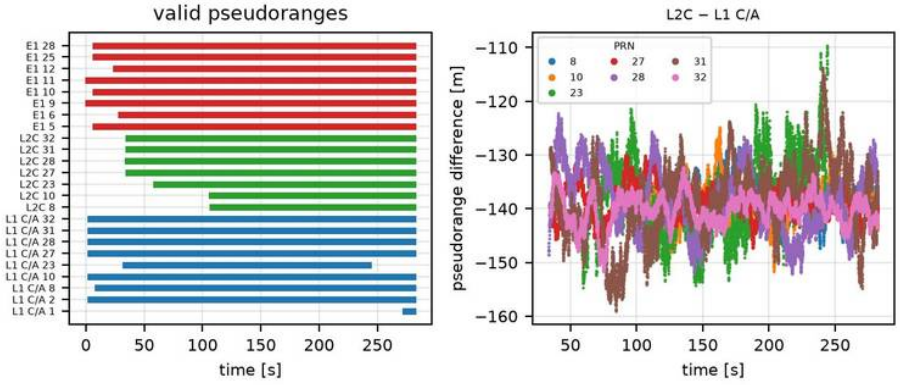


Figure 56: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L2C – L1 C/A	8	8789	59	-137.54	3.60	-0.0130
L2C – L1 C/A	10	8829	59	-138.91	3.87	0.0045
L2C – L1 C/A	23	9286	62	-138.17	6.94	0.0399
L2C – L1 C/A	27	12381	83	-138.90	3.03	-0.0087
L2C – L1 C/A	28	12437	83	-138.47	5.77	-0.0274
L2C – L1 C/A	31	12393	83	-138.52	7.00	0.0405
L2C – L1 C/A	32	12387	83	-140.16	2.89	0.0019

Table 50: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

13.2.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

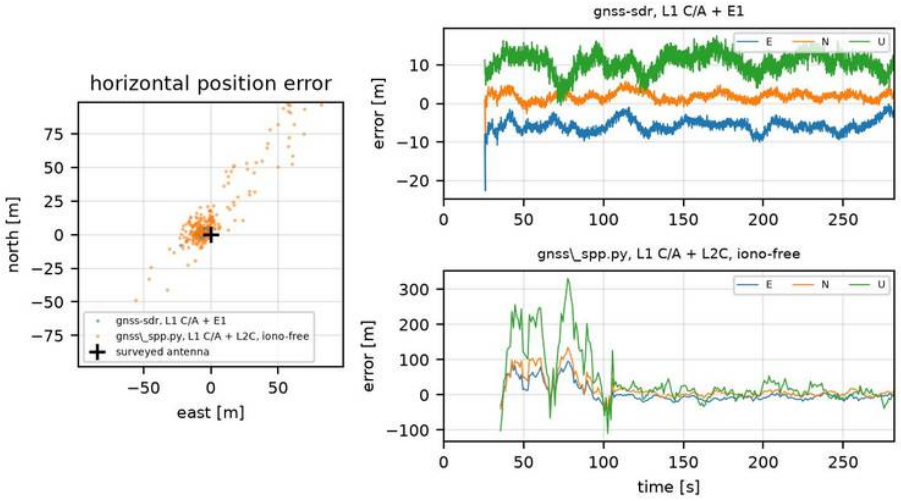


Figure 57: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

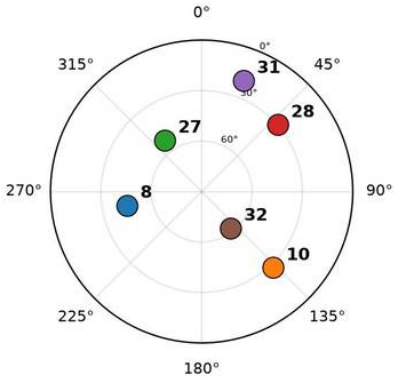


Figure 58: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, L1 C/A + E1**	**gnss_spp_lib.py, L1 C/A + L2C, iono-free**
Solutions	13328	247
Run with fix [%]	91	88
Mean E / N / U [m]	-5.8 / 1.8 / 10.6	3.3 / 16.8 / 42.9
Std E / N / U [m]	1.6 / 1.2 / 2.5	26.5 / 32.0 / 78.1
RMS 2-D [m]	6.4	44.9
RMS 3-D [m]	12.6	99.8
95% horiz. [m]	8.4	116.1

Table 51: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

14 Records on 24/10/2026

14.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-24 19:00:07 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	0	2115024
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`, `e1_8bits4msps_e5a_8bits20msps.conf`	

Table 52: Metadata for recording `x310_20261024T190000Z`, captured starting 2026-10-24T19:00:07.446070Z.

14.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

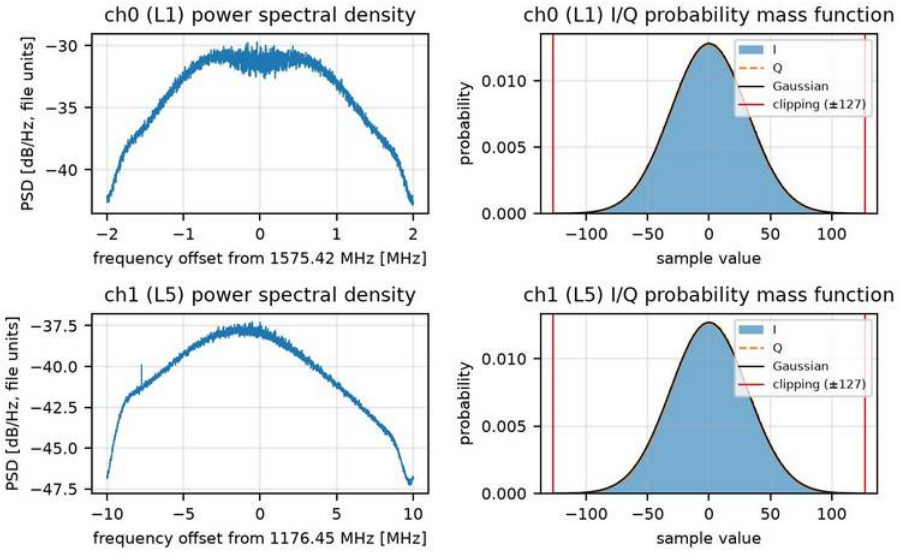


Figure 59: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Bytes per complex sample	2 (1 for I, 1 for Q)	2 (1 for I, 1 for Q)
Full scale	±127	±127
Gain [dB]	25	25
Std I / Q	31.2 / 31.2	31.4 / 31.4
Mean I / Q	-0.10 / -0.16	-0.01 / -0.01
Min / max I	-127 / 127	-127 / 127
Min / max Q	-127 / 127	-127 / 127
Headroom[17]	4.1	4.0
Clipped [%]	0.004	0.005
Lost samples	0	2,115,024

Table 53: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[17] Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

14.1.2 Signal conditioner

Channel	ch0 (L1 C/A)	ch1 (L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 54: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

14.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

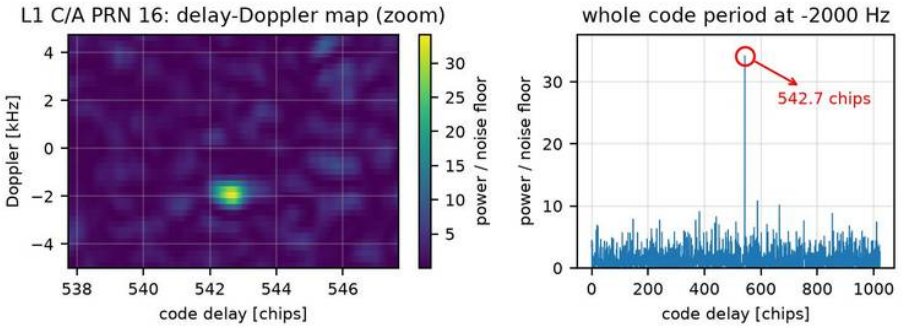


Figure 60: Acquisition, L1 C/A: delay–Doppler search of PRN 16; one sharp peak (test statistic 49 vs threshold 47) at -2000 Hz and 542.7 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[18]
L1 C/A	2	1	16	1.0
L5	2016	0	–	–

Table 55: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[18] Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

14.1.4 Tracking

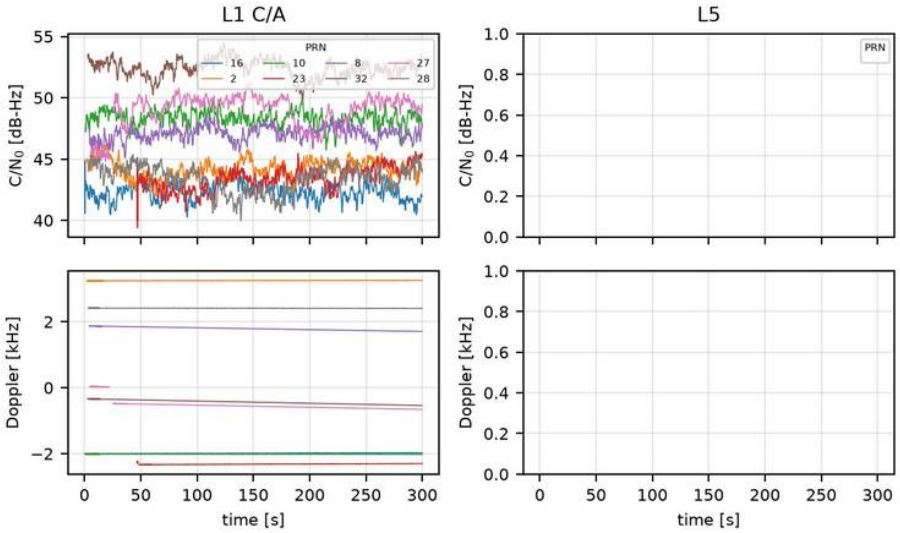


Figure 61: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

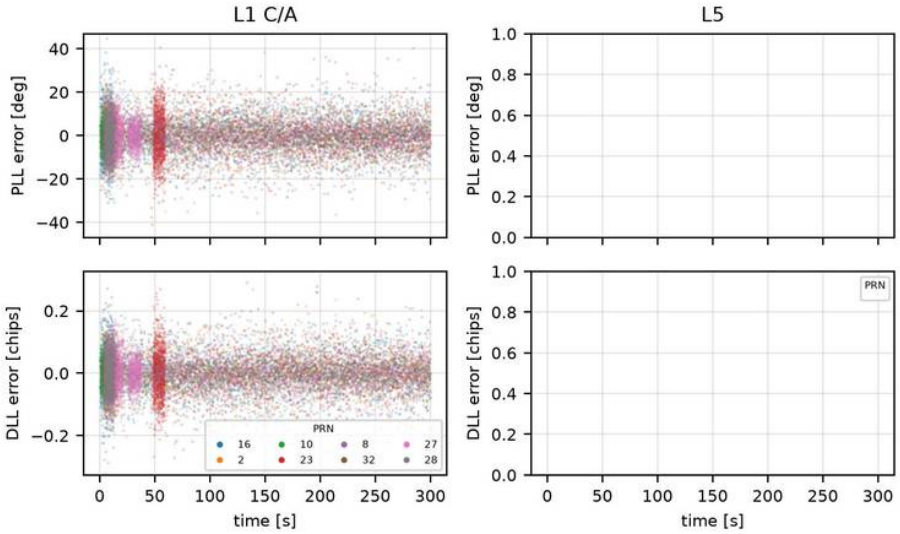


Figure 62: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

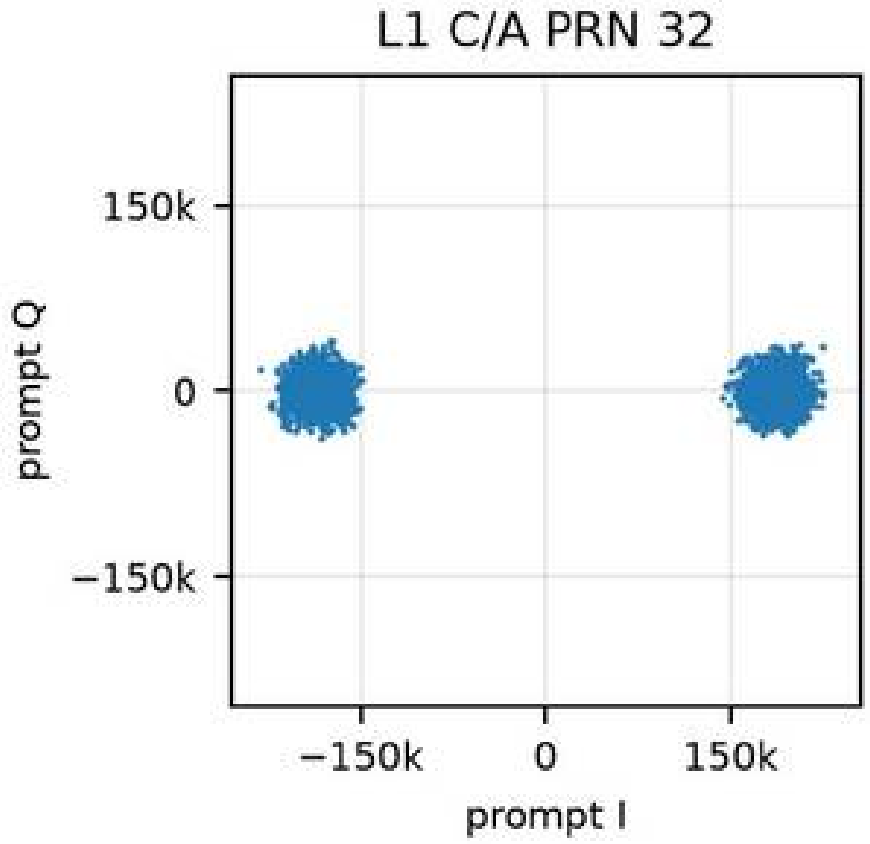


Figure 63: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	16	42.3	100	0	300	11.1	0.079
L1 C/A	1	2	44.7	97	0	299	8.4	0.061
L1 C/A	2	10	48.4	98	0	300	5.5	0.039
L1 C/A	3	23	43.3	95	0	253	11.1	0.072
L1 C/A	4	8	46.8	92	0	298	6.6	0.050
L1 C/A	5	32	52.7	94	0	299	3.5	0.024
L1 C/A	6	27	47.4	88	1	296	6.2	0.046
L1 C/A	7	28	44.2	90	0	299	9.3	0.067

Table 56: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

14.1.5 Telemetry decoder

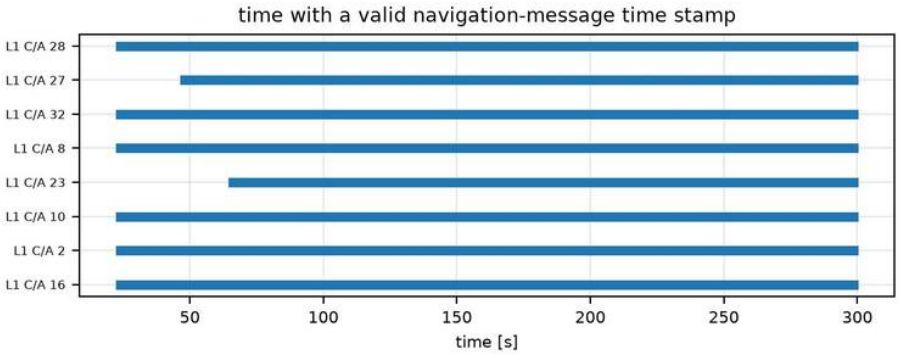


Figure 64: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
L1 C/A	2	45	yes	22.6	586848.0
L1 C/A	8	46	yes	22.6	586848.0
L1 C/A	10	46	yes	22.6	586848.0
L1 C/A	16	46	yes	22.6	586848.0
L1 C/A	23	39	yes	64.6	586890.0
L1 C/A	27	43	yes	46.6	586872.0
L1 C/A	28	45	yes	22.6	586848.0
L1 C/A	32	47	yes	22.6	586848.0

Table 57: Telemetry decoder: L1 C/A LNAV subframes, L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

14.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

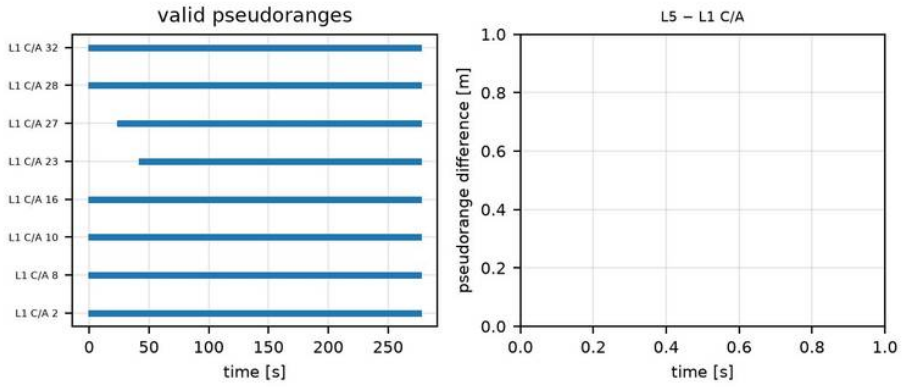


Figure 65: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Not produced: epochs with pseudoranges on both bands (L5 − L1 C/A).

14.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

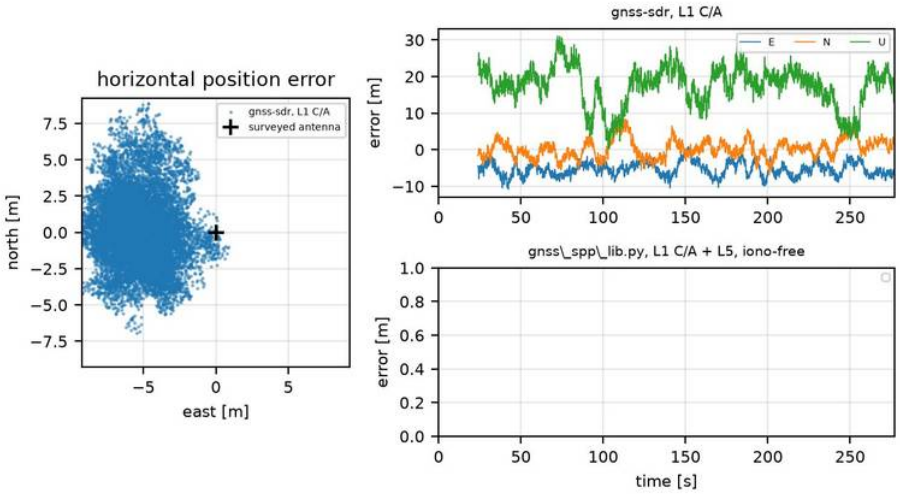


Figure 66: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

Solution	**gnss-sdr, L1 C/A**	**gnss_spp_lib.py, L1 C/A + L5, iono-free**
Solutions	13160	0
Run with fix [%]	92	–
Mean E / N / U [m]	-5.3 / 0.3 / 17.5	–
Std E / N / U [m]	1.9 / 2.5 / 5.4	–
RMS 2-D [m]	6.2	–
RMS 3-D [m]	19.3	–
95% horiz. [m]	8.7	–

Table 58: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

14.1.8 Signal conditioner (Galileo E1 + E5a)

Channel	ch0 (E1)	ch1 (E5a)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 59: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

14.1.9 Acquisition (Galileo E1 + E5a)

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

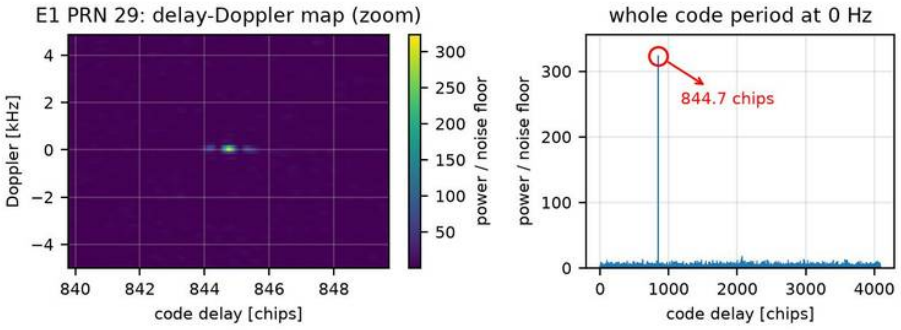


Figure 67: Acquisition, E1: delay–Doppler search of PRN 29; one sharp peak (test statistic 450 vs threshold 52) at 0 Hz and 844.7 chips.

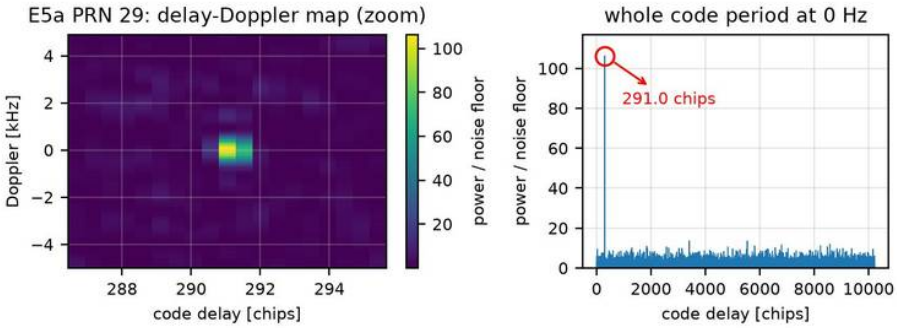


Figure 68: Acquisition, E5a: delay–Doppler search of PRN 29; one sharp peak (test statistic 149 vs threshold 49) at 27 Hz and 291.0 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[19]
E1	4	1	29	8.7
E5a	1	1	29	3.0

Table 60: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[19] Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

14.1.10 Tracking (Galileo E1 + E5a)

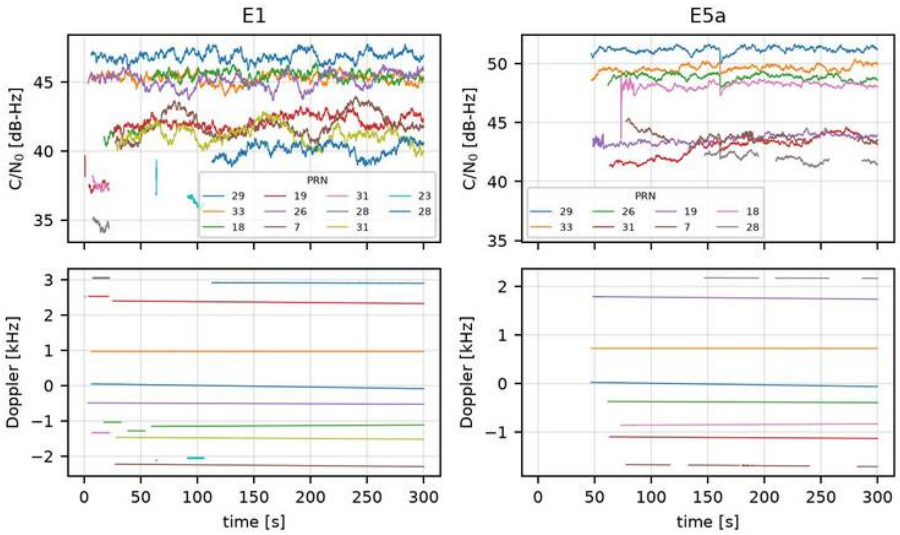


Figure 69: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

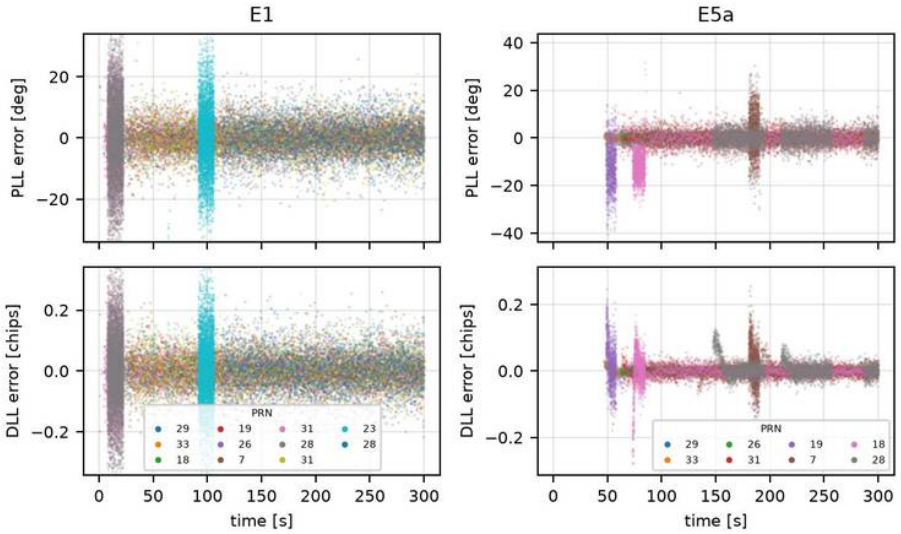


Figure 70: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

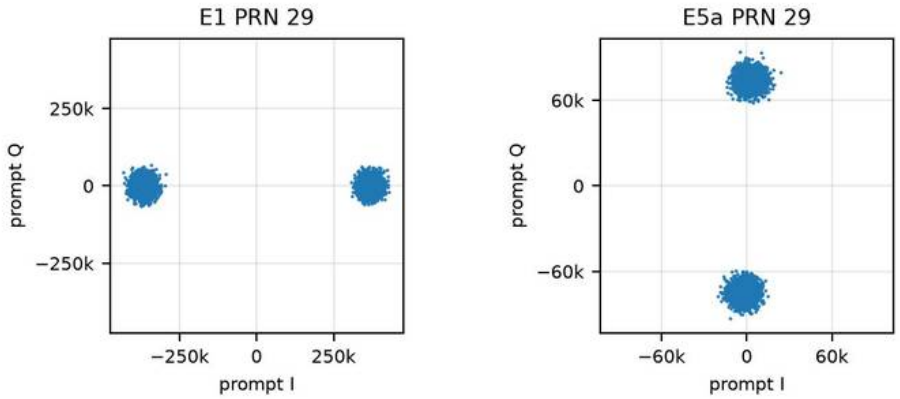


Figure 71: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
E1	0	29	46.9	98	0	299	3.5	0.036
E1	1	33	45.3	98	0	299	3.7	0.038
E1	2	18	45.4	91	3	299	4.4	0.046
E1	3	19	42.0	98	1	299	6.3	0.065
E1	4	26	45.2	99	0	299	4.1	0.043
E1	5	7	41.8	98	1	277	5.7	0.061
E1	5	31	37.5	71	1	21	10.3	0.105
E1	6	28	34.9	68	1	21	14.2	0.139
E1	6	31	41.1	98	1	276	6.5	0.070
E1	7	23	39.8	14	5	105	12.1	0.116
E1	7	28	40.1	97	5	193	7.0	0.076
E5a	8	29	51.2	52	0	274	1.3	0.007
E5a	9	33	49.4	52	0	274	1.5	0.008
E5a	10	26	48.4	31	1	274	1.7	0.008
E5a	11	31	41.8	51	0	257	3.2	0.017
E5a	12	19	43.4	91	0	254	9.0	0.050
E5a	13	7	43.4	25	4	240	7.3	0.050
E5a	14	18	48.2	100	0	227	7.4	0.044
E5a	15	28	42.1	14	2	158	3.6	0.028

Table 61: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

14.1.11 Telemetry decoder (Galileo E1 + E5a)

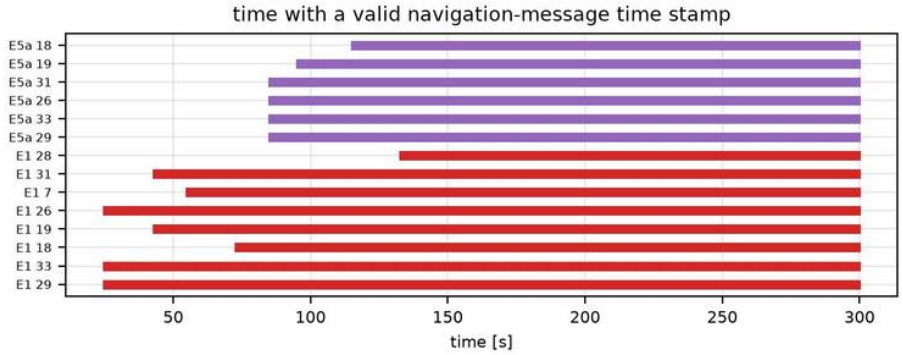


Figure 72: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	7	24	yes	54.7	586880.0
E1	18	22	yes	72.7	586898.0
E1	19	24	yes	42.7	586868.0
E1	26	27	yes	24.7	586850.0
E1	28	16	yes	132.7	586958.0
E1	29	25	yes	24.7	586850.0
E1	31	25	yes	42.7	586868.0
E1	33	26	yes	24.7	586850.0
E5a	18	10	yes	114.9	586940.3
E5a	19	11	yes	94.9	586920.3
E5a	26	12	yes	84.9	586910.3
E5a	29	12	yes	84.9	586910.3
E5a	31	12	yes	84.9	586910.3
E5a	33	12	yes	84.9	586910.3

Table 62: Telemetry decoder: E1 I/NAV words, E5a F/NAV pages decoded, ephemeris obtained, and the first valid time of week.

14.1.12 Observables (Galileo E1 + E5a)

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

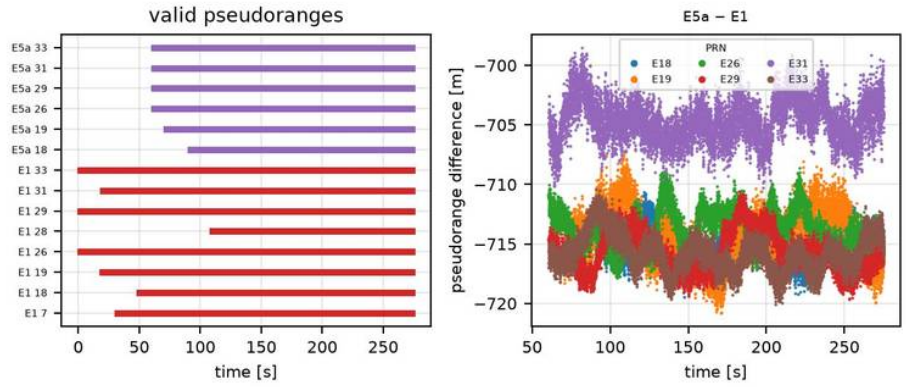


Figure 73: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
E5a − E1	18	9240	62	-715.18	1.29	-0.0008
E5a − E1	19	10241	68	-714.24	2.20	-0.0021
E5a − E1	26	10728	72	-713.77	1.56	-0.0003
E5a − E1	29	10727	72	-715.68	1.65	-0.0031
E5a − E1	31	10726	72	-704.59	1.77	0.0011
E5a − E1	33	10727	72	-715.80	1.51	-0.0056

Table 63: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

14.1.13 PVT (Galileo E1 + E5a)

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

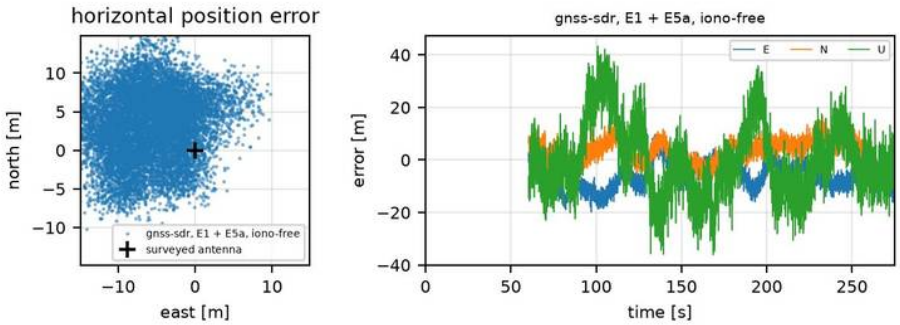


Figure 74: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution.

Solution	**gnss-sdr, E1 + E5a, iono-free**
Solutions	10909
Run with fix [%]	79
Mean E / N / U [m]	-6.0 / 3.2 / 0.0
Std E / N / U [m]	4.8 / 4.5 / 13.5
RMS 2-D [m]	9.4
RMS 3-D [m]	16.4
95% horiz. [m]	15.1

Table 64: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

14.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-24 19:30:07 UTC	
Frequency	1575.420 MHz (L1)	1227.600 MHz (L2)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	4.000 MS/s
Sample type	signed complex 16-bit samples (I+Q), little-endian, as received from the X310 (sc16)	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	4800.0 MB	4800.0 MB
Samples written	1,200,000,000	1,200,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca16bits4msps_l2c16bits4msps.conf`	

Table 65: Metadata for recording `x310_20261024T193000Z`, captured starting 2026-10-24T19:30:07.284275Z.

14.2.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

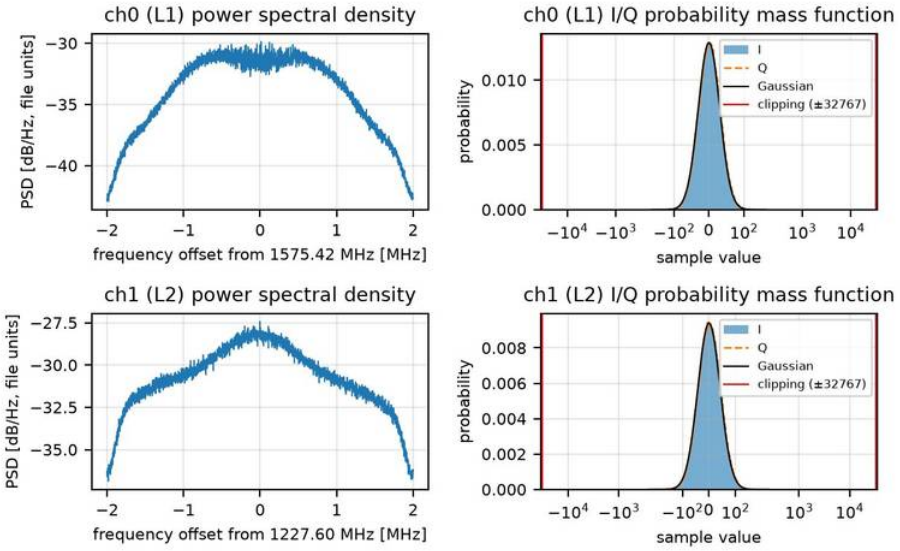


Figure 75: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1227.60
Rate [MS/s]	4	4
Format	ci16_le	ci16_le
Bytes per complex sample	4 (2 for I, 2 for Q)	4 (2 for I, 2 for Q)
Full scale	±32767	±32767
Gain [dB]	25	25
Std I / Q	30.9 / 31.0	42.5 / 42.4
Mean I / Q	-0.15 / 0.08	-0.17 / -0.04
Min / max I	-146 / 146	-216 / 196
Min / max Q	-152 / 146	-204 / 217
Headroom[20]	1058.6	772.1
Clipped [%]	0.000	0.000
Lost samples	0	0

Table 66: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[20]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

14.2.2 Signal conditioner

Channel	ch0 (L1 C/A, E1)	ch1 (L2C)
File format	ci16_le	ci16_le
Source item type	ishort	ishort
Data type adapter	Ishort_To_Complex	Ishort_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	Pass_Through	Pass_Through
Rate [MS/s]	4 → 4	4 → 4

Table 67: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

14.2.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

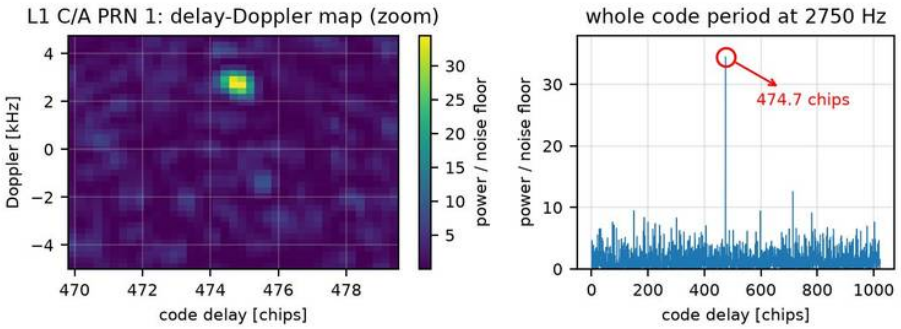


Figure 76: Acquisition, L1 C/A: delay–Doppler search of PRN 1; one sharp peak (test statistic 49 vs threshold 44) at 2750 Hz and 474.7 chips.

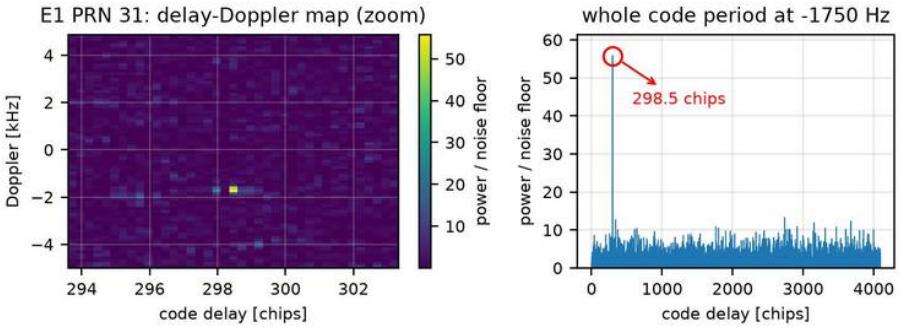


Figure 77: Acquisition, E1: delay–Doppler search of PRN 31; one sharp peak (test statistic 79 vs threshold 48) at -1750 Hz and 298.5 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[21]
L1 C/A	485	1	1	1.1
L2C	not dumped (search grid too large); acquired satellites: see the tracking table			
E1	4	1	31	1.6

Table 68: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[21] Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

14.2.4 Tracking

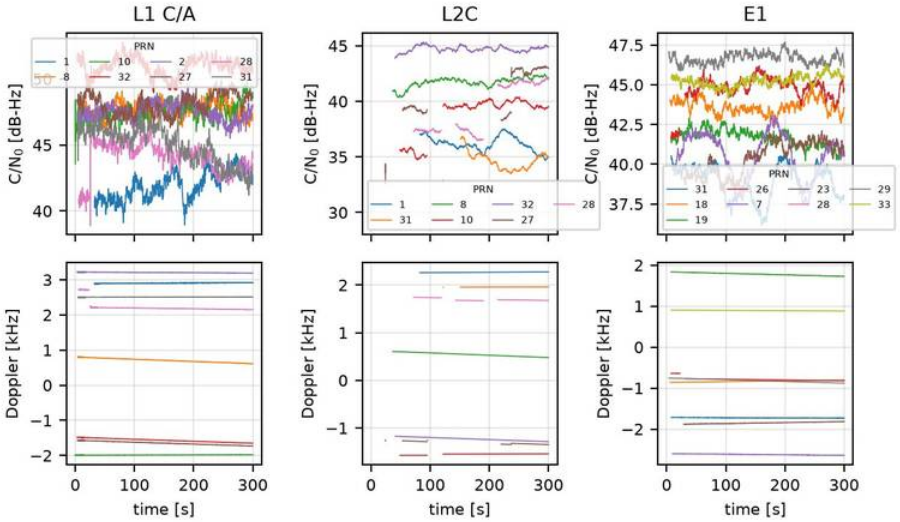


Figure 78: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

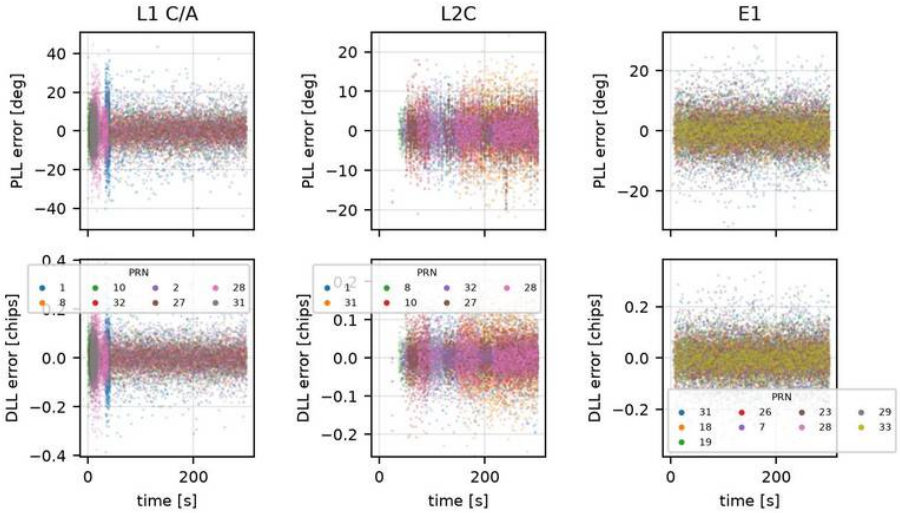


Figure 79: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

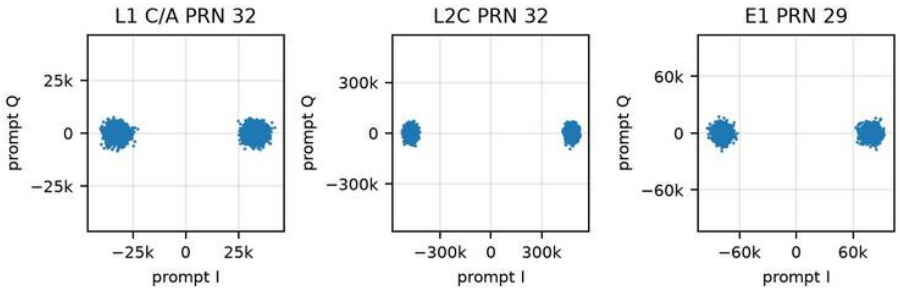


Figure 80: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	1	41.0	88	0	270	12.5	0.095
L1 C/A	1	8	47.8	95	0	297	5.9	0.047
L1 C/A	2	10	47.2	100	0	300	6.4	0.049
L1 C/A	3	32	51.3	96	0	298	4.0	0.031
L1 C/A	4	2	47.4	96	0	298	6.1	0.046
L1 C/A	5	27	48.2	92	0	298	5.6	0.043
L1 C/A	6	28	43.1	90	1	296	10.8	0.083
L1 C/A	7	31	45.9	100	0	295	7.8	0.061
L2C	8	1	36.2	86	0	253	5.3	0.060
L2C	9	31	35.0	55	3	276	6.6	0.069
L2C	10	8	41.8	95	0	277	3.3	0.030
L2C	11	10	39.5	81	1	276	4.3	0.045
L2C	12	32	44.7	94	0	276	2.7	0.022
L2C	14	27	39.6	45	3	276	5.6	0.036
L2C	15	28	37.7	71	2	252	4.4	0.042
E1	16	31	39.2	98	0	298	8.5	0.097
E1	17	18	43.6	99	0	299	4.8	0.056
E1	18	19	41.7	98	0	299	6.0	0.070
E1	19	26	44.9	97	1	294	4.4	0.049
E1	20	7	40.8	98	0	296	6.9	0.079
E1	21	23	40.6	99	1	275	7.0	0.081
E1	21	28	42.5	0	1	21	–	–
E1	22	29	46.6	100	0	298	3.4	0.040
E1	23	33	45.3	98	0	297	3.5	0.041

Table 69: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

14.2.5 Telemetry decoder

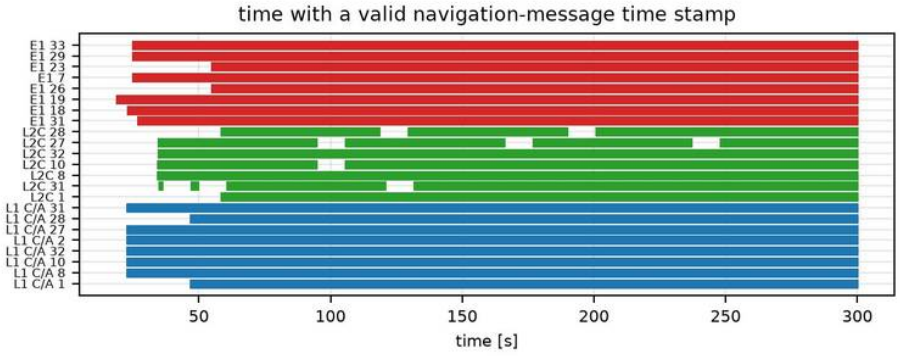


Figure 81: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	7	26	yes	24.9	588650.0
E1	18	27	yes	22.8	588648.0
E1	19	27	yes	18.8	588644.0
E1	23	24	yes	54.9	588680.0
E1	26	24	yes	54.8	588680.0
E1	29	26	yes	24.8	588650.0
E1	31	26	yes	26.9	588652.0
E1	33	26	yes	24.8	588650.0
L1 C/A	1	43	yes	46.8	588672.0
L1 C/A	2	47	yes	22.8	588648.0
L1 C/A	8	47	yes	22.8	588648.0
L1 C/A	10	46	yes	22.8	588648.0
L1 C/A	27	47	yes	22.8	588648.0
L1 C/A	28	42	yes	46.8	588672.0
L1 C/A	31	47	yes	22.8	588648.0
L1 C/A	32	47	yes	22.8	588648.0
L2C	1	14	yes	58.4	0.0
L2C	8	13	yes	34.4	0.0
L2C	10	9	yes	34.5	0.0
L2C	27	2	no	34.7	0.0
L2C	28	5	yes	58.5	0.0
L2C	31	8	yes	35.0	0.0
L2C	32	13	yes	34.5	0.0

Table 70: Telemetry decoder: L1 C/A LNAV subframes, L2C CNAV messages, E1 I/NAV words decoded, ephemeris obtained, and the first valid time of week.

14.2.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

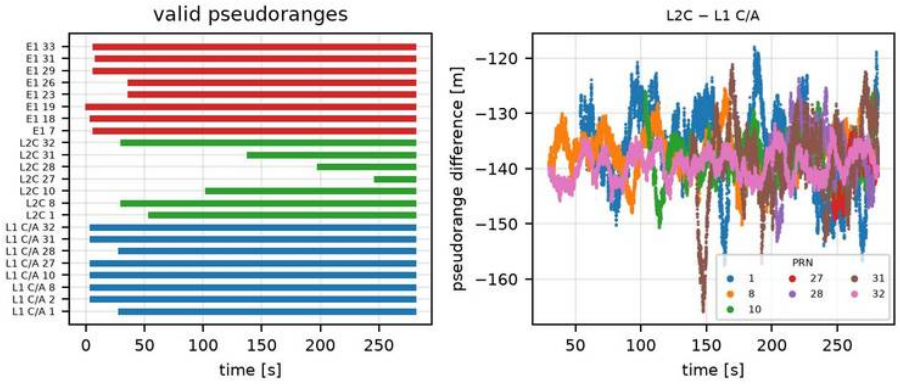


Figure 82: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L2C – L1 C/A	1	11364	76	-136.67	7.21	-0.0353
L2C – L1 C/A	8	12562	84	-137.27	3.68	-0.0021
L2C – L1 C/A	10	8932	60	-138.51	4.05	-0.0000
L2C – L1 C/A	27	1736	12	-140.56	4.07	0.1949
L2C – L1 C/A	28	4170	28	-137.26	5.58	-0.0237
L2C – L1 C/A	31	7158	48	-138.45	7.12	0.0509
L2C – L1 C/A	32	12556	84	-139.30	2.55	0.0083

Table 71: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

14.2.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

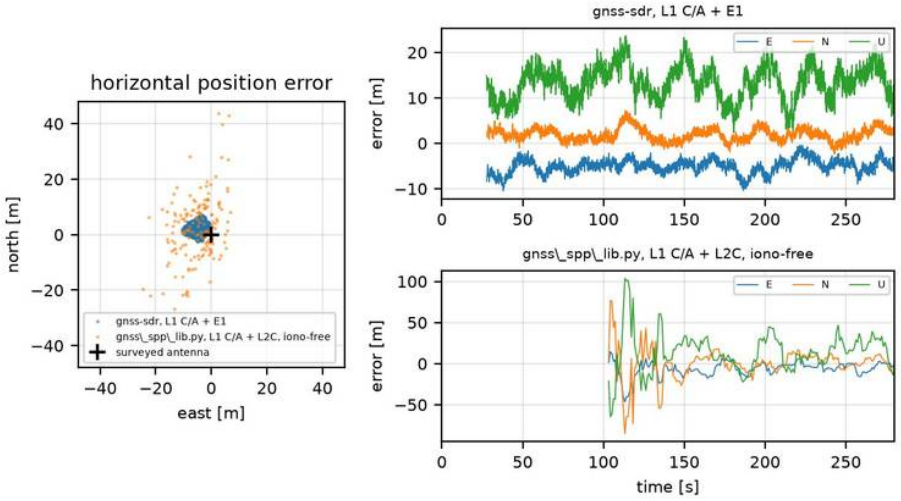


Figure 83: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

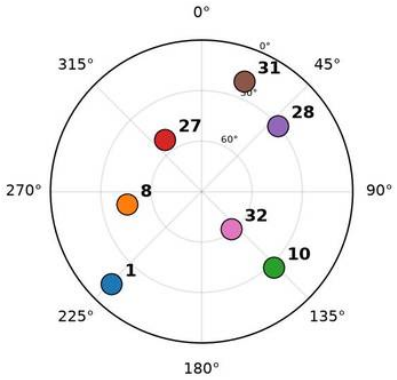


Figure 84: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, L1 C/A + E1**	**gnss_spp_lib.py, L1 C/A + L2C, iono-free**
Solutions	13152	178
Run with fix [%]	91	64
Mean E / N / U [m]	-5.0 / 1.9 / 13.3	-6.4 / 0.3 / 13.9
Std E / N / U [m]	1.6 / 1.4 / 3.7	9.2 / 19.9 / 24.5
RMS 2-D [m]	5.8	22.8
RMS 3-D [m]	15.0	36.2
95% horiz. [m]	8.2	55.6

Table 72: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

15 Records on 25/10/2026

15.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-25 19:00:07 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	0	1134464
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`, `e1_8bits4msps_e5a_8bits20msps.conf`	

Table 73: Metadata for recording `x310_20261025T190000Z`, captured starting 2026-10-25T19:00:07.043233Z.

15.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

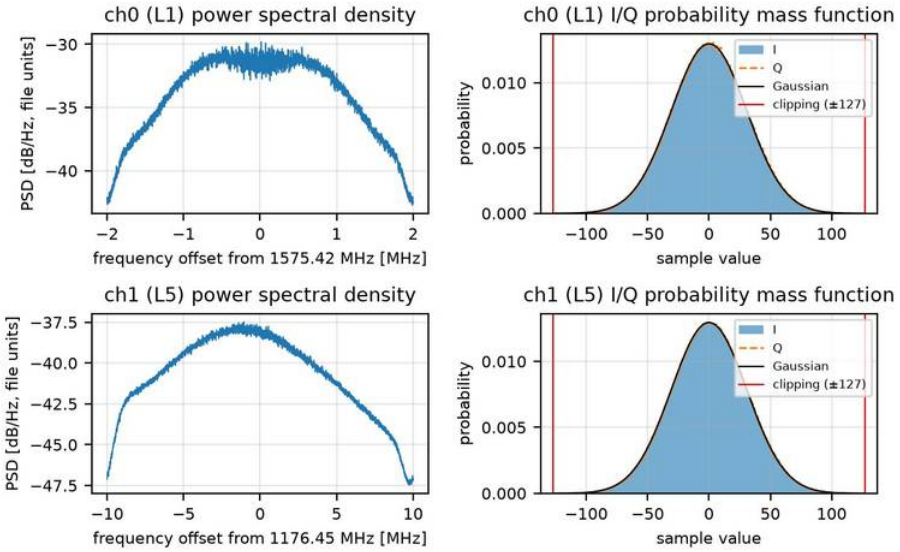


Figure 85: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Bytes per complex sample	2 (1 for I, 1 for Q)	2 (1 for I, 1 for Q)
Full scale	±127	±127
Gain [dB]	25	25
Std I / Q	30.7 / 30.7	30.8 / 30.8
Mean I / Q	-0.19 / 0.06	-0.02 / 0.01
Min / max I	-127 / 127	-127 / 127
Min / max Q	-127 / 127	-127 / 127
Headroom[22]	4.1	4.1
Clipped [%]	0.004	0.004
Lost samples	0	1,134,464

Table 74: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[22]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

15.1.2 Signal conditioner

Channel	ch0 (L1 C/A)	ch1 (L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 75: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

15.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

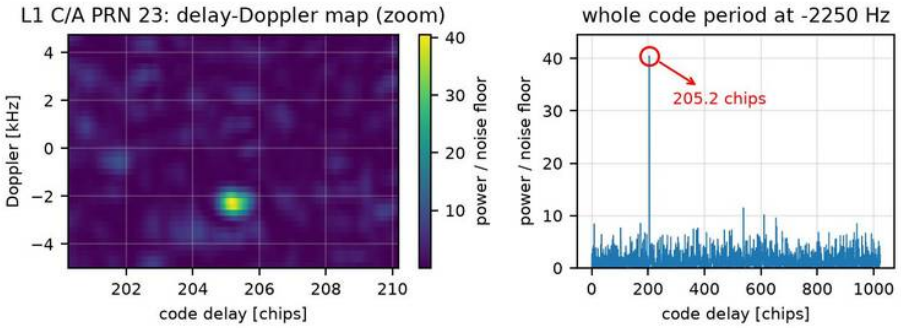


Figure 86: Acquisition, L1 C/A: delay–Doppler search of PRN 23; one sharp peak (test statistic 55 vs threshold 47) at -2250 Hz and 205.2 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[23]
L1 C/A	3	1	23	1.2
L5	13137	0	–	–

Table 76: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[23] Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

15.1.4 Tracking

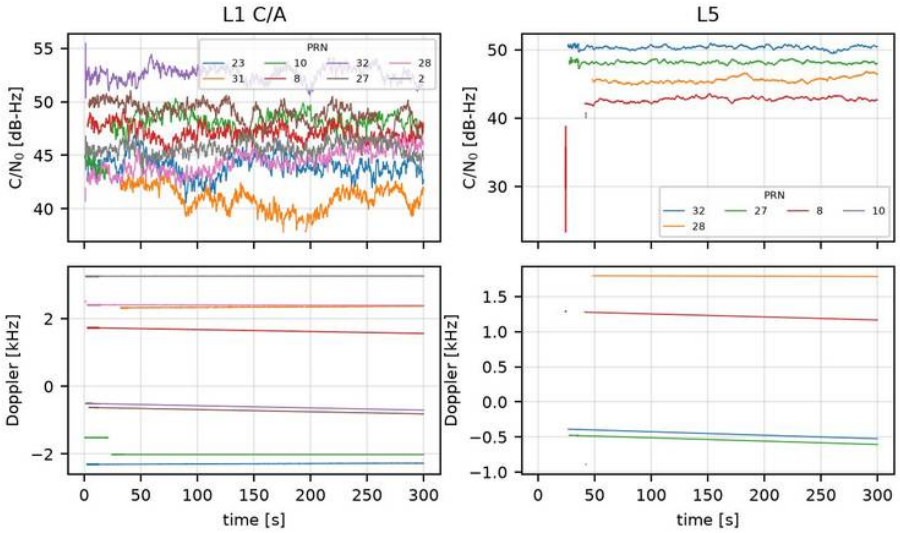


Figure 87: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

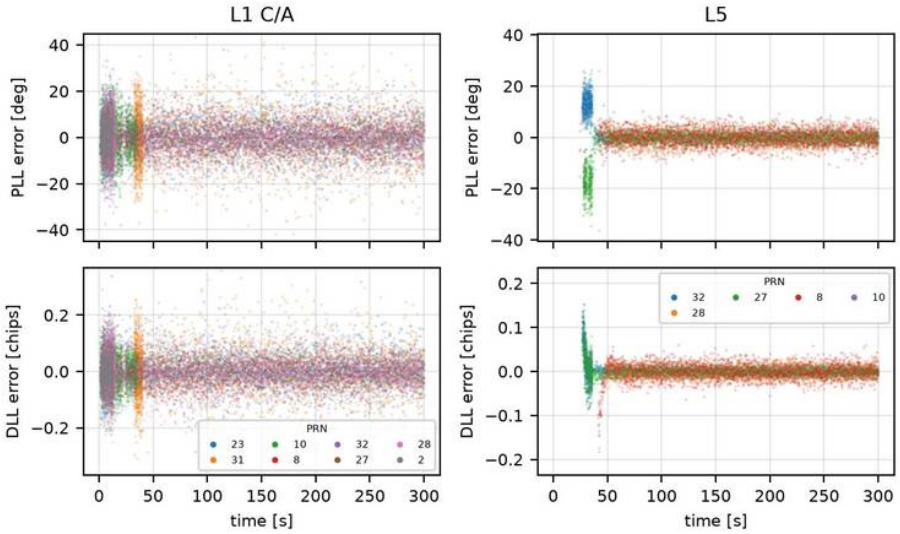


Figure 88: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

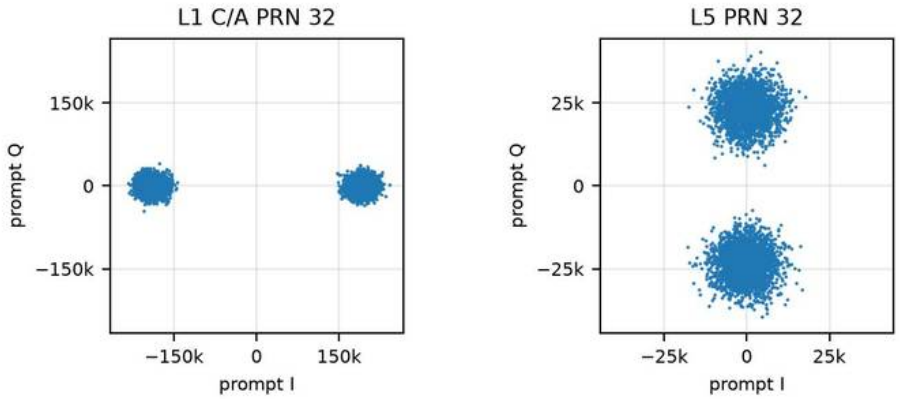


Figure 89: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	23	43.9	94	0	299	9.2	0.067
L1 C/A	1	31	41.8	84	0	271	12.4	0.094
L1 C/A	2	10	46.9	97	1	297	7.5	0.054
L1 C/A	3	8	47.6	96	0	298	6.1	0.046
L1 C/A	4	32	52.5	100	0	299	3.6	0.025
L1 C/A	5	27	49.7	88	0	299	4.9	0.036
L1 C/A	6	28	43.3	93	0	299	9.4	0.069
L1 C/A	7	2	45.5	100	0	299	7.5	0.055
L5	9	32	50.4	95	0	275	6.4	0.026
L5	10	28	45.3	68	0	272	2.5	0.013
L5	11	27	48.2	86	0	275	7.2	0.021
L5	14	8	42.7	69	0	276	4.7	0.031
L5	15	10	28.8	0	1	16	48.3	0.070

Table 77: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

15.1.5 Telemetry decoder

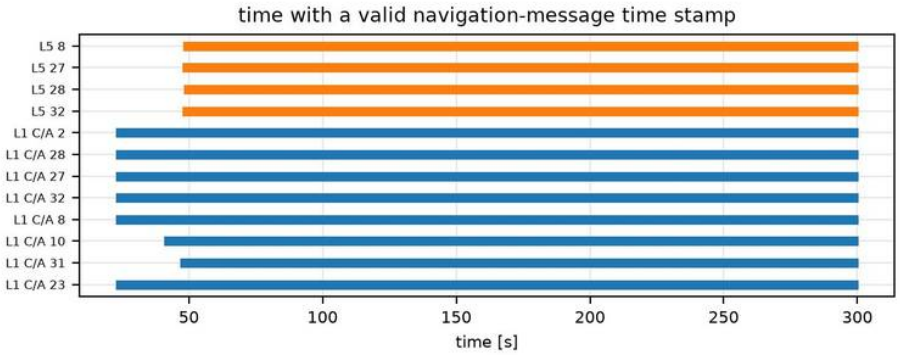


Figure 90: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
L1 C/A	2	47	yes	23.0	68448.0
L1 C/A	8	47	yes	23.0	68448.0
L1 C/A	10	44	yes	41.0	68466.0
L1 C/A	23	46	yes	23.0	68448.0
L1 C/A	27	47	yes	23.0	68448.0
L1 C/A	28	47	yes	23.0	68448.0
L1 C/A	31	41	yes	47.0	68472.0
L1 C/A	32	47	yes	23.0	68448.0
L5	8	25	yes	48.2	68473.1
L5	27	26	yes	47.8	68472.8
L5	28	32	yes	48.4	68473.4
L5	32	26	yes	47.8	68472.8

Table 78: Telemetry decoder: L1 C/A LNAV subframes, L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

15.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

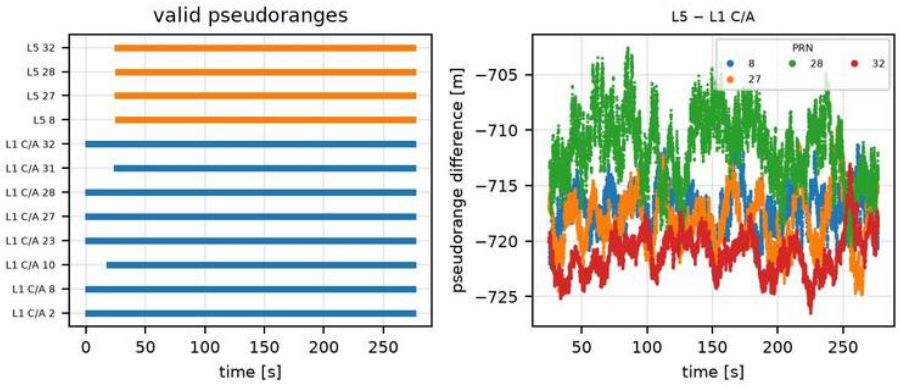


Figure 91: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L5 − L1 C/A	8	12423	83	-716.96	2.04	0.0011
L5 − L1 C/A	27	12441	83	-718.12	2.55	0.0021
L5 − L1 C/A	28	12412	83	-711.59	3.06	-0.0100
L5 − L1 C/A	32	12442	83	-721.11	1.85	0.0058

Table 79: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

15.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

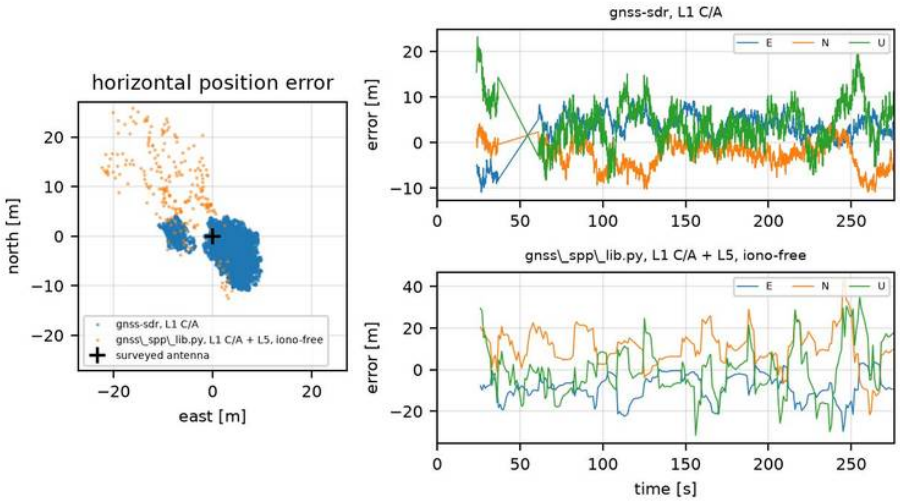


Figure 92: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

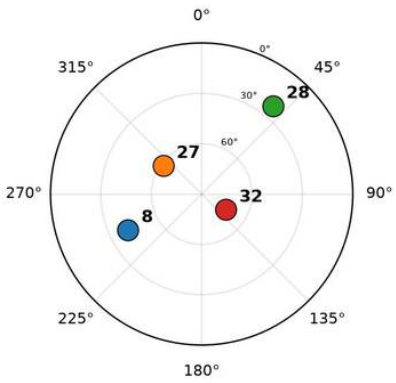


Figure 93: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	gnss-sdr, L1 C/A	gnss_spp_lib.py, L1 C/A + L5, iono-free
Solutions	11859	247
Run with fix [%]	84	89
Mean E / N / U [m]	3.5 / -2.7 / 4.2	-8.3 / 10.5 / -0.1
Std E / N / U [m]	3.3 / 2.9 / 4.8	6.5 / 9.3 / 12.6
RMS 2-D [m]	6.2	17.6
RMS 3-D [m]	8.9	21.6
95% horiz. [m]	10.1	29.1

Table 80: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

15.1.8 Signal conditioner (Galileo E1 + E5a)

Channel	ch0 (E1)	ch1 (E5a)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 81: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

15.1.9 Acquisition (Galileo E1 + E5a)

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

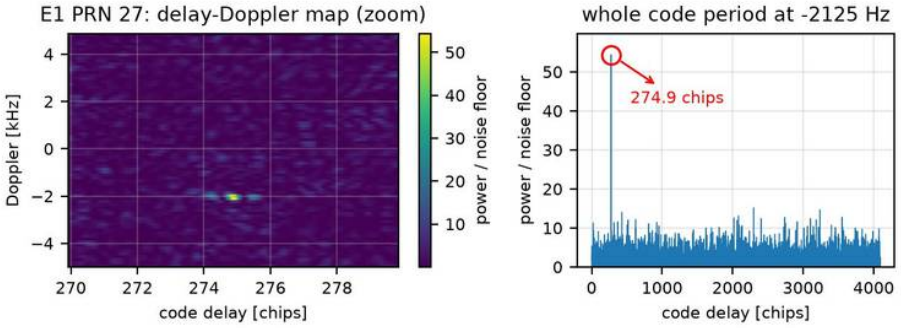


Figure 94: Acquisition, E1: delay–Doppler search of PRN 27; one sharp peak (test statistic 76 vs threshold 52) at -2125 Hz and 274.9 chips.

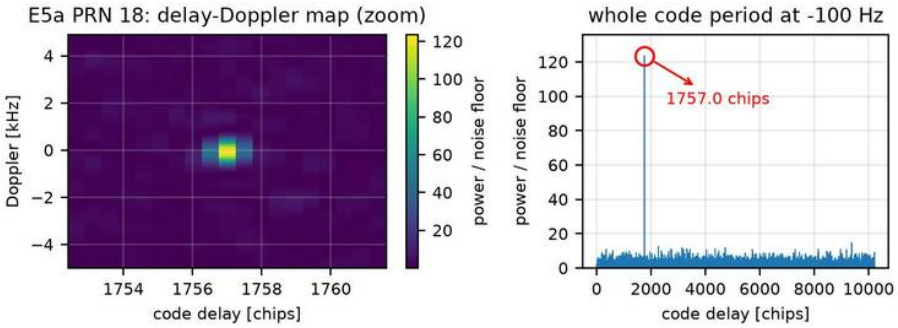


Figure 95: Acquisition, E5a: delay–Doppler search of PRN 18; one sharp peak (test statistic 171 vs threshold 49) at -1556 Hz and 1757.0 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[24]
E1	4	1	27	1.5
E5a	1	1	18	3.5

Table 82: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[24]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

15.1.10 Tracking (Galileo E1 + E5a)

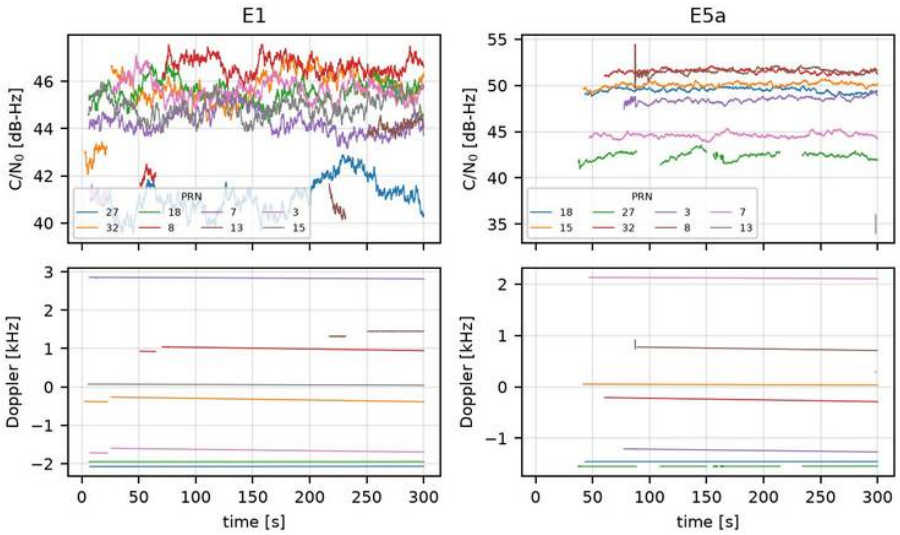


Figure 96: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

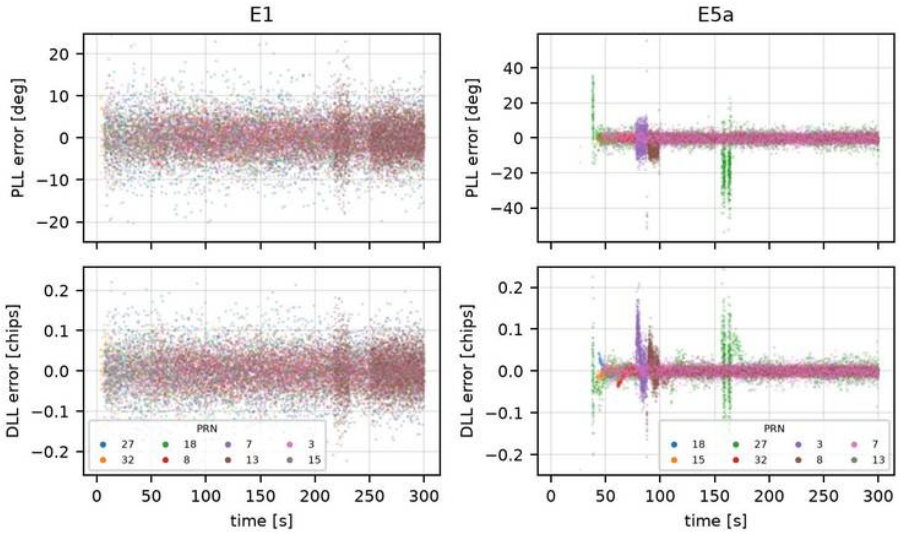


Figure 97: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

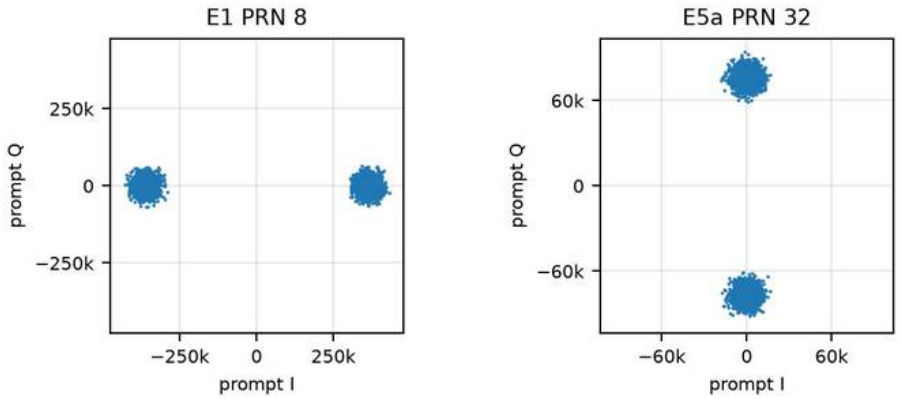


Figure 98: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
E1	0	27	41.0	98	0	299	6.6	0.070
E1	1	32	45.8	98	1	299	3.7	0.040
E1	2	18	45.6	98	0	299	4.0	0.042
E1	3	8	46.5	82	3	298	3.8	0.039
E1	4	7	44.2	98	0	298	4.3	0.046
E1	5	13	43.8	21	12	300	4.8	0.052
E1	6	3	45.6	97	1	298	4.2	0.043
E1	7	15	44.9	98	0	300	3.9	0.042
E5a	8	18	49.5	53	0	276	1.6	0.008
E5a	9	15	50.1	53	0	274	1.6	0.012
E5a	10	27	42.4	25	3	269	10.7	0.043
E5a	11	32	51.4	51	0	256	1.3	0.007
E5a	12	3	29.7	38	2	256	4.1	0.041
E5a	13	8	51.4	93	0	213	6.2	0.026
E5a	14	7	44.6	52	0	274	2.8	0.016
E5a	15	13	29.0	0	2	34	28.5	0.177

Table 83: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

15.1.11 Telemetry decoder (Galileo E1 + E5a)

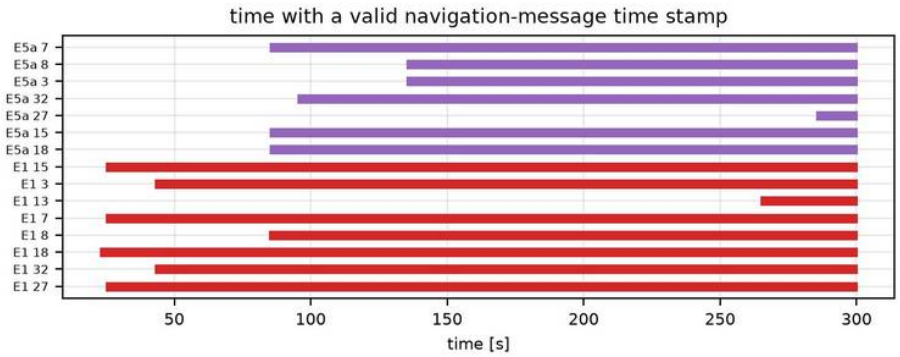


Figure 99: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	3	25	yes	43.1	68468.0
E1	7	26	yes	25.1	68450.0
E1	8	21	yes	85.1	68510.0
E1	13	3	yes	265.1	68690.0
E1	15	26	yes	25.1	68450.0
E1	18	27	yes	23.1	68448.0
E1	27	26	yes	25.1	68450.0
E1	32	25	yes	43.1	68468.0
E5a	3	9	yes	135.3	68560.3
E5a	7	12	yes	85.3	68510.3
E5a	8	9	yes	135.3	68560.3
E5a	15	12	yes	85.3	68510.3
E5a	18	11	yes	85.3	68510.3
E5a	27	–	yes	285.3	68710.3
E5a	32	11	yes	95.3	68520.3

Table 84: Telemetry decoder: E1 I/NAV words, E5a F/NAV pages decoded, ephemeris obtained, and the first valid time of week.

15.1.12 Observables (Galileo E1 + E5a)

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

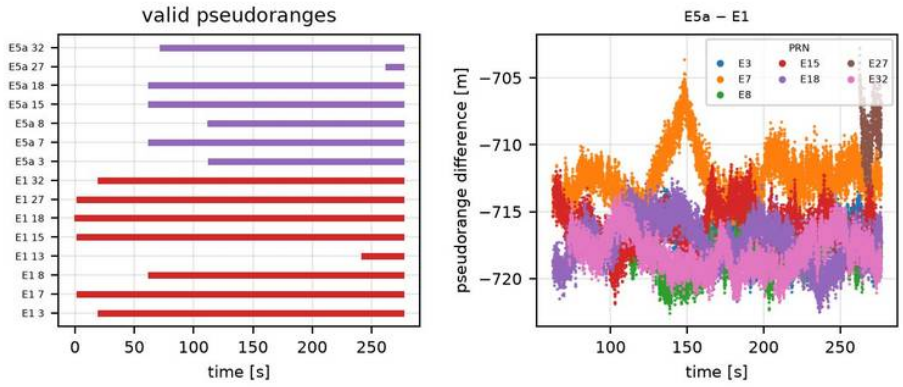


Figure 100: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
E5a − E1	3	8206	55	-717.05	1.31	-0.0028
E5a − E1	7	10398	69	-712.39	1.87	0.0037
E5a − E1	8	8207	55	-718.19	1.48	0.0094
E5a − E1	15	10422	70	-716.17	1.51	0.0016
E5a − E1	18	10398	69	-717.25	1.64	-0.0080
E5a − E1	27	720	5	-708.73	1.89	0.0081
E5a − E1	32	9987	67	-718.07	1.39	-0.0022

Table 85: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

15.1.13 PVT (Galileo E1 + E5a)

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

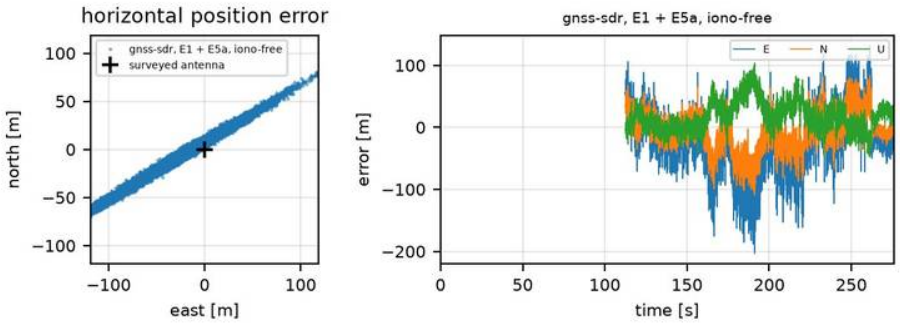


Figure 101: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution.

Solution	**gnss-sdr, E1 + E5a, iono-free**
Solutions	8535
Run with fix [%]	60
Mean E / N / U [m]	-25.3 / -7.1 / 17.7
Std E / N / U [m]	51.5 / 30.6 / 23.4
RMS 2-D [m]	65.4
RMS 3-D [m]	71.7
95% horiz. [m]	133.9

Table 86: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

15.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-25 19:52:09 UTC	
Frequency	1575.420 MHz (L1)	1227.600 MHz (L2)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	4.000 MS/s
Sample type	signed complex 16-bit samples (I+Q), little-endian, as received from the X310 (sc16)	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	4800.0 MB	4800.0 MB
Samples written	1,200,000,000	1,200,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca16bits4msps_l2c16bits4msps.conf`	

Table 87: Metadata for recording `x310_20261025T195201Z`, captured starting 2026-10-25T19:52:09.046382Z.

15.2.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

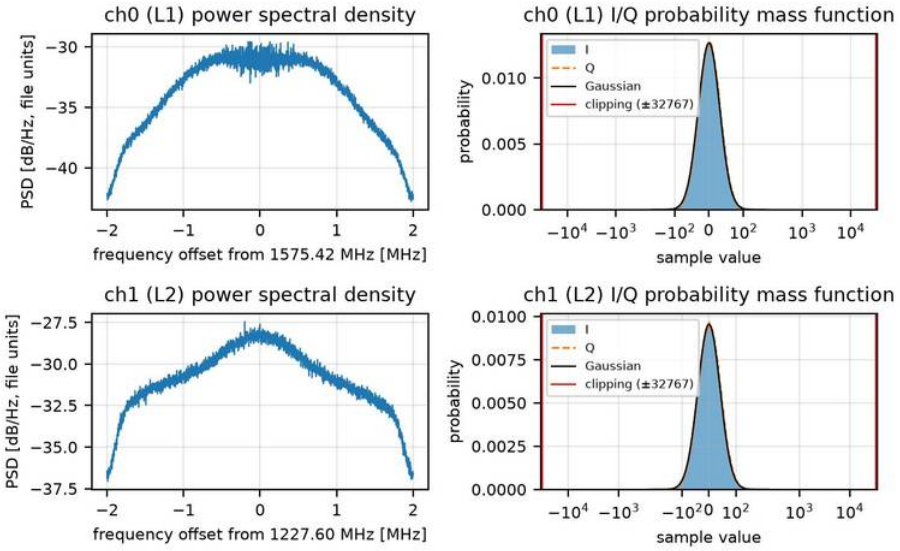


Figure 102: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1227.60
Rate [MS/s]	4	4
Format	ci16_le	ci16_le
Bytes per complex sample	4 (2 for I, 2 for Q)	4 (2 for I, 2 for Q)
Full scale	±32767	±32767
Gain [dB]	25	25
Std I / Q	31.5 / 31.5	41.8 / 41.9
Mean I / Q	-0.03 / 0.01	0.20 / -0.00
Min / max I	-150 / 162	-197 / 207
Min / max Q	-149 / 177	-208 / 205
Headroom[25]	1040.8	783.7
Clipped [%]	0.000	0.000
Lost samples	0	0

Table 88: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[25] Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

15.2.2 Signal conditioner

Channel	ch0 (L1 C/A, E1)	ch1 (L2C)
File format	ci16_le	ci16_le
Source item type	ishort	ishort
Data type adapter	Ishort_To_Complex	Ishort_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	Pass_Through	Pass_Through
Rate [MS/s]	4 → 4	4 → 4

Table 89: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

15.2.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

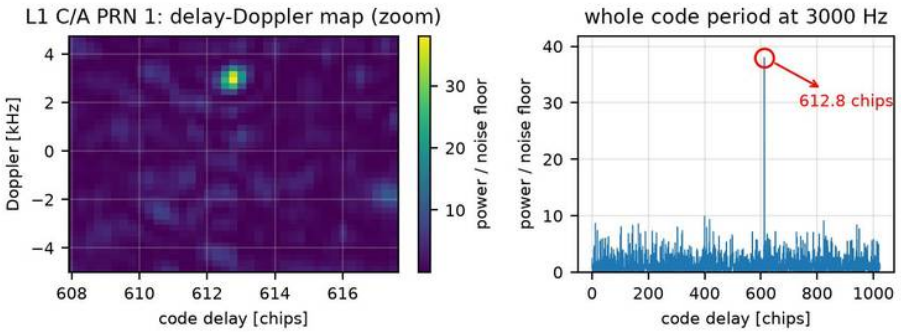


Figure 103: Acquisition, L1 C/A: delay–Doppler search of PRN 1; one sharp peak (test statistic 51 vs threshold 44) at 3000 Hz and 612.8 chips.

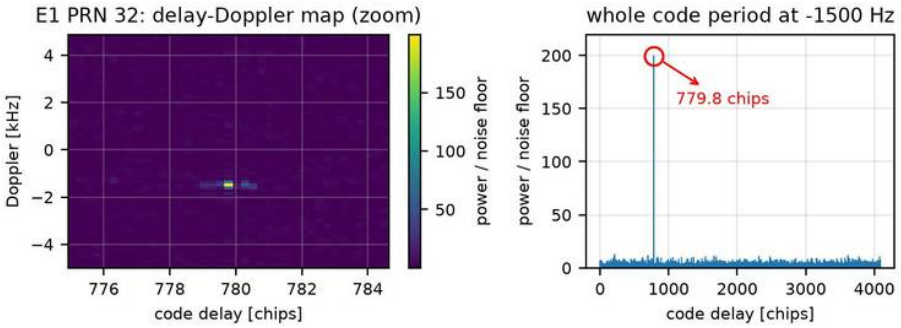


Figure 104: Acquisition, E1: delay–Doppler search of PRN 32; one sharp peak (test statistic 277 vs threshold 48) at -1500 Hz and 779.8 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[26]
L1 C/A	1	1	1	1.2
L2C	not dumped (search grid too large); acquired satellites: see the tracking table			
E1	5	1	32	5.7

Table 90: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[26]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

15.2.4 Tracking

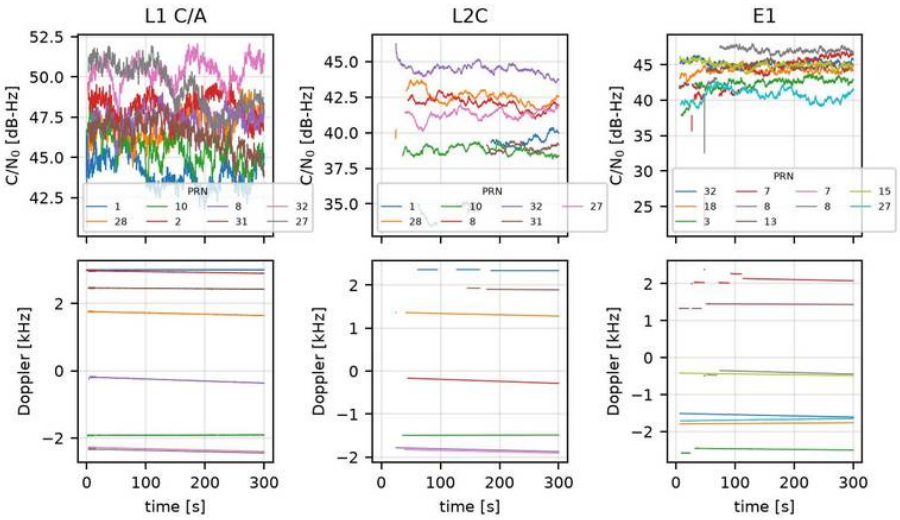


Figure 105: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

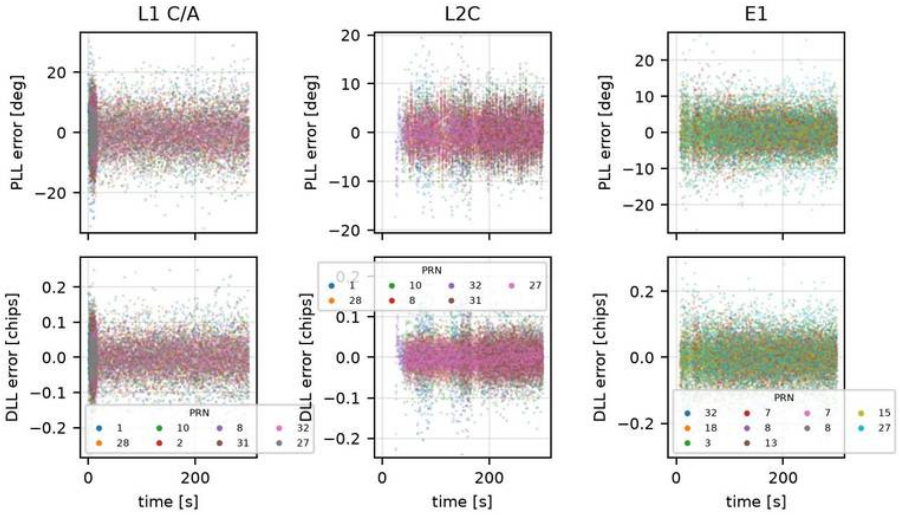


Figure 106: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

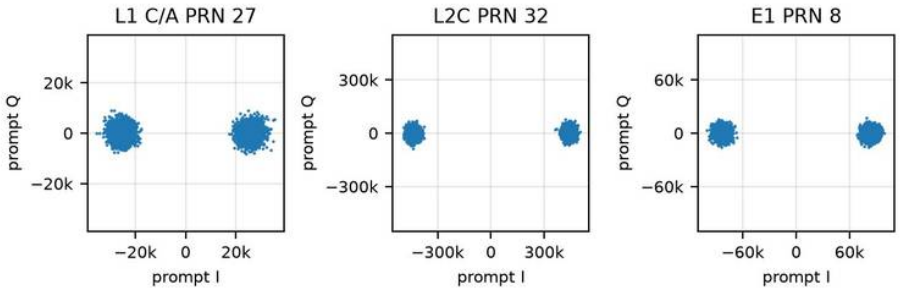


Figure 107: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	1	44.3	100	0	300	8.9	0.071
L1 C/A	1	28	46.4	97	0	298	6.6	0.052
L1 C/A	2	10	46.1	94	0	300	7.2	0.055
L1 C/A	3	2	48.4	97	0	298	5.7	0.042
L1 C/A	4	8	47.2	95	0	296	6.4	0.050
L1 C/A	5	31	46.6	95	0	298	6.9	0.052
L1 C/A	6	32	50.2	95	0	298	4.4	0.033
L1 C/A	7	27	50.7	95	0	298	4.5	0.034
L2C	8	1	36.7	68	2	276	5.0	0.056
L2C	9	28	42.5	94	0	276	3.8	0.030
L2C	10	10	38.7	96	0	276	4.3	0.043
L2C	12	8	42.1	93	0	276	3.3	0.029
L2C	13	32	44.5	100	0	276	3.0	0.025
L2C	14	31	37.3	52	2	276	4.4	0.044
L2C	15	27	41.3	95	0	275	3.5	0.033
E1	16	32	45.2	99	0	297	3.8	0.045
E1	17	18	44.1	98	0	299	4.7	0.052
E1	18	3	42.5	96	1	294	5.7	0.065
E1	19	7	44.6	87	5	273	5.2	0.052
E1	19	8	45.3	0	5	21	–	–
E1	20	13	44.7	94	2	296	4.1	0.046
E1	21	7	44.0	0	3	21	–	–
E1	21	8	46.9	89	3	273	4.2	0.040
E1	22	15	45.0	98	0	299	3.9	0.046
E1	23	27	40.7	98	0	298	6.9	0.079

Table 91: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

15.2.5 Telemetry decoder

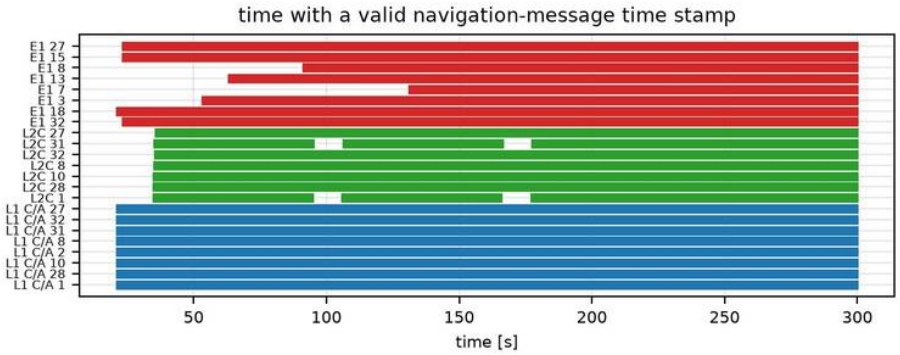


Figure 108: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	3	23	yes	53.1	71600.0
E1	7	16	yes	131.1	71678.0
E1	8	21	yes	91.1	71638.0
E1	13	22	yes	63.1	71610.0
E1	15	26	yes	23.1	71570.0
E1	18	26	yes	21.1	71568.0
E1	27	27	yes	23.1	71570.0
E1	32	27	yes	23.1	71570.0
L1 C/A	1	46	yes	21.0	71568.0
L1 C/A	2	33	yes	21.0	71568.0
L1 C/A	8	36	yes	21.0	71568.0
L1 C/A	10	47	yes	21.0	71568.0
L1 C/A	27	39	yes	21.0	71568.0
L1 C/A	28	46	yes	21.0	71568.0
L1 C/A	31	47	yes	21.0	71568.0
L1 C/A	32	44	yes	21.0	71568.0
L2C	1	7	yes	34.8	0.0
L2C	8	13	yes	35.0	0.0
L2C	10	13	yes	34.9	0.0
L2C	27	13	yes	35.6	0.0
L2C	28	16	yes	34.8	0.0
L2C	31	6	yes	35.2	0.0
L2C	32	13	yes	35.4	0.0

Table 92: Telemetry decoder: L1 C/A LNAV subframes, L2C CNAV messages, E1 I/NAV words decoded, ephemeris obtained, and the first valid time of week.

15.2.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

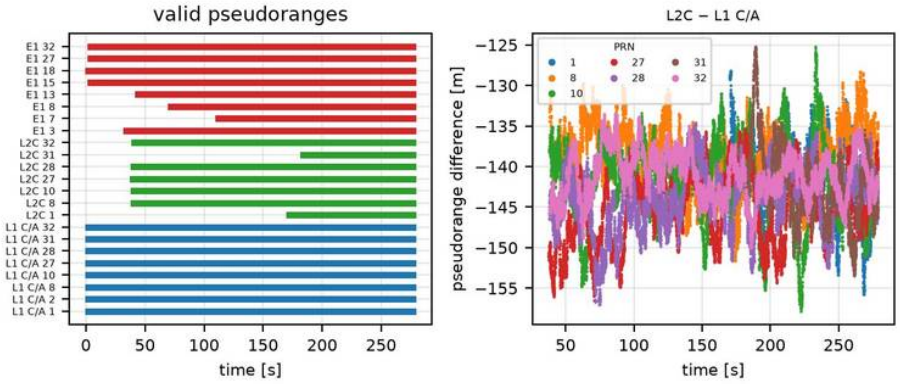


Figure 109: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L2C – L1 C/A	1	5420	36	-141.16	4.36	-0.0391
L2C – L1 C/A	8	12020	80	-138.93	3.76	-0.0024
L2C – L1 C/A	10	12026	80	-141.77	4.86	-0.0059
L2C – L1 C/A	27	12052	80	-145.24	4.16	0.0175
L2C – L1 C/A	28	12030	80	-143.94	3.79	0.0182
L2C – L1 C/A	31	4826	32	-142.15	4.64	-0.0814
L2C – L1 C/A	32	12000	80	-141.08	2.83	-0.0038

Table 93: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

15.2.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

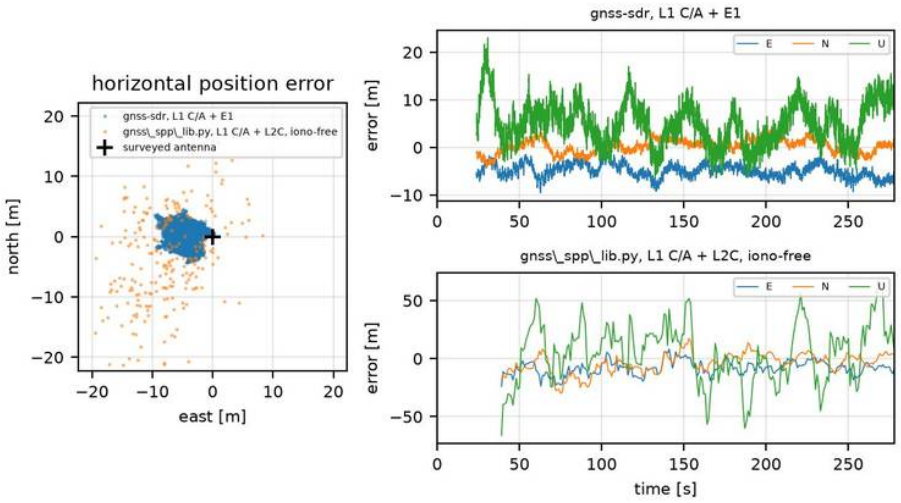


Figure 110: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

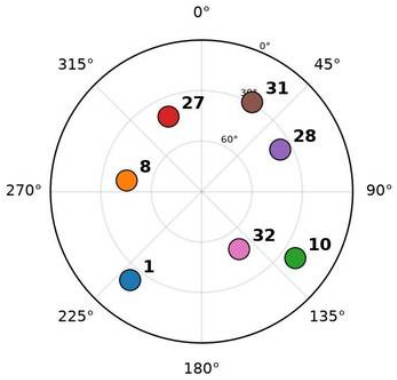


Figure 111: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, L1 C/A + E1**	**gnss_spp_lib.py, L1 C/A + L2C, iono-free**
Solutions	13243	240
Run with fix [%]	92	86
Mean E / N / U [m]	-4.9 / 0.3 / 5.1	-8.1 / -4.1 / 5.0
Std E / N / U [m]	1.4 / 1.5 / 4.5	6.0 / 8.6 / 25.4
RMS 2-D [m]	5.3	13.9
RMS 3-D [m]	8.6	29.4
95% horiz. [m]	7.3	24.5

Table 94: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

16 Records on 26/10/2026

16.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-26 19:00:08 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	0	3287000
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`, `e1_8bits4msps_e5a_8bits20msps.conf`	

Table 95: Metadata for recording `x310_20261026T190001Z`, captured starting 2026-10-26T19:00:08.297684Z.

16.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

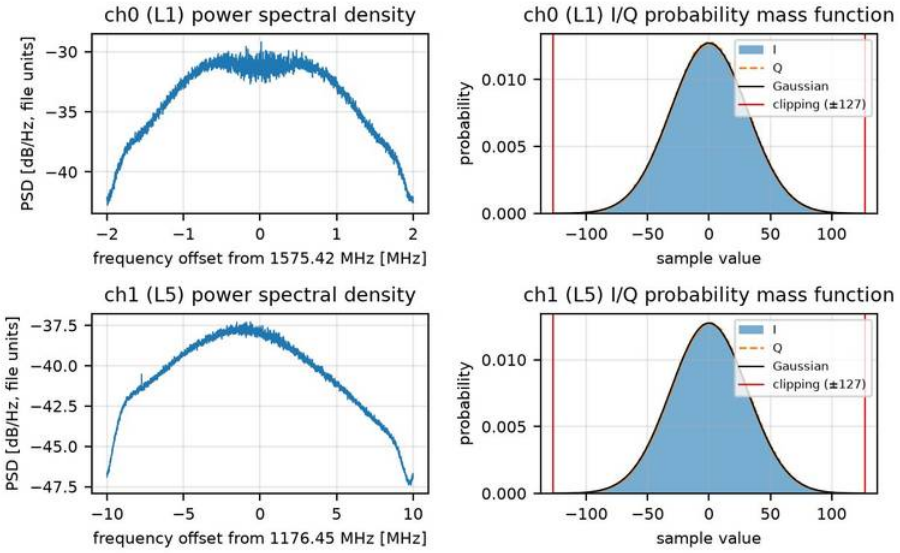


Figure 112: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Bytes per complex sample	2 (1 for I, 1 for Q)	2 (1 for I, 1 for Q)
Full scale	±127	±127
Gain [dB]	25	25
Std I / Q	31.4 / 31.4	31.3 / 31.3
Mean I / Q	0.02 / -0.15	-0.03 / 0.04
Min / max I	-127 / 127	-127 / 127
Min / max Q	-127 / 127	-127 / 127
Headroom[27]	4.0	4.1
Clipped [%]	0.004	0.005
Lost samples	0	3,287,000

Table 96: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[27]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

16.1.2 Signal conditioner

Channel	ch0 (L1 C/A)	ch1 (L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 97: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

16.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

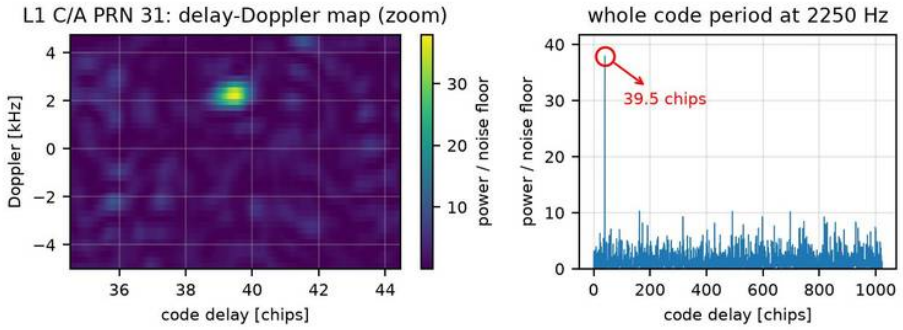


Figure 113: Acquisition, L1 C/A: delay–Doppler search of PRN 31; one sharp peak (test statistic 51 vs threshold 47) at 2250 Hz and 39.5 chips.

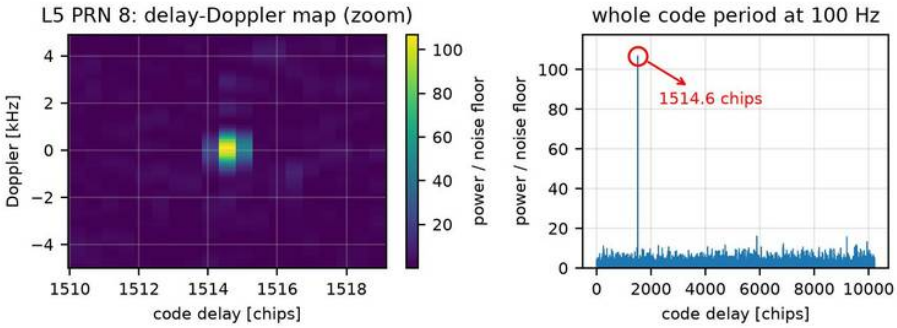


Figure 114: Acquisition, L5: delay–Doppler search of PRN 8; one sharp peak (test statistic 148 vs threshold 49) at 1278 Hz and 1514.6 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[28]
L1 C/A	3039	1	31	1.1
L5	1	1	8	3.0

Table 98: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[28]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

16.1.4 Tracking

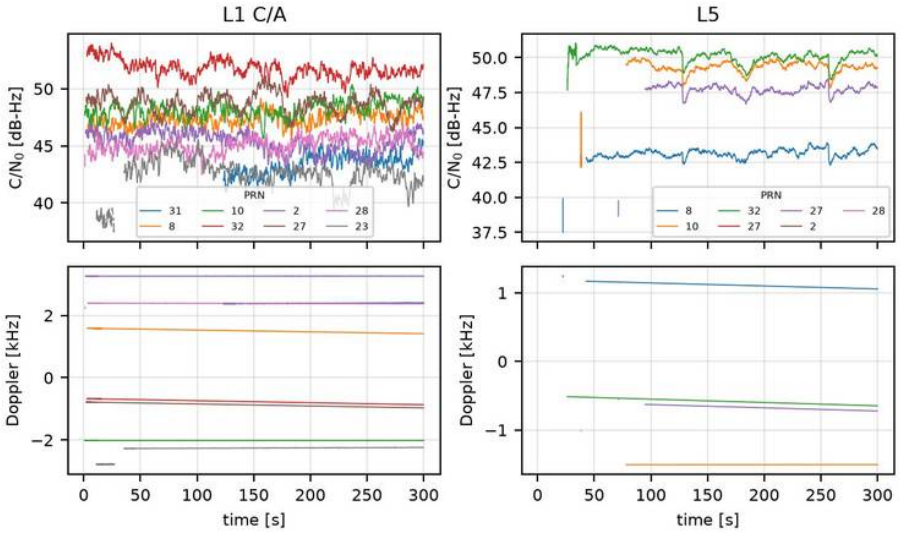


Figure 115: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

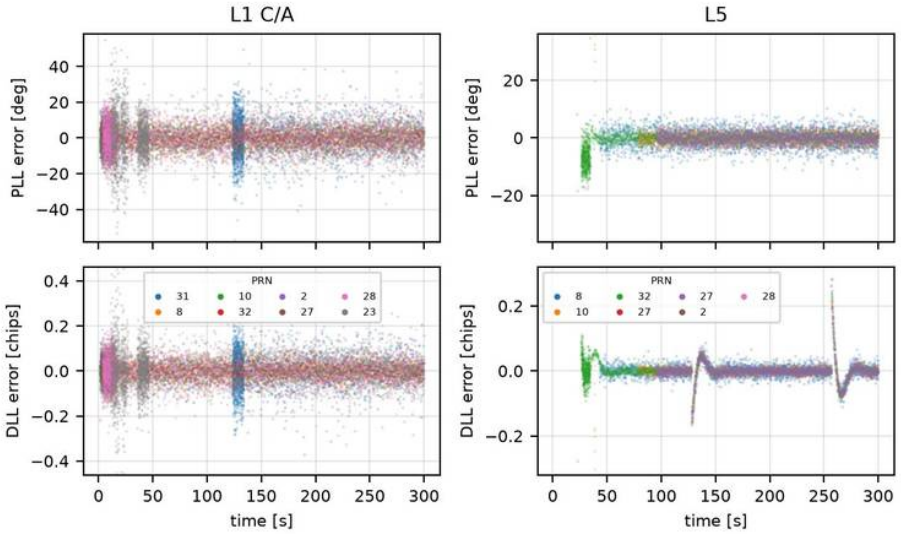


Figure 116: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

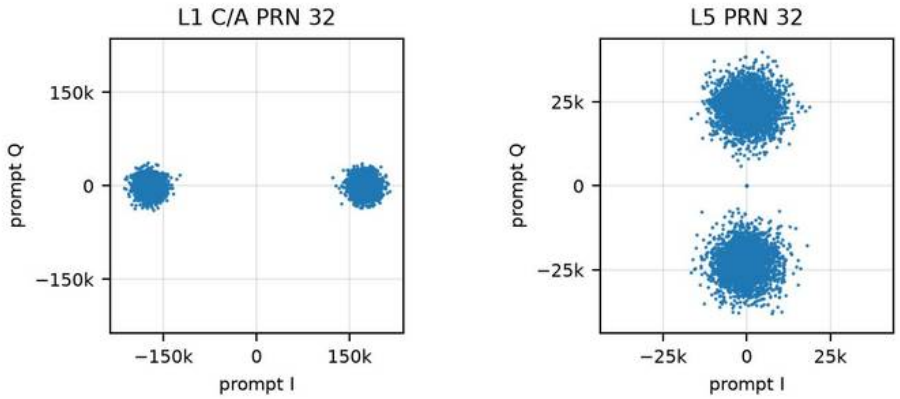


Figure 117: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	31	42.5	88	0	179	10.5	0.079
L1 C/A	1	8	47.2	95	0	298	6.3	0.047
L1 C/A	2	10	48.0	96	0	300	5.7	0.041
L1 C/A	3	32	52.6	95	0	299	3.5	0.025
L1 C/A	4	2	46.0	98	0	299	7.4	0.055
L1 C/A	5	27	49.0	95	0	299	5.1	0.037
L1 C/A	6	28	44.9	90	0	299	8.4	0.060
L1 C/A	7	23	42.0	77	1	287	13.4	0.097
L5	8	8	43.1	68	0	278	3.6	0.031
L5	9	10	48.1	34	2	278	2.9	0.029
L5	10	32	50.3	91	0	277	4.2	0.029
L5	12	27	28.9	0	2	32	–	–
L5	14	27	47.6	42	1	245	2.4	0.031
L5	15	2	28.7	0	2	16	–	–
L5	15	28	29.0	0	2	16	–	–

Table 99: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

16.1.5 Telemetry decoder

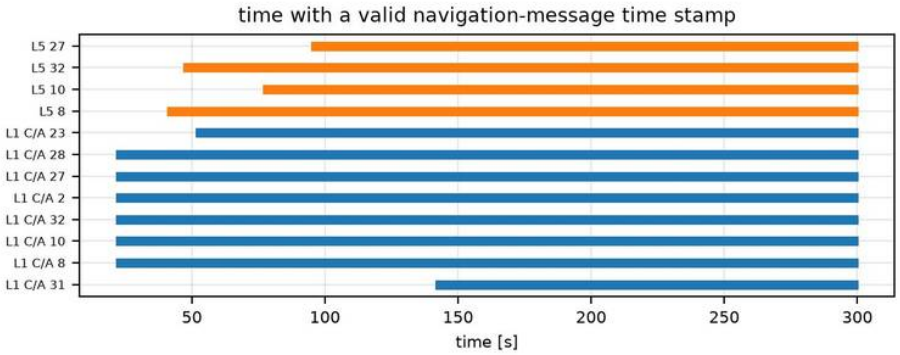


Figure 118: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
L1 C/A	2	46	yes	21.8	154848.0
L1 C/A	8	47	yes	21.8	154848.0
L1 C/A	10	47	yes	21.8	154848.0
L1 C/A	23	42	yes	51.8	154878.0
L1 C/A	27	47	yes	21.8	154848.0
L1 C/A	28	46	yes	21.8	154848.0
L1 C/A	31	27	yes	141.8	154968.0
L1 C/A	32	47	yes	21.8	154848.0
L5	8	27	yes	41.0	154867.3
L5	10	23	yes	76.9	154903.1
L5	27	21	yes	95.1	154921.4
L5	32	26	yes	46.9	154873.1

Table 100: Telemetry decoder: L1 C/A LNAV subframes, L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

16.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

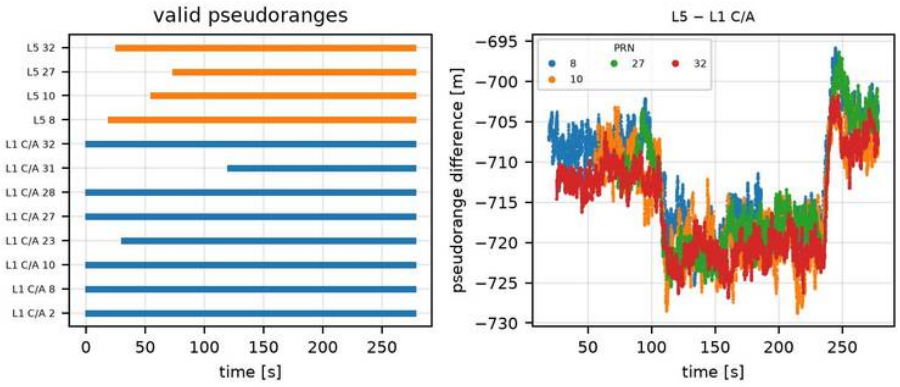


Figure 119: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L5 − L1 C/A	8	12444	83	-712.46	6.49	-0.0031
L5 − L1 C/A	10	10772	72	-715.60	6.47	0.0076
L5 − L1 C/A	27	9977	67	-714.78	7.17	0.0509
L5 − L1 C/A	32	12166	81	-715.94	6.04	0.0013

Table 101: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

16.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

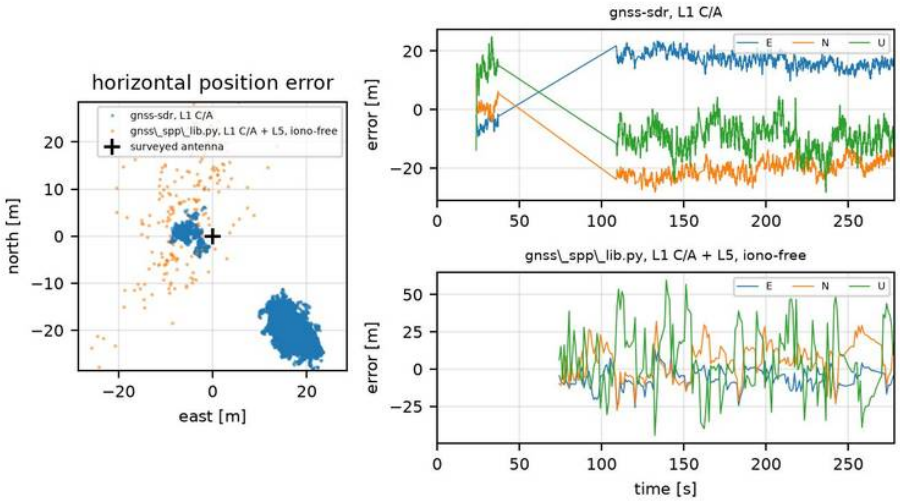


Figure 120: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

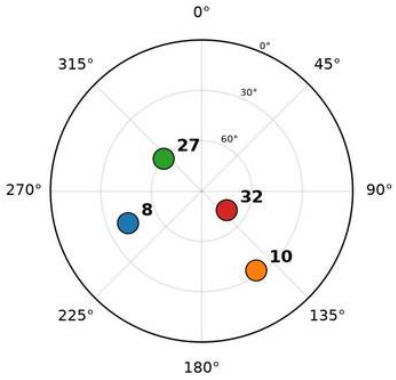


Figure 121: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	gnss-sdr, L1 C/A	gnss_spp_lib.py, L1 C/A + L5, iono-free
Solutions	8879	199
Run with fix [%]	66	72
Mean E / N / U [m]	15.2 / -18.6 / -7.8	-6.2 / 5.3 / 6.3
Std E / N / U [m]	6.3 / 6.1 / 7.9	6.7 / 11.8 / 21.9
RMS 2-D [m]	25.6	15.8
RMS 3-D [m]	27.9	27.7
95% horiz. [m]	31.8	28.5

Table 102: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

16.1.8 Signal conditioner (Galileo E1 + E5a)

Channel	ch0 (E1)	ch1 (E5a)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 103: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

16.1.9 Acquisition (Galileo E1 + E5a)

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

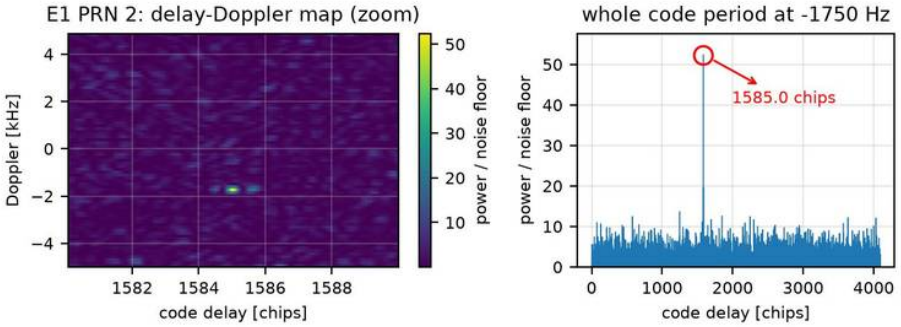


Figure 122: Acquisition, E1: delay–Doppler search of PRN 2; one sharp peak (test statistic 73 vs threshold 52) at -1750 Hz and 1585.0 chips.

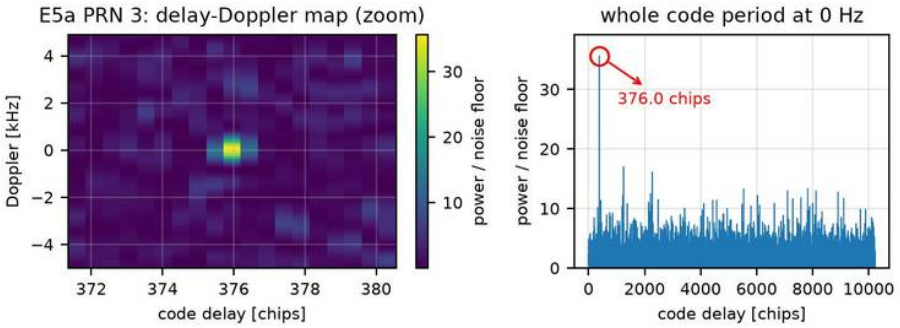


Figure 123: Acquisition, E5a: delay–Doppler search of PRN 3; one sharp peak (test statistic 49 vs threshold 49) at 2223 Hz and 376.0 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[29]
E1	7	1	2	1.4
E5a	74	1	3	1.0

Table 104: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[29] Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

16.1.10 Tracking (Galileo E1 + E5a)

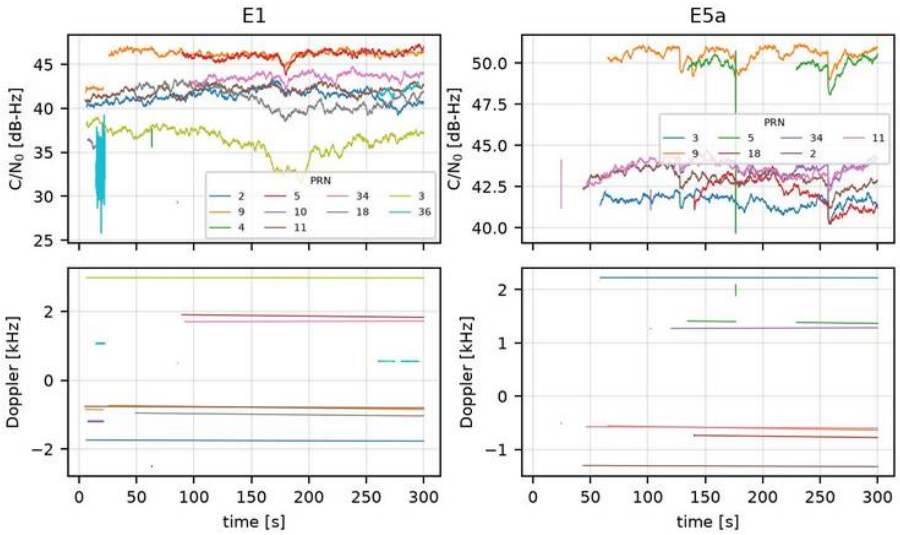


Figure 124: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

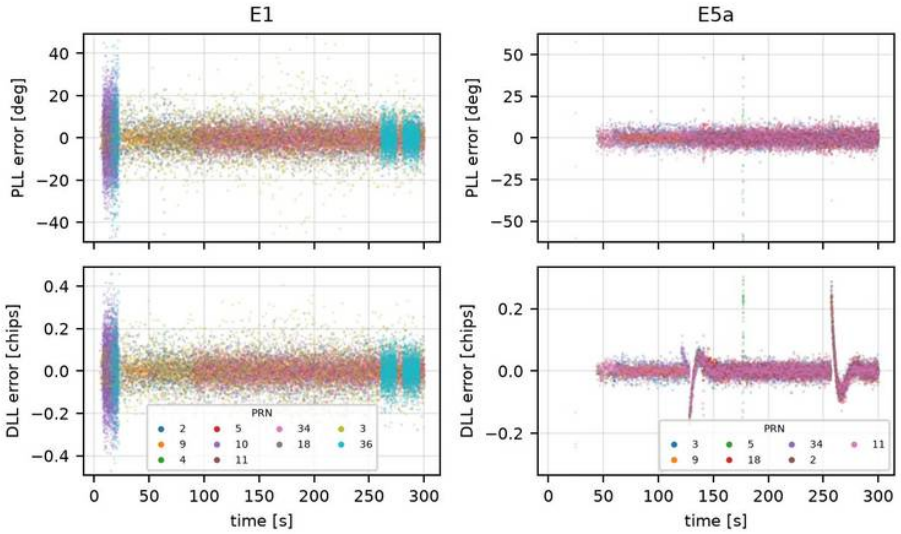


Figure 125: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

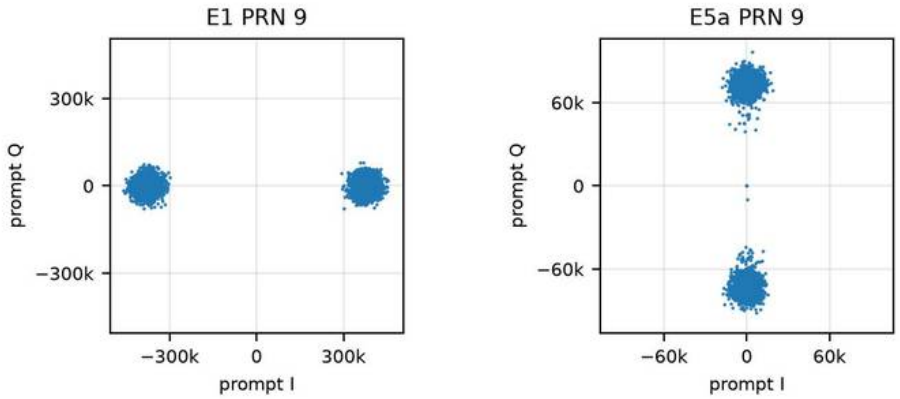


Figure 126: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
E1	0	2	41.4	98	0	298	6.2	0.066
E1	1	9	46.2	97	1	300	3.3	0.037
E1	2	4	38.5	1	4	63	57.4	0.106
E1	2	5	46.1	98	4	216	3.6	0.040
E1	2	10	36.2	64	4	21	12.3	0.125
E1	3	11	42.1	98	0	300	6.1	0.062
E1	4	34	43.3	70	4	299	4.9	0.052
E1	5	18	41.0	84	2	299	6.5	0.071
E1	6	3	36.7	98	0	298	11.9	0.121
E1	7	36	44.5	12	14	299	9.2	0.089
E5a	8	3	41.6	51	0	263	3.8	0.033
E5a	9	9	50.7	50	0	258	1.4	0.027
E5a	10	5	34.1	10	3	185	8.0	0.043
E5a	11	18	42.1	42	0	171	4.4	0.033
E5a	12	34	43.7	44	0	198	2.9	0.035
E5a	13	2	42.8	53	0	275	3.2	0.029
E5a	15	11	43.1	52	0	275	4.5	0.030

Table 105: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

16.1.11 Telemetry decoder (Galileo E1 + E5a)

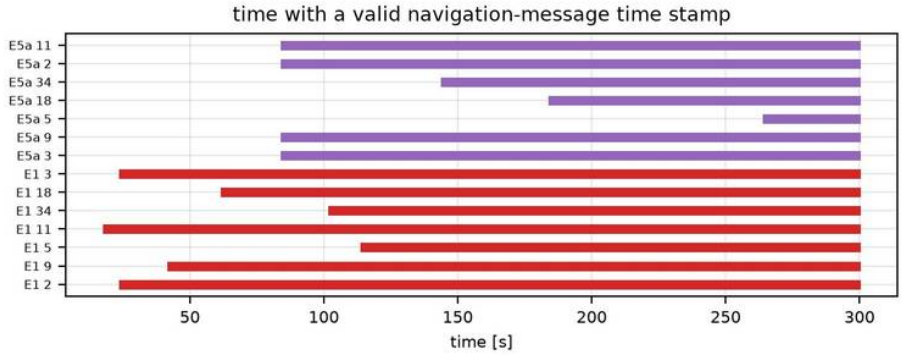


Figure 127: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	2	27	yes	23.8	154850.0
E1	3	25	yes	23.8	154850.0
E1	5	18	yes	113.8	154940.0
E1	9	25	yes	41.8	154868.0
E1	11	27	yes	17.8	154844.0
E1	18	24	yes	61.8	154888.0
E1	34	19	yes	101.8	154928.0
E5a	2	12	yes	84.1	154910.3
E5a	3	12	yes	84.1	154910.3
E5a	5	1	yes	264.0	155090.3
E5a	9	12	yes	84.0	154910.3
E5a	11	12	yes	84.0	154910.3
E5a	18	4	yes	184.0	155010.3
E5a	34	8	yes	144.1	154970.3

Table 106: Telemetry decoder: E1 I/NAV words, E5a F/NAV pages decoded, ephemeris obtained, and the first valid time of week.

16.1.12 Observables (Galileo E1 + E5a)

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

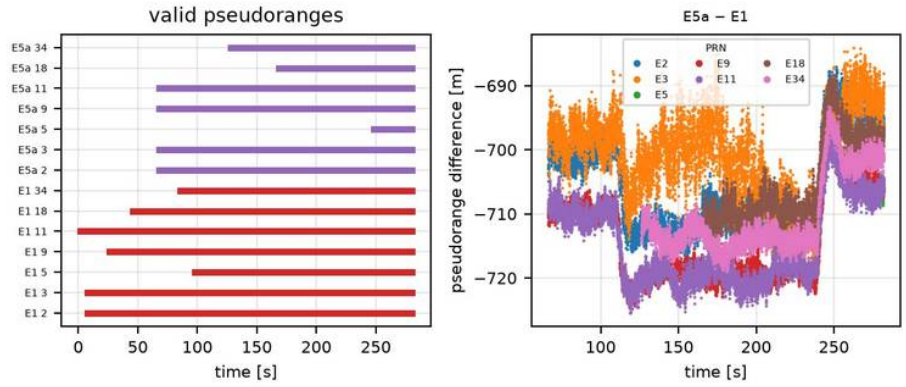


Figure 128: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
E5a − E1	2	9649	64	-705.48	7.29	0.0272
E5a − E1	3	9649	64	-700.47	6.77	0.0028
E5a − E1	5	1769	12	-703.48	2.96	-0.2477
E5a − E1	9	9649	64	-714.02	6.79	0.0259
E5a − E1	11	9649	64	-714.54	6.51	0.0227
E5a − E1	18	5271	35	-704.75	6.76	0.1565
E5a − E1	34	7027	47	-710.08	6.21	0.0963

Table 107: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

16.1.13 PVT (Galileo E1 + E5a)

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

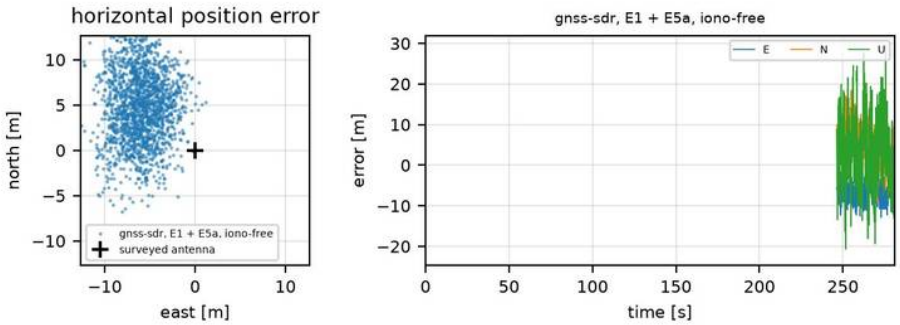


Figure 129: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution.

Solution	**gnss-sdr, E1 + E5a, iono-free**
Solutions	1732
Run with fix [%]	13
Mean E / N / U [m]	-6.0 / 4.9 / 3.0
Std E / N / U [m]	2.3 / 4.1 / 8.1
RMS 2-D [m]	9.0
RMS 3-D [m]	12.5
95% horiz. [m]	13.7

Table 108: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

16.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-26 19:30:07 UTC	
Frequency	1575.420 MHz (L1)	1227.600 MHz (L2)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	4.000 MS/s
Sample type	signed complex 16-bit samples (I+Q), little-endian, as received from the X310 (sc16)	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	4800.0 MB	4800.0 MB
Samples written	1,200,000,000	1,200,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca16bits4msps_l2c16bits4msps.conf`	

Table 109: Metadata for recording `x310_20261026T193001Z`, captured starting 2026-10-26T19:30:07.793328Z.

16.2.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

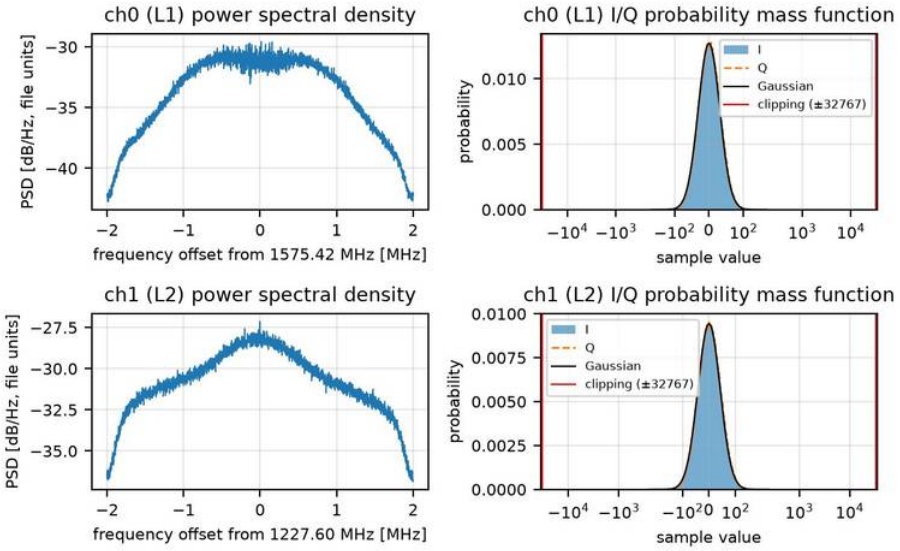


Figure 130: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1227.60
Rate [MS/s]	4	4
Format	ci16_le	ci16_le
Bytes per complex sample	4 (2 for I, 2 for Q)	4 (2 for I, 2 for Q)
Full scale	±32767	±32767
Gain [dB]	25	25
Std I / Q	31.4 / 31.4	42.3 / 42.3
Mean I / Q	-0.07 / -0.04	0.22 / 0.09
Min / max I	-162 / 151	-222 / 203
Min / max Q	-149 / 149	-202 / 210
Headroom[30]	1043.8	774.6
Clipped [%]	0.000	0.000
Lost samples	0	0

Table 110: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[30]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

16.2.2 Signal conditioner

Channel	ch0 (L1 C/A, E1)	ch1 (L2C)
File format	ci16_le	ci16_le
Source item type	ishort	ishort
Data type adapter	Ishort_To_Complex	Ishort_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	Pass_Through	Pass_Through
Rate [MS/s]	4 → 4	4 → 4

Table 111: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

16.2.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

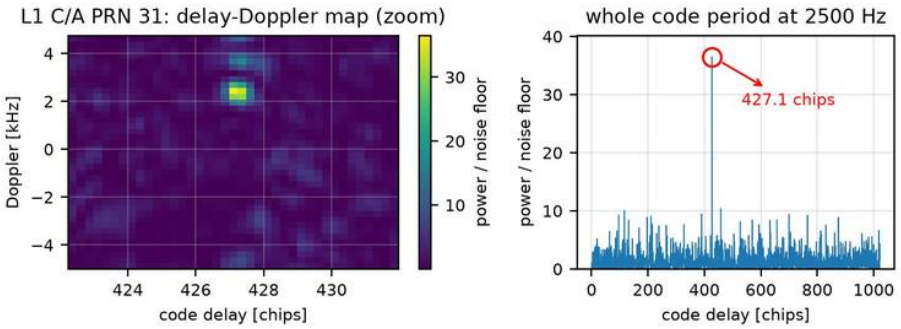


Figure 131: Acquisition, L1 C/A: delay–Doppler search of PRN 31; one sharp peak (test statistic 51 vs threshold 44) at 2500 Hz and 427.1 chips.

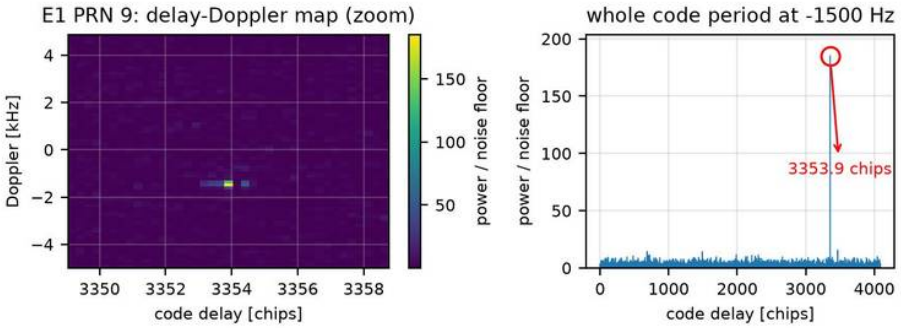


Figure 132: Acquisition, E1: delay–Doppler search of PRN 9; one sharp peak (test statistic 263 vs threshold 48) at -1500 Hz and 3353.9 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[31]
L1 C/A	26	1	31	1.2
L2C	not dumped (search grid too large); acquired satellites: see the tracking table			
E1	2	2	9	5.4

Table 112: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[31] Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

16.2.4 Tracking

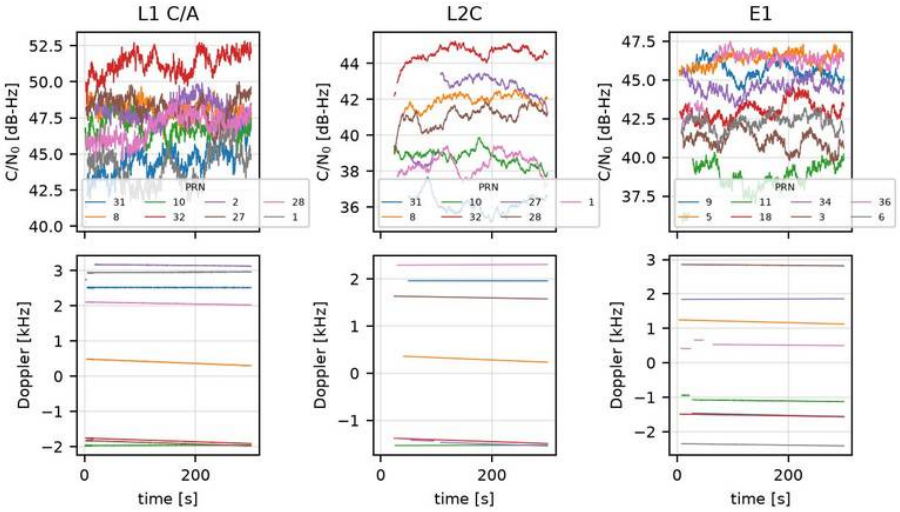


Figure 133: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

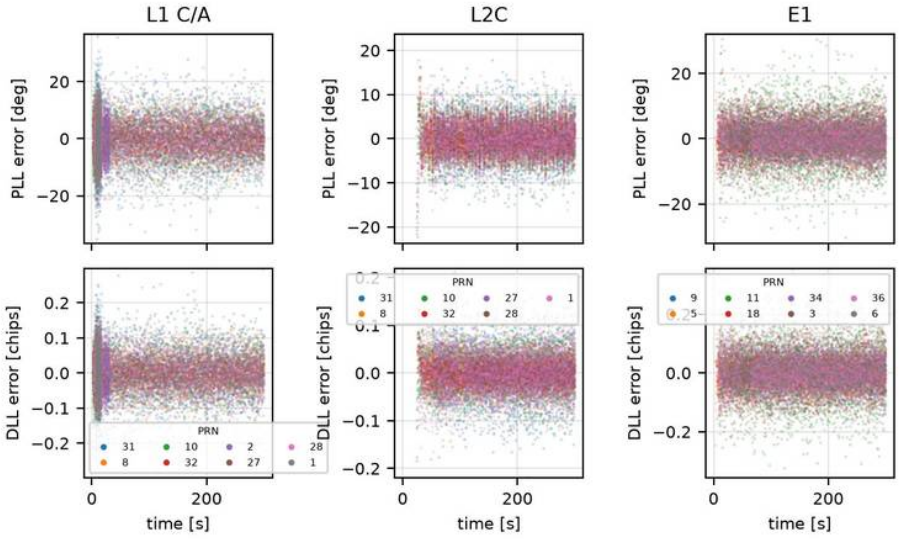


Figure 134: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

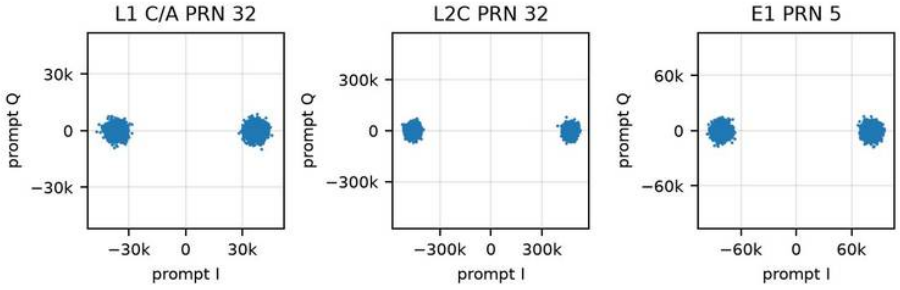


Figure 135: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	31	43.5	95	0	296	9.3	0.071
L1 C/A	1	8	48.4	96	0	297	5.5	0.044
L1 C/A	2	10	46.7	95	0	300	6.6	0.050
L1 C/A	3	32	50.8	98	0	298	4.1	0.031
L1 C/A	4	2	48.0	57	1	298	5.5	0.043
L1 C/A	5	27	48.2	94	0	298	5.5	0.041
L1 C/A	6	28	45.9	94	0	298	6.9	0.053
L1 C/A	7	1	44.3	90	0	297	9.3	0.069
L2C	8	31	36.2	91	0	276	5.1	0.056
L2C	9	8	42.0	94	0	276	3.2	0.030
L2C	10	10	38.8	99	0	276	4.2	0.044
L2C	11	32	44.7	99	0	276	3.5	0.023
L2C	12	27	42.7	85	1	276	3.1	0.031
L2C	13	28	41.1	100	0	276	3.6	0.035
L2C	14	1	38.4	98	0	276	4.3	0.046
E1	16	9	45.5	91	1	299	3.7	0.041
E1	17	5	46.4	100	0	296	3.7	0.041
E1	18	11	38.8	95	1	299	8.9	0.096
E1	19	18	43.0	99	0	299	5.3	0.060
E1	20	34	44.5	98	0	297	4.3	0.050
E1	21	3	41.1	98	0	297	6.6	0.075
E1	22	36	46.4	91	3	295	3.9	0.044
E1	23	6	42.3	99	0	296	5.4	0.064

Table 113: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

16.2.5 Telemetry decoder

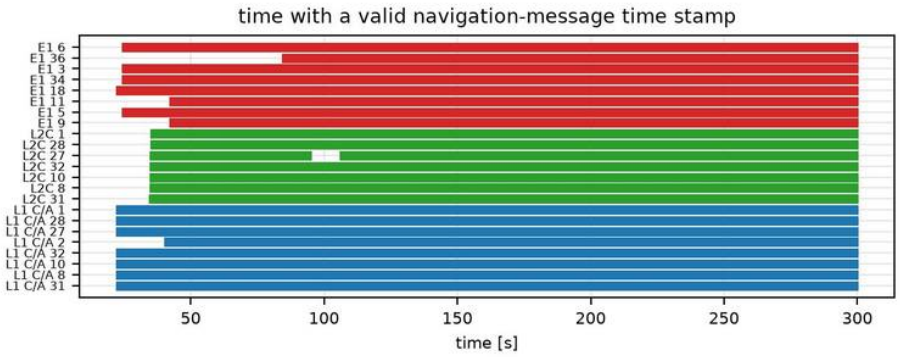


Figure 136: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	3	26	yes	24.3	156650.0
E1	5	26	yes	24.3	156650.0
E1	6	26	yes	24.3	156650.0
E1	9	25	yes	42.3	156668.0
E1	11	25	yes	42.3	156668.0
E1	18	26	yes	22.3	156648.0
E1	34	26	yes	24.3	156650.0
E1	36	21	yes	84.3	156710.0
L1 C/A	1	47	yes	22.3	156648.0
L1 C/A	2	44	yes	40.3	156666.0
L1 C/A	8	46	yes	22.3	156648.0
L1 C/A	10	46	yes	22.3	156648.0
L1 C/A	27	46	yes	22.3	156648.0
L1 C/A	28	46	yes	22.3	156648.0
L1 C/A	31	47	yes	22.3	156648.0
L1 C/A	32	46	yes	22.3	156648.0
L2C	1	15	yes	35.1	0.0
L2C	8	13	yes	34.7	0.0
L2C	10	13	yes	34.8	0.0
L2C	27	9	yes	34.9	0.0
L2C	28	15	yes	35.0	0.0
L2C	31	13	yes	34.7	0.0
L2C	32	13	yes	34.9	0.0

Table 114: Telemetry decoder: L1 C/A LNAV subframes, L2C CNAV messages, E1 I/NAV words decoded, ephemeris obtained, and the first valid time of week.

16.2.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

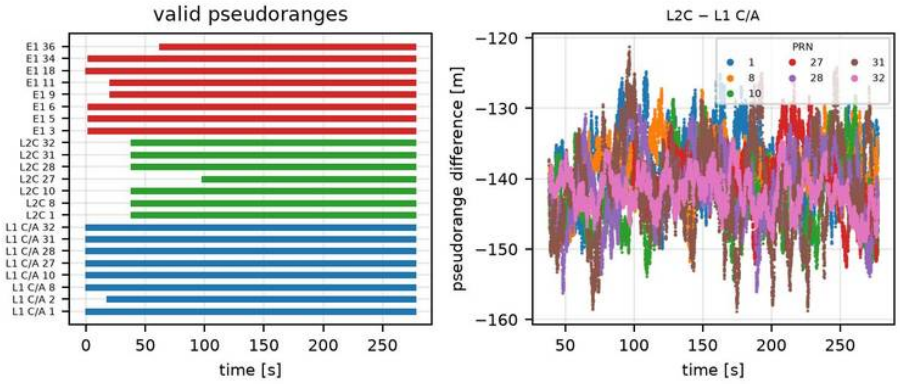


Figure 137: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L2C – L1 C/A	1	11953	80	-138.46	4.58	0.0012
L2C – L1 C/A	8	11970	80	-138.63	3.59	0.0075
L2C – L1 C/A	10	11967	80	-142.09	4.87	0.0069
L2C – L1 C/A	27	8972	60	-139.85	4.53	-0.0158
L2C – L1 C/A	28	11957	80	-140.93	4.98	0.0068
L2C – L1 C/A	31	11973	80	-141.25	6.69	0.0048
L2C – L1 C/A	32	11964	80	-141.84	2.75	-0.0096

Table 115: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

16.2.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

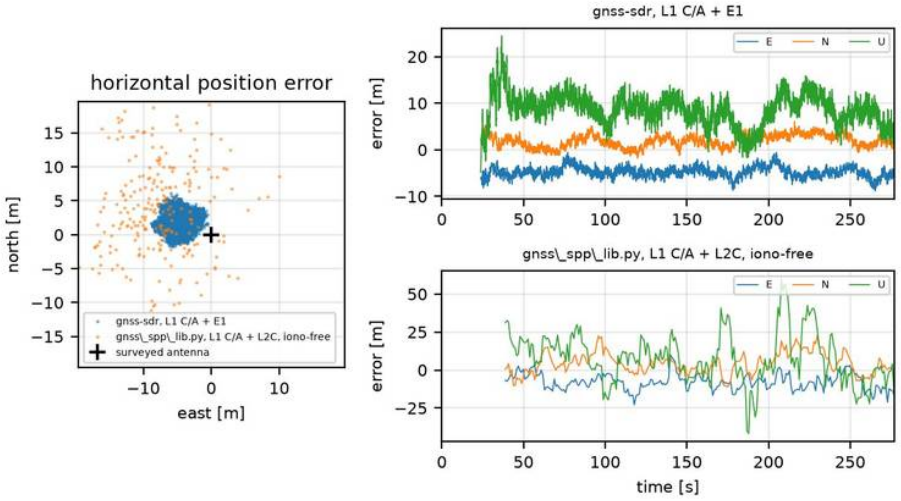


Figure 138: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp_lib.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

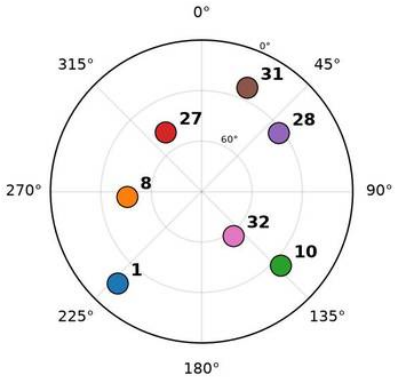


Figure 139: PVT: sky plot of the GPS satellites in the gnss_spp_lib.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, L1 C/A + E1**	**gnss_spp_lib.py, L1 C/A + L2C, iono-free**
Solutions	13178	239
Run with fix [%]	92	86
Mean E / N / U [m]	-4.8 / 1.9 / 8.6	-7.8 / 4.2 / 8.0
Std E / N / U [m]	1.1 / 1.3 / 3.3	5.3 / 7.0 / 16.6
RMS 2-D [m]	5.4	12.5
RMS 3-D [m]	10.7	22.3
95% horiz. [m]	7.2	20.8

Table 116: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp_lib.py computes one per second).

Part IV
Permanent scintillating data

This part holds the recordings classified with moderate or strong scintillation (Section 9), which are kept in the permanent NAS storage. To keep the report small enough to be shared, each recording is shown with its metadata and its scintillation outputs only; the full GNSS-SDR outputs are shown while the recording is in Part III.

Table 117 shows the days of the permanently stored recordings, with the same colors as the temporary calendar: green means 2 recordings, blue means 1 recording, and no color means no recording was stored.

No recording stored yet.

Table 117: Permanent recording calendar.

No recording has been classified and stored permanently yet.

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[1] B. H. Briggs and I. A. Parkin. "On the Variation of Radio Star and Satellite Scintillations with Zenith Angle". In: *Journal of Atmospheric and Terrestrial Physics* 25.6 (1963), pp. 339–366. DOI: 10.1016/0021-9169(63)90150-8.

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[3] Rafael Lopes et al. "Linear Time-Invariant Filtering for Real-Time Monitoring of Ionospheric Scintillation in GNSS Receivers". In: *GPS Solutions* 27 (2023). DOI: 10.1007/s10291-023-01470-0.

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[6] Friederike Wendler. *Scintillation Processing with the Software Receiver: A Short Introduction*. Tech. rep. Deutsches Zentrum für Luft- und Raumfahrt (DLR), July 2016.